Backup — Z24 full results gallery

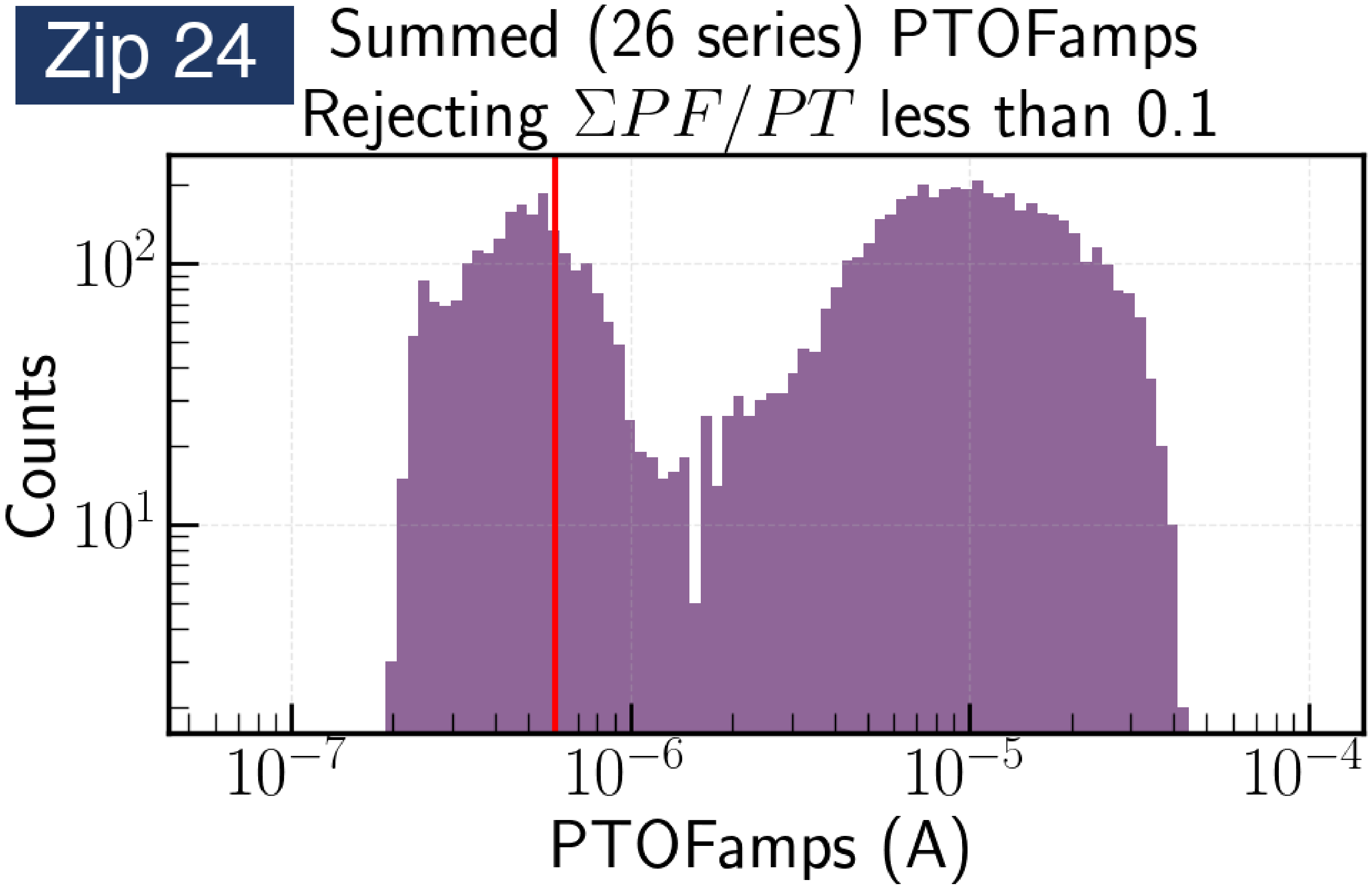
every diagnostic and result figure of the pipeline, in processing order

- Dataset: the K-line selection window
- Example grids: LP + fit and raw vs fit, event × channel
- Fit diagnostics: NRMSE, fitted pretrigger, τ_rise / τ_fall distributions
- Aligned fans: no cut → NRMSE ≤ 0.4 → + τ_rise ≤ 0.3 ms; kept vs rejected; measured traces
- Special populations: NRMSE-rejected events, well-fit slow-rise events
- Templates: PCA nxm0–4 per channel, summed PT / PS1 / PS2

Z24 · K-line selection window

figure: kline_zip24.png

Summed PTOFamps spectrum of Z24; red line = Prof. Saab's 10.37 keV K-line position; the ×/÷ 1.35 window around it is the event selection; everything inside is cached raw.

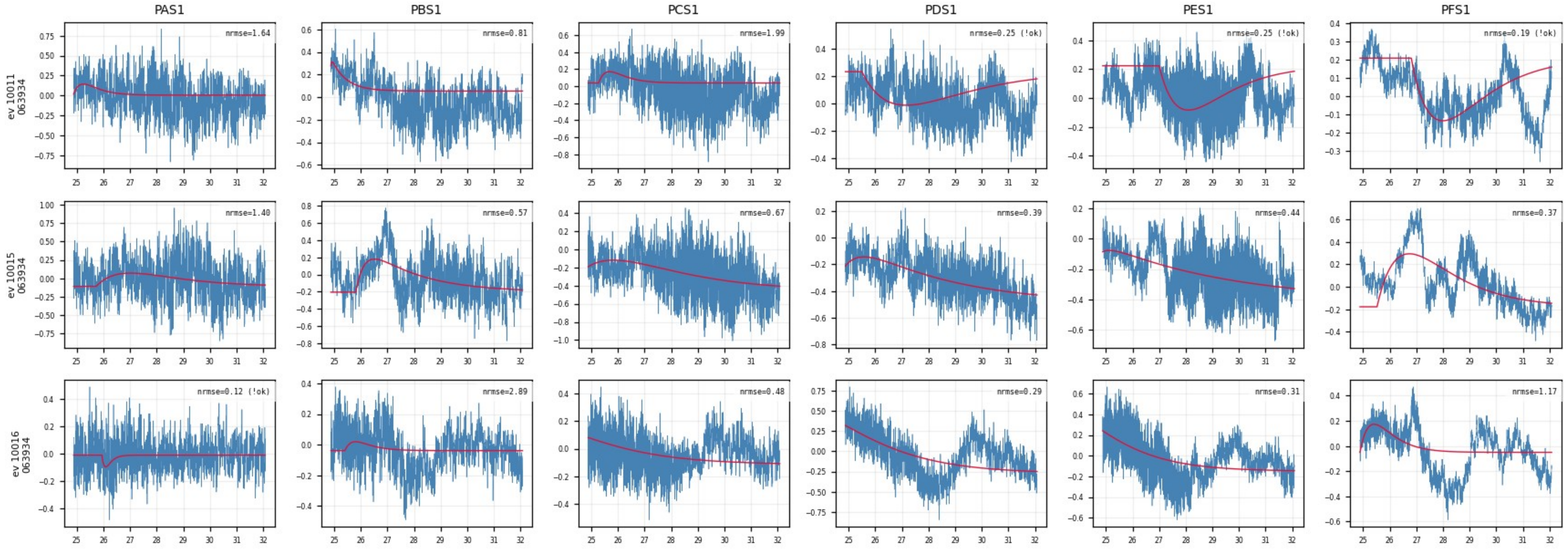


Z24 PTOFamps spectrum with the K-line selection window

Z24 · fit_examples — LP trace vs 2-exp fit (1/10)

figure: zip24_fit_examples.png — event rows 1–3 of 15, left half of the channels

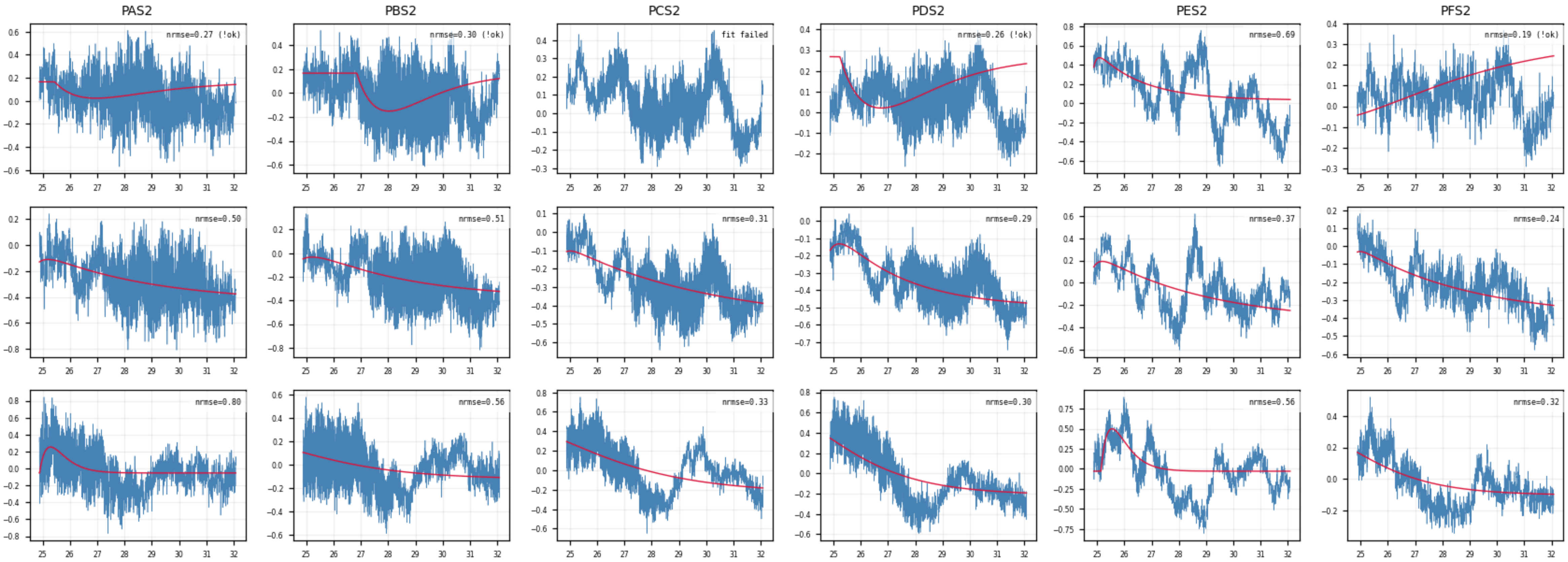
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (2/10)

figure: zip24_fit_examples.png — event rows 1–3 of 15, right half of the channels

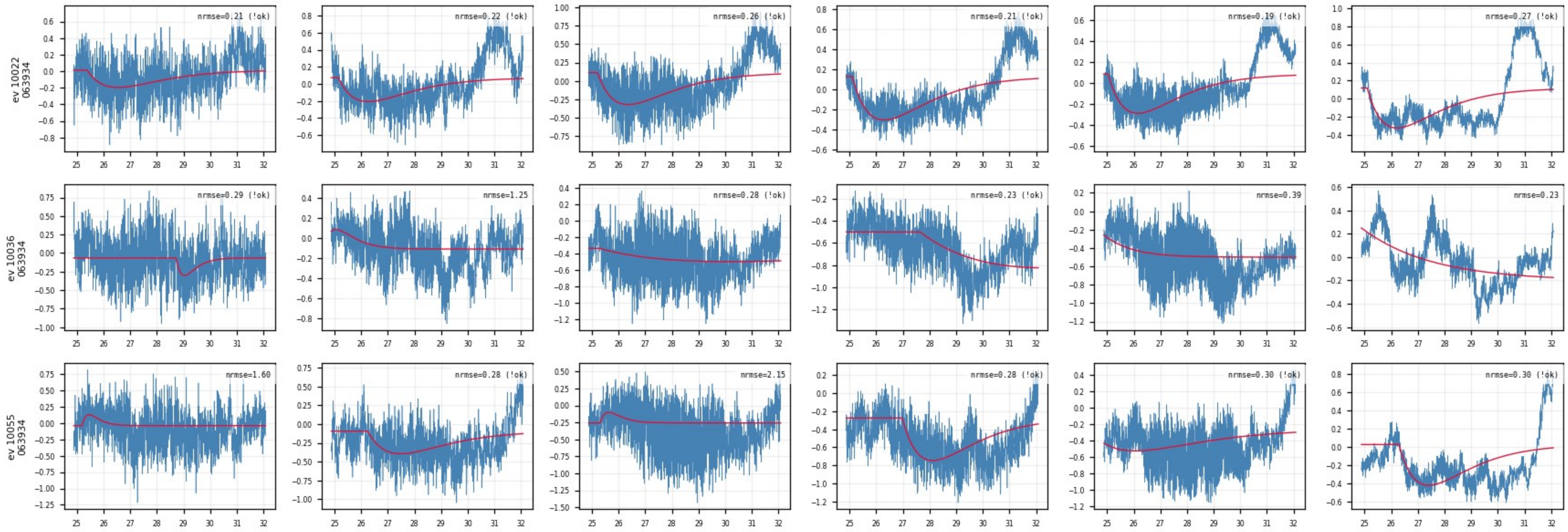
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (3/10)

figure: zip24_fit_examples.png — event rows 4–6 of 15, left half of the channels

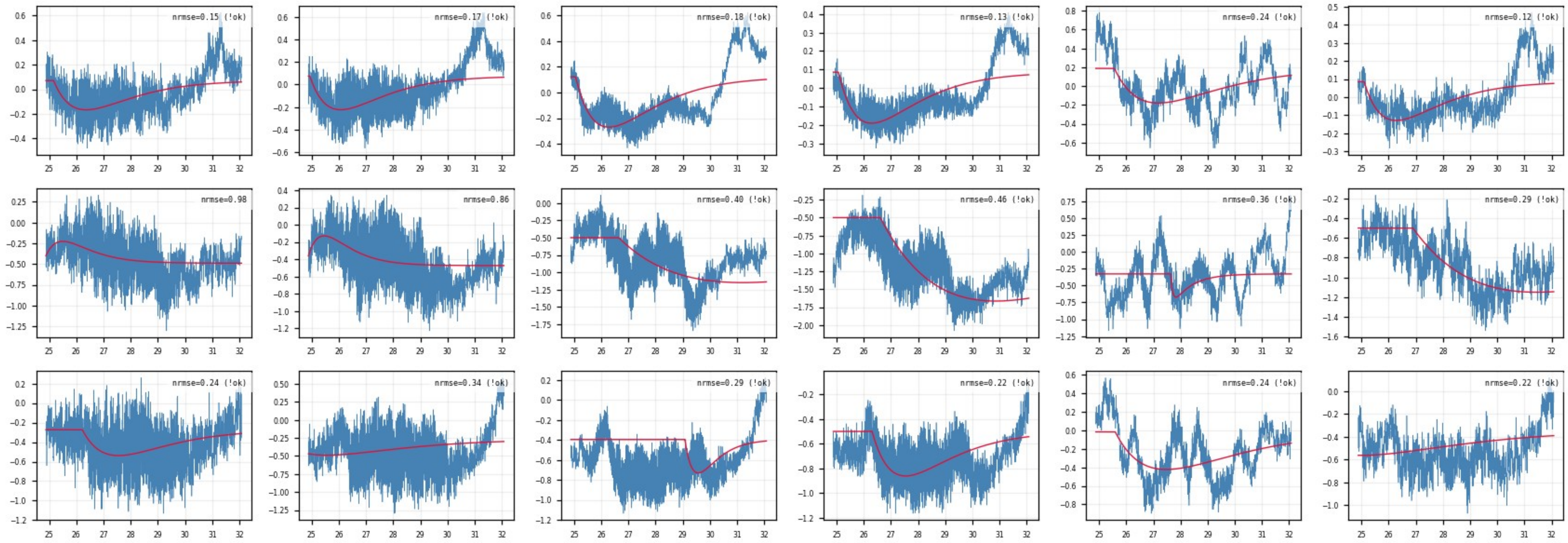
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (4/10)

figure: zip24_fit_examples.png — event rows 4–6 of 15, right half of the channels

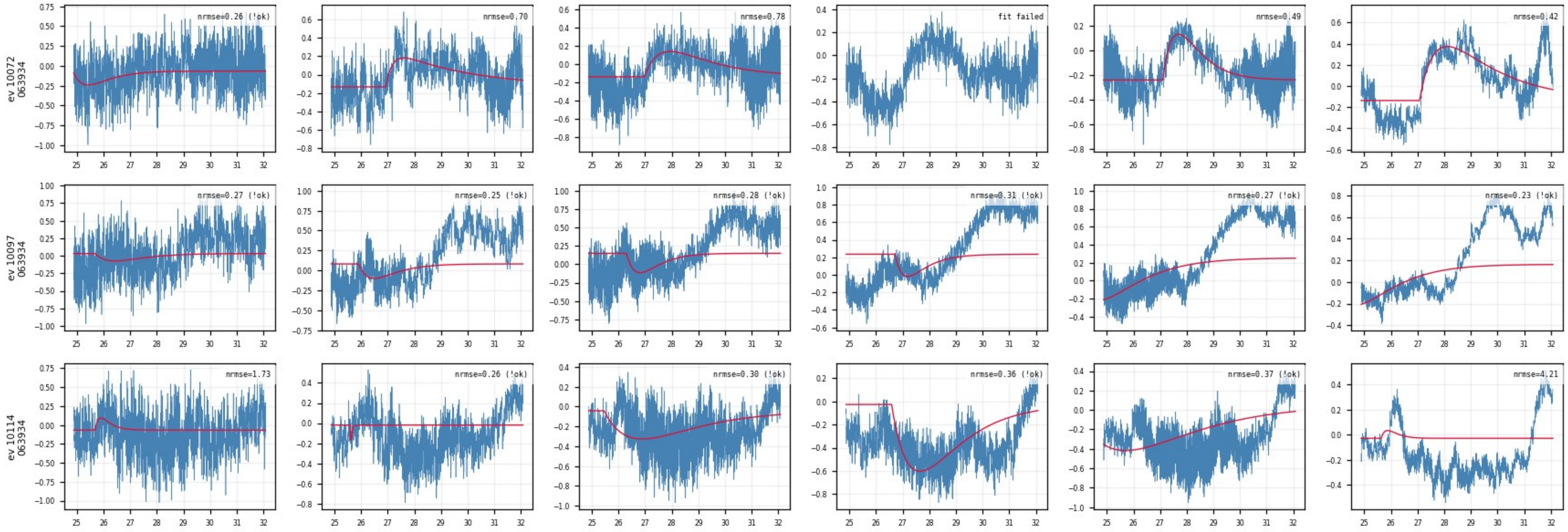
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (5/10)

figure: zip24_fit_examples.png — event rows 7–9 of 15, left half of the channels

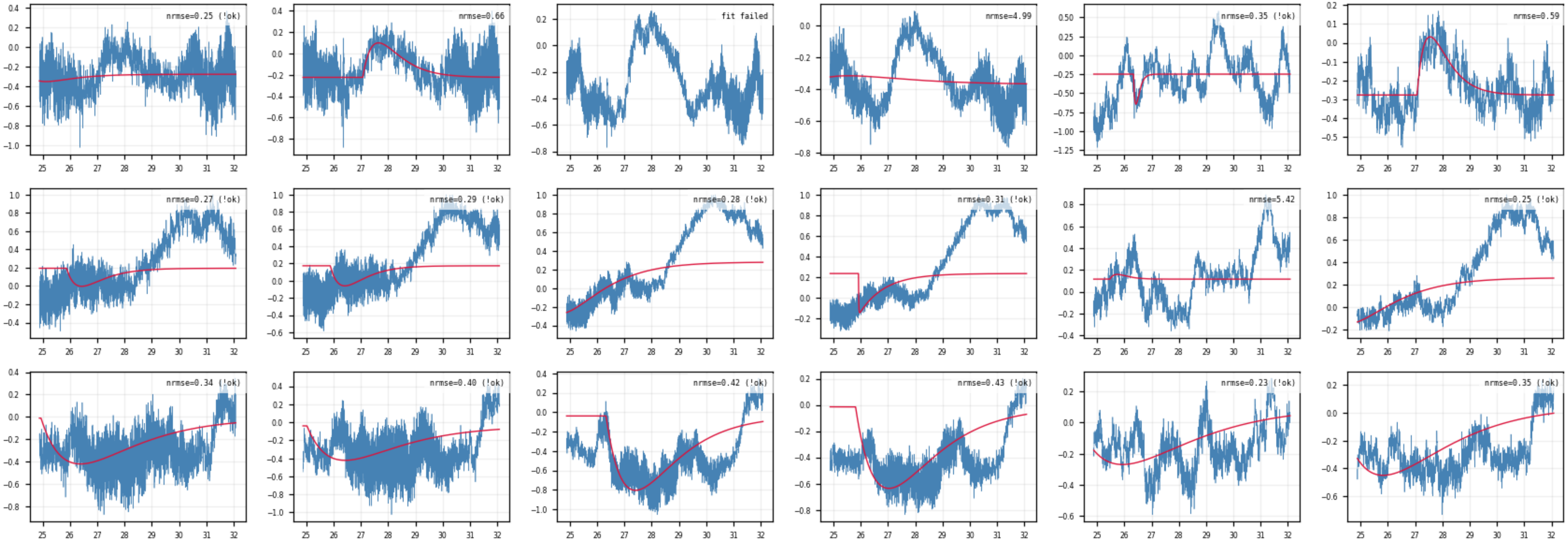
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (6/10)

figure: zip24_fit_examples.png — event rows 7–9 of 15, right half of the channels

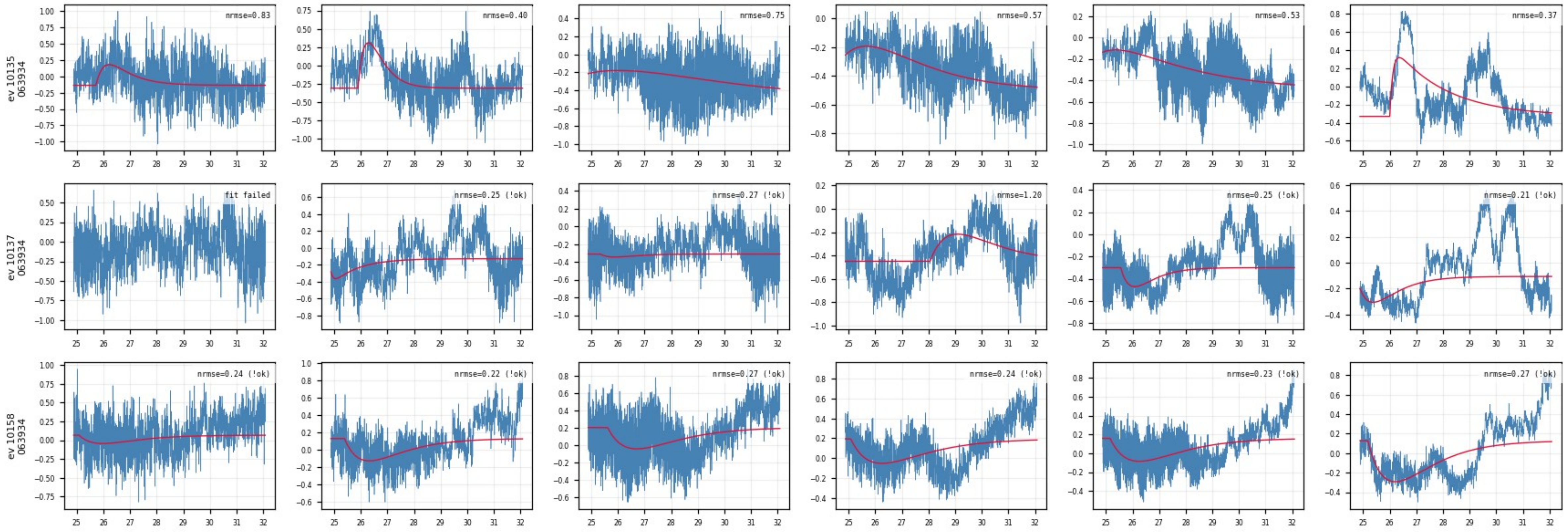
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (7/10)

figure: zip24_fit_examples.png — event rows 10–12 of 15, left half of the channels

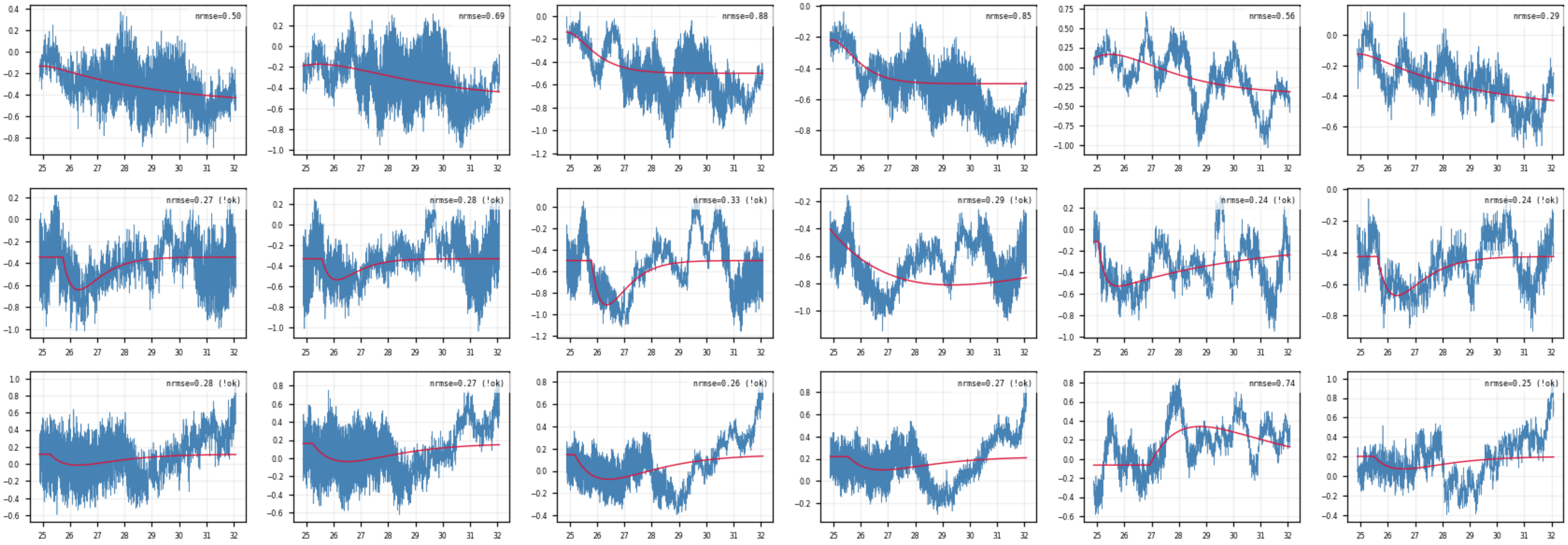
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (8/10)

figure: zip24_fit_examples.png — event rows 10–12 of 15, right half of the channels

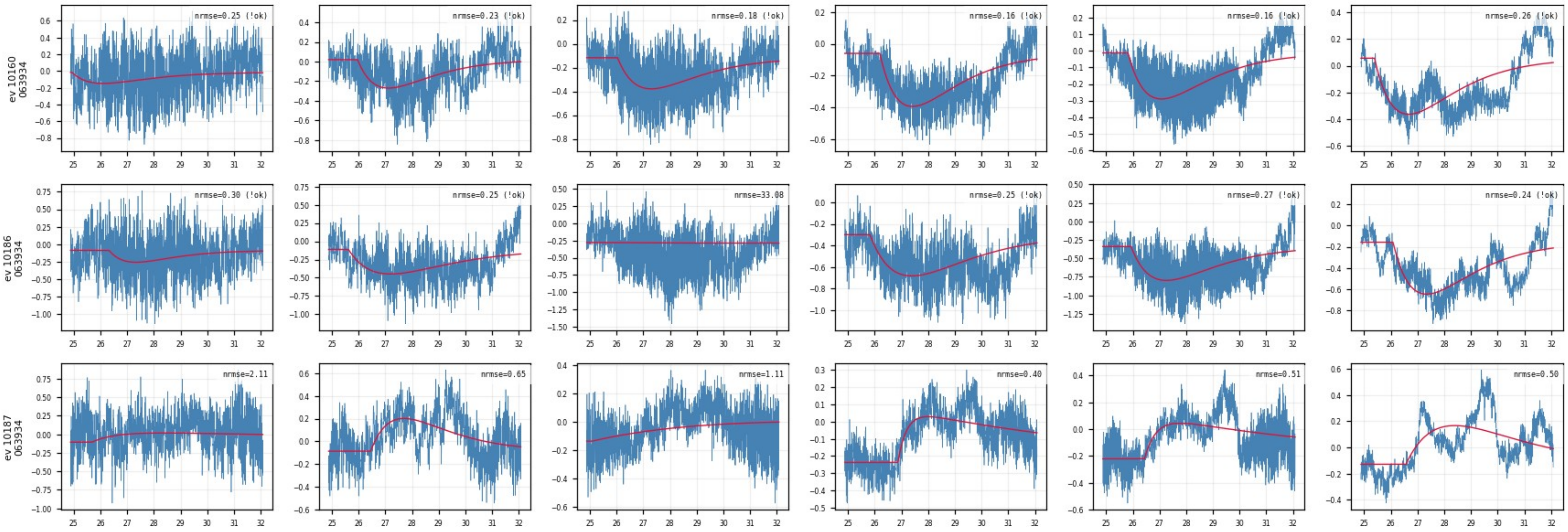
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (9/10)

figure: zip24_fit_examples.png — event rows 13–15 of 15, left half of the channels

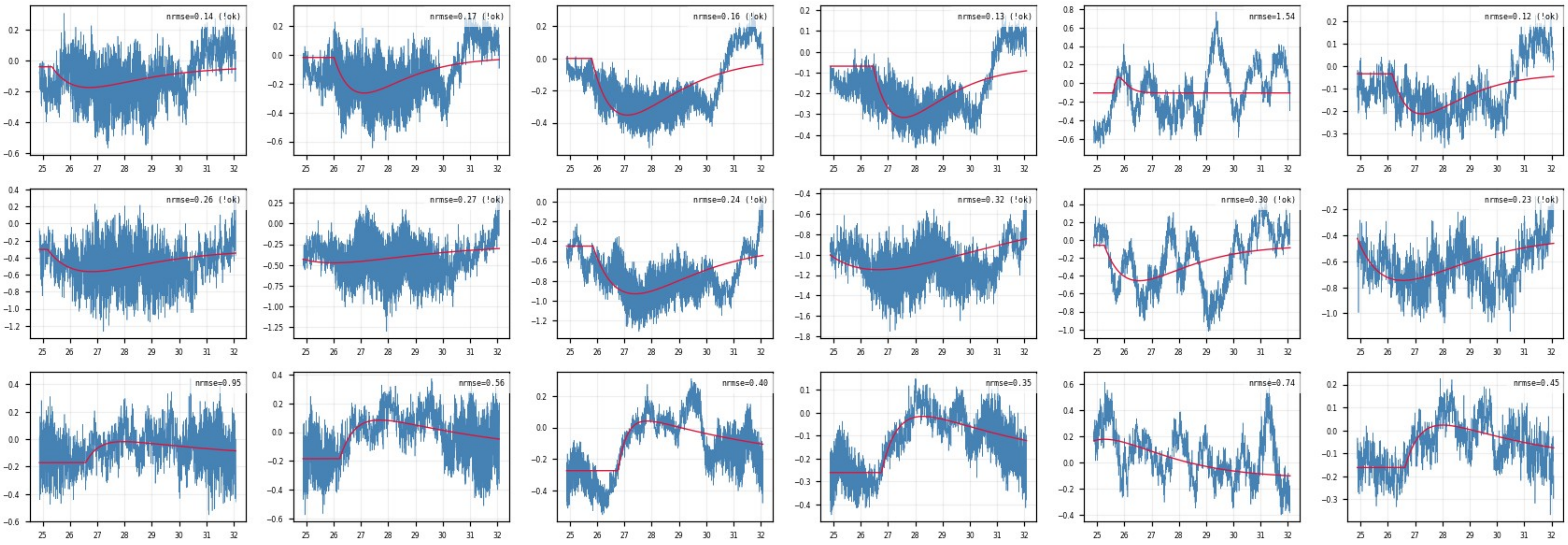
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · fit_examples — LP trace vs 2-exp fit (10/10)

figure: zip24_fit_examples.png — event rows 13–15 of 15, right half of the channels

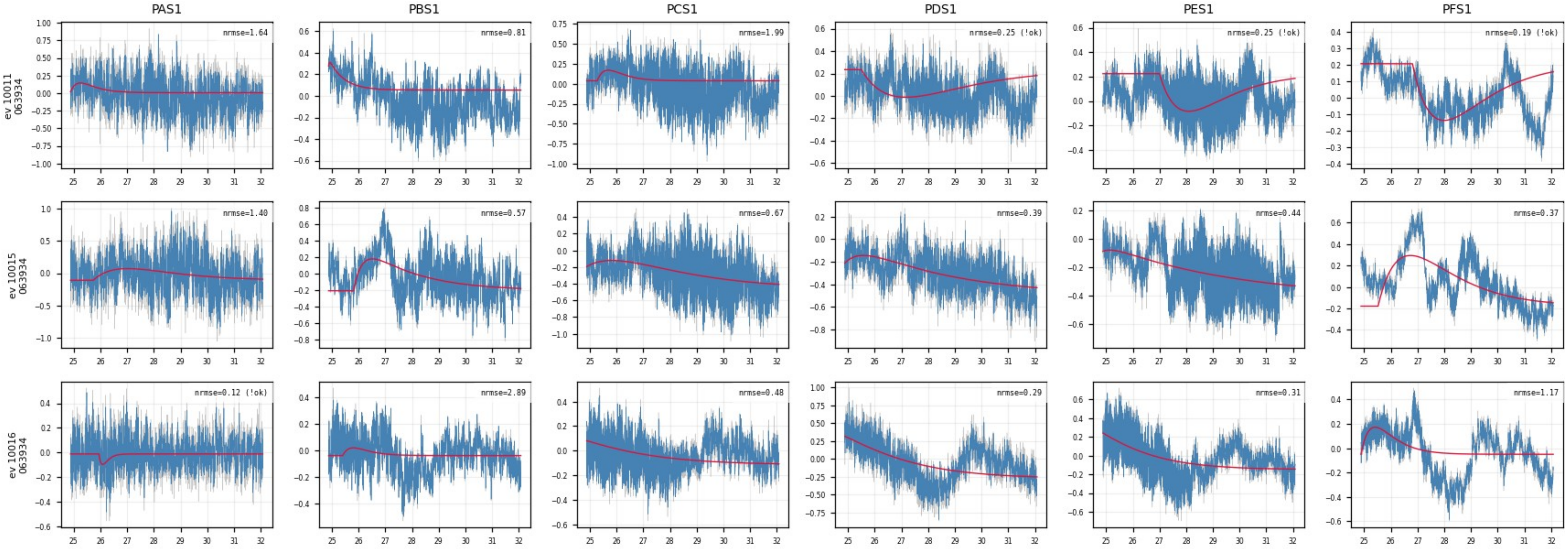
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z24 · raw_vs_fit_examples — raw trace vs fit (1/10)

figure: zip24_raw_vs_fit_examples.png — event rows 1–3 of 15, left half of the channels

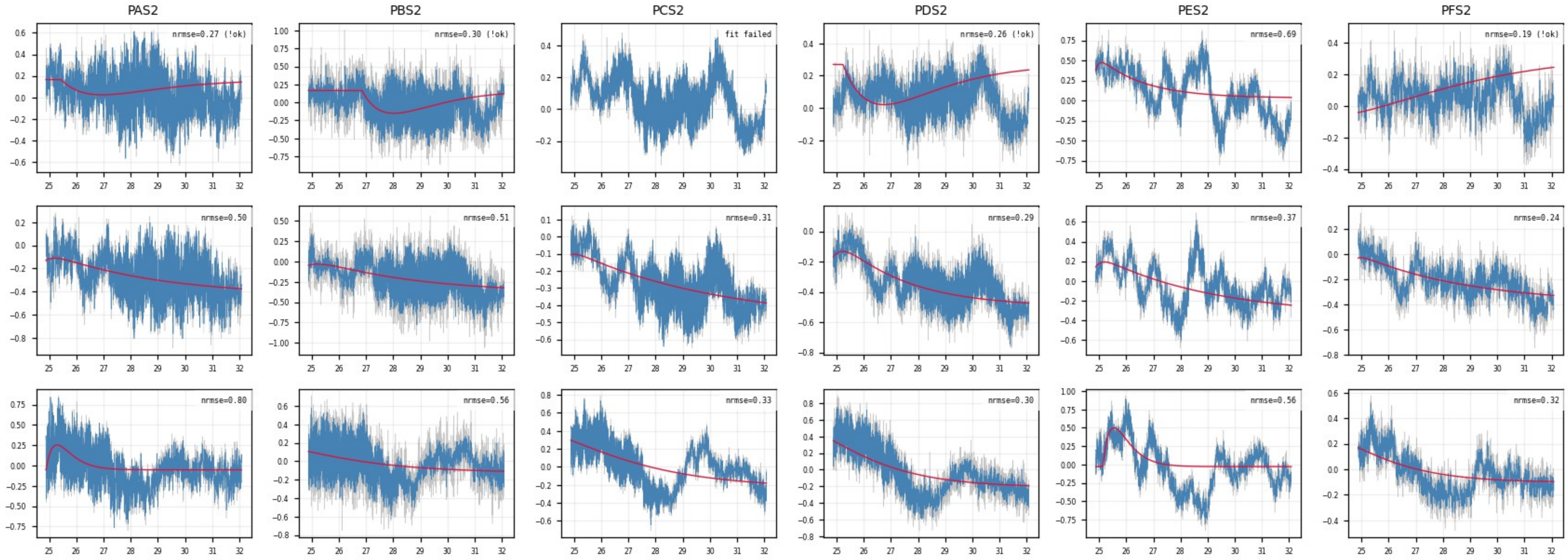
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (2/10)

figure: zip24_raw_vs_fit_examples.png — event rows 1–3 of 15, right half of the channels

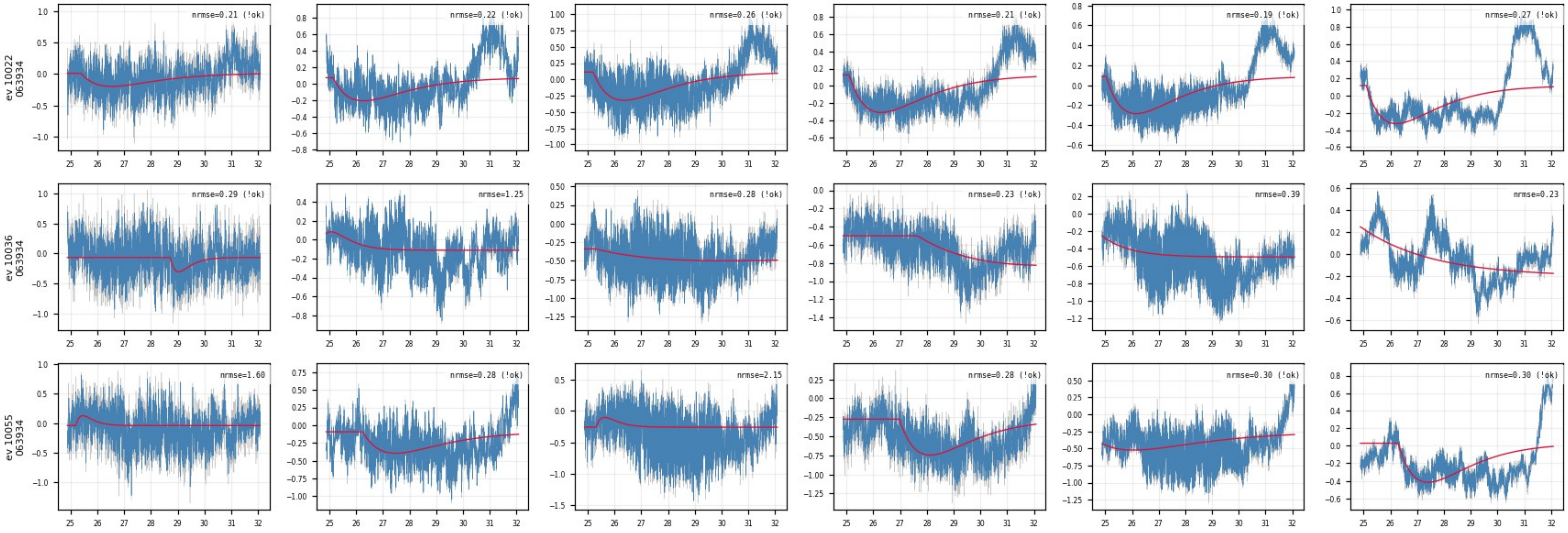
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (3/10)

figure: zip24_raw_vs_fit_examples.png — event rows 4–6 of 15, left half of the channels

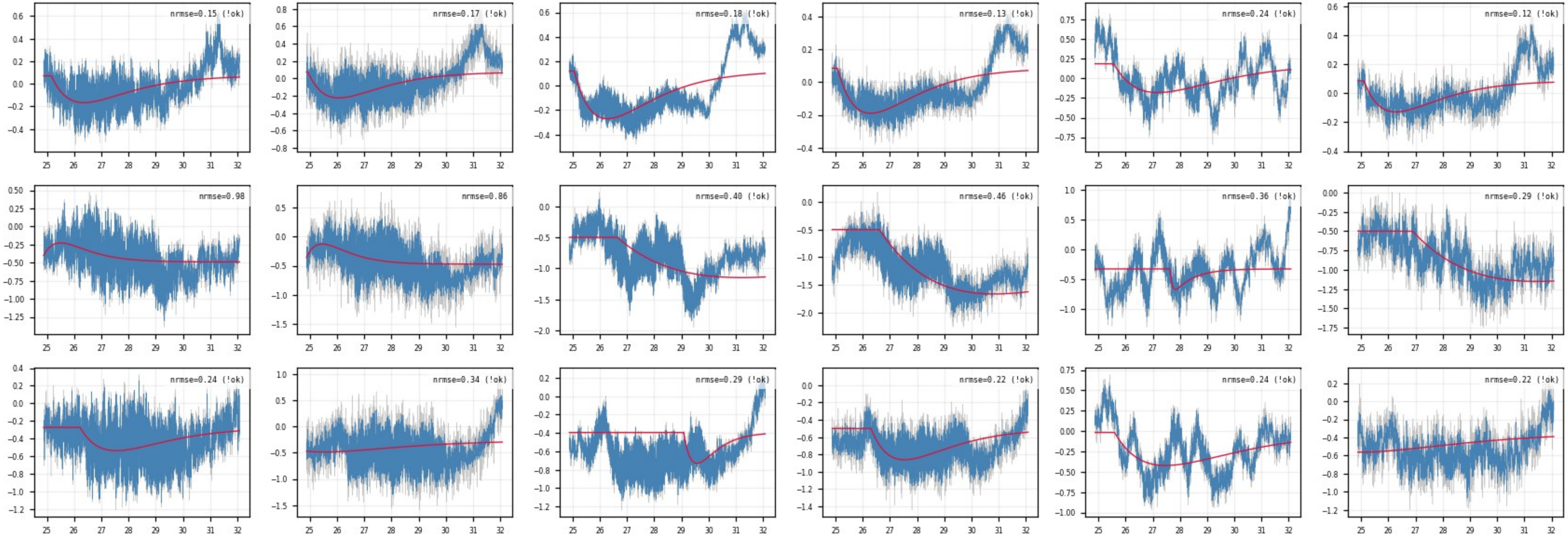
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (4/10)

figure: zip24_raw_vs_fit_examples.png — event rows 4–6 of 15, right half of the channels

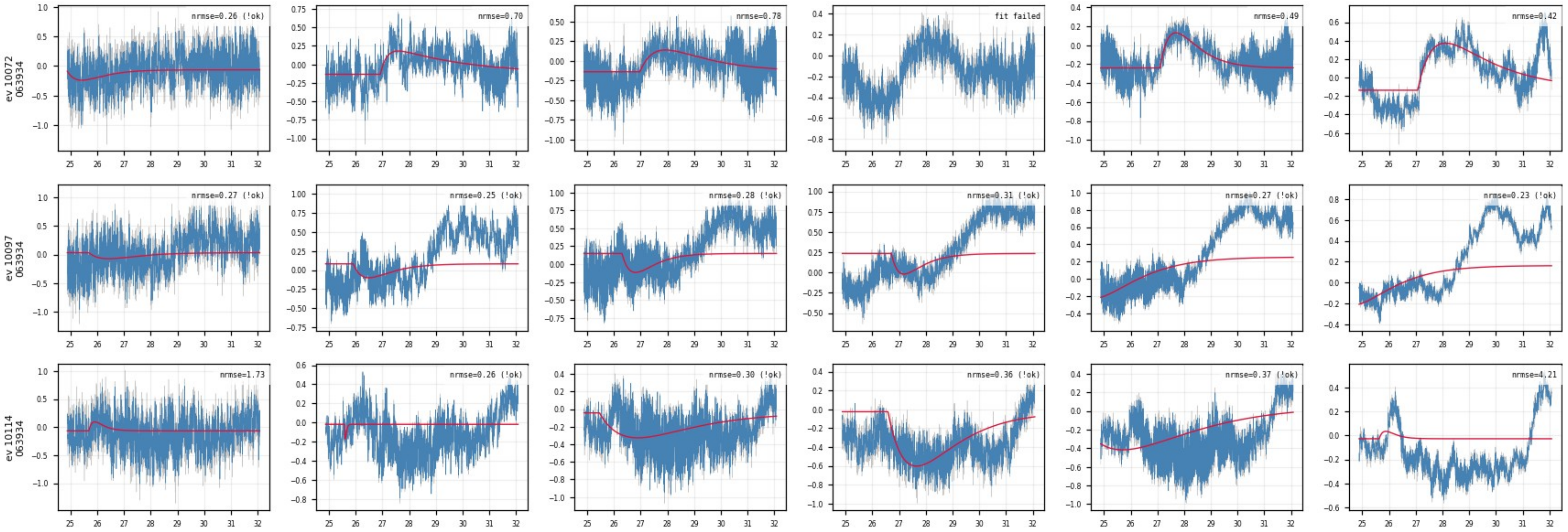
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (5/10)

figure: zip24_raw_vs_fit_examples.png — event rows 7–9 of 15, left half of the channels

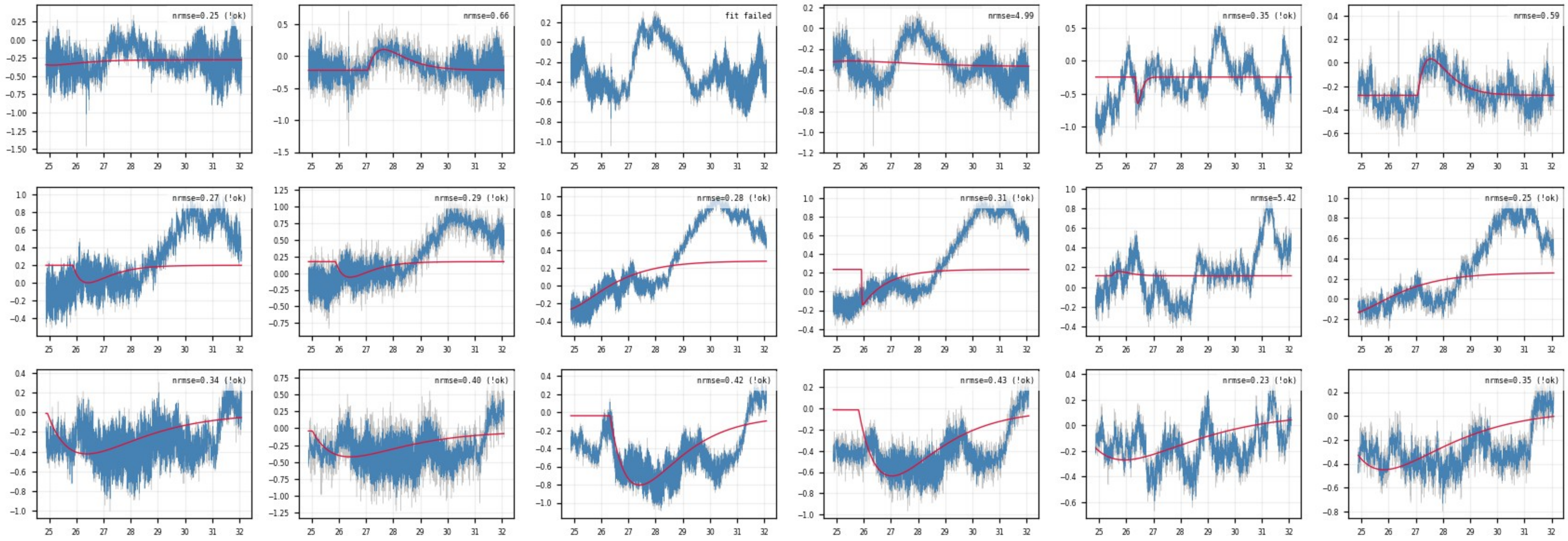
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (6/10)

figure: zip24_raw_vs_fit_examples.png — event rows 7–9 of 15, right half of the channels

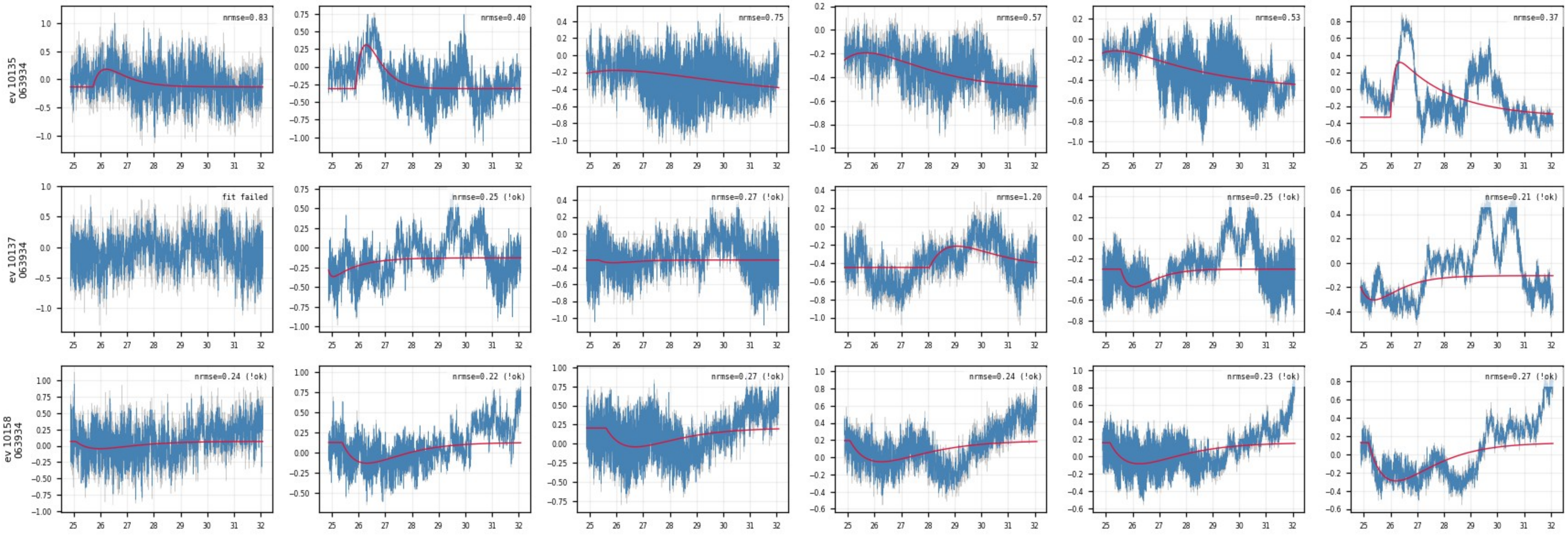
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (7/10)

figure: zip24_raw_vs_fit_examples.png — event rows 10–12 of 15, left half of the channels

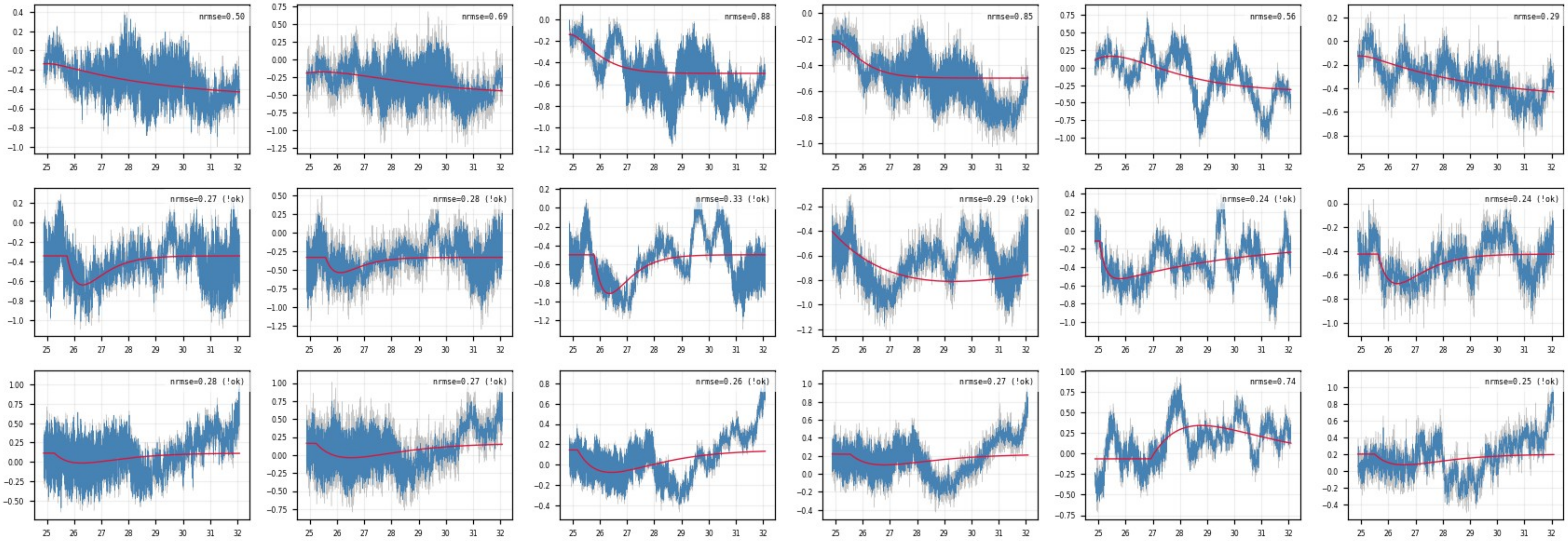
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (8/10)

figure: zip24_raw_vs_fit_examples.png — event rows 10–12 of 15, right half of the channels

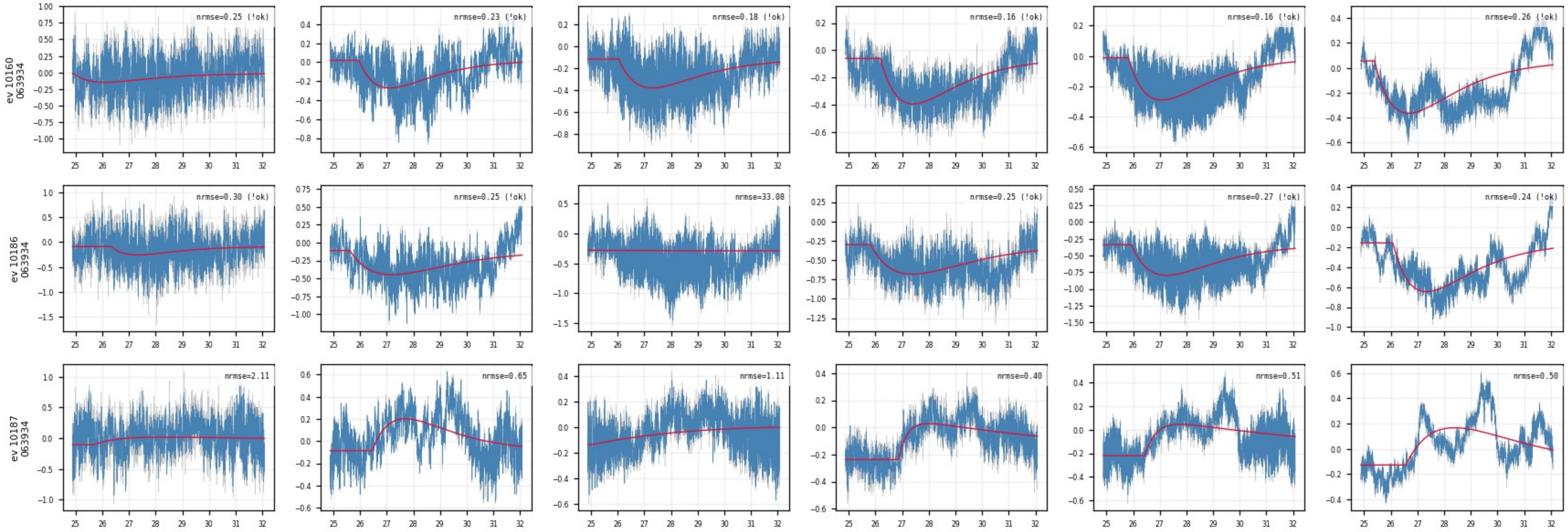
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (9/10)

figure: zip24_raw_vs_fit_examples.png — event rows 13–15 of 15, left half of the channels

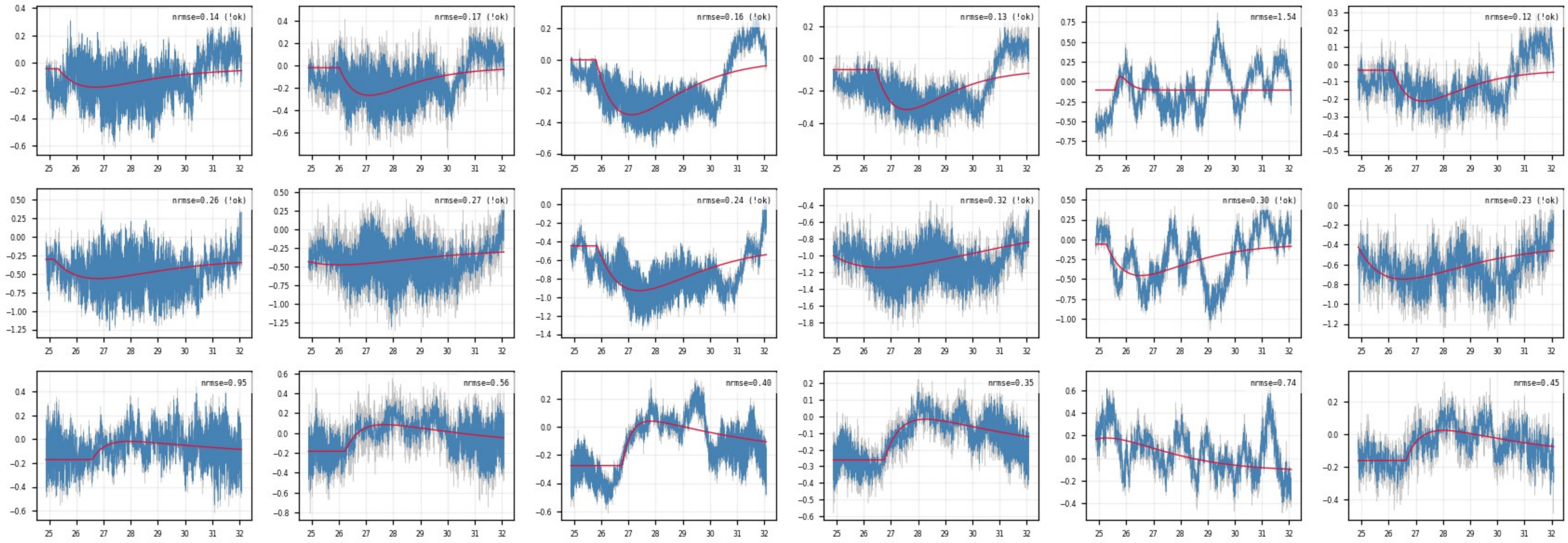
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z24 · raw_vs_fit_examples — raw trace vs fit (10/10)

figure: zip24_raw_vs_fit_examples.png — event rows 13–15 of 15, right half of the channels

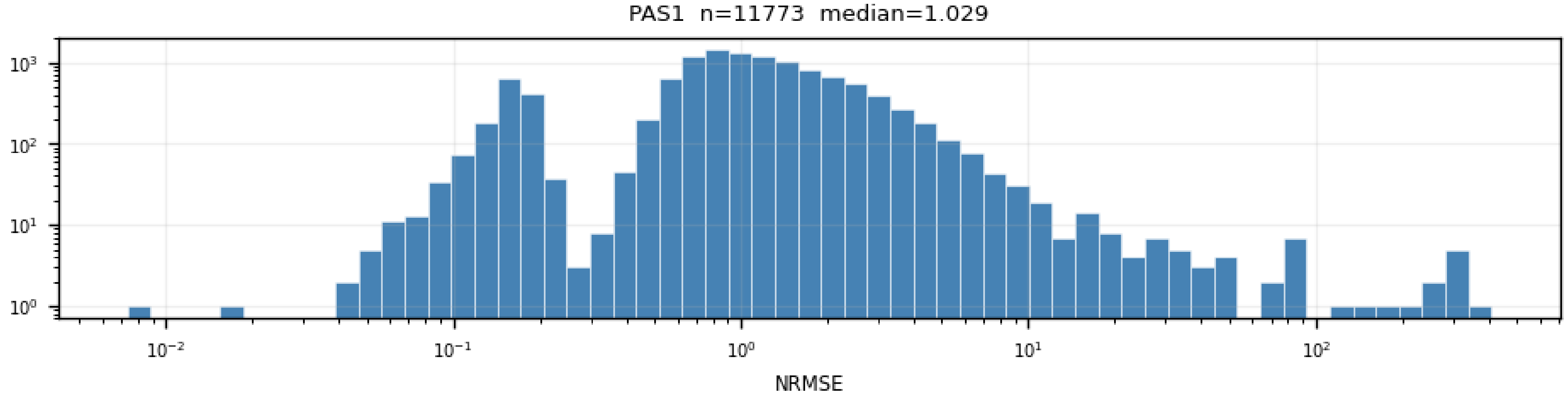
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

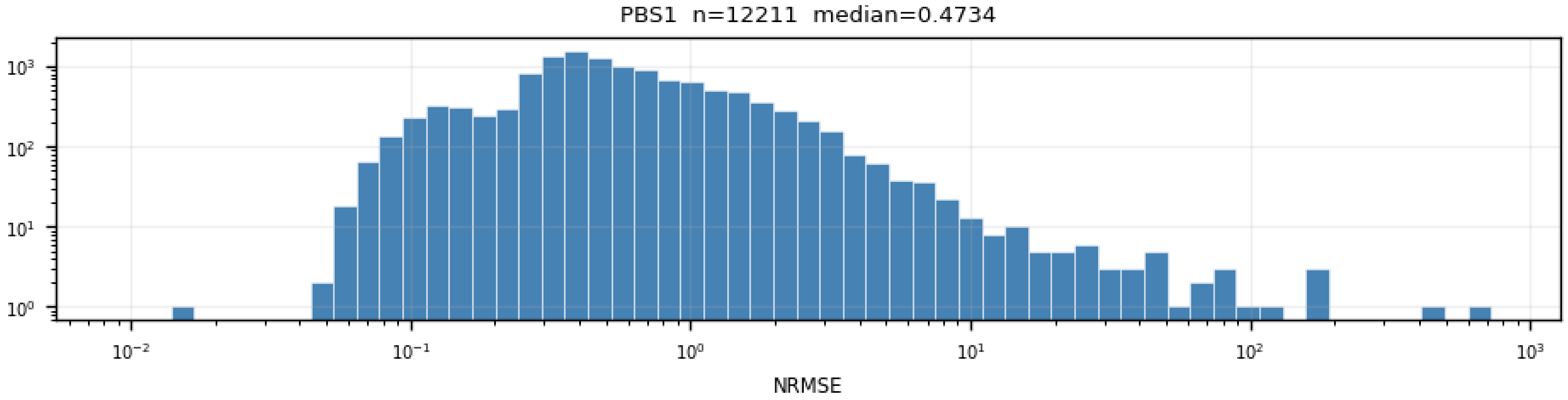


Z24 · nrmse — NRMSE distribution per channel (1/6)

figure: zip24_nrmse.png — channel panels 1–2 of 12, top to bottom

NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits vs noise triggers; the valley at ≈0.4 sets quality cut 1.

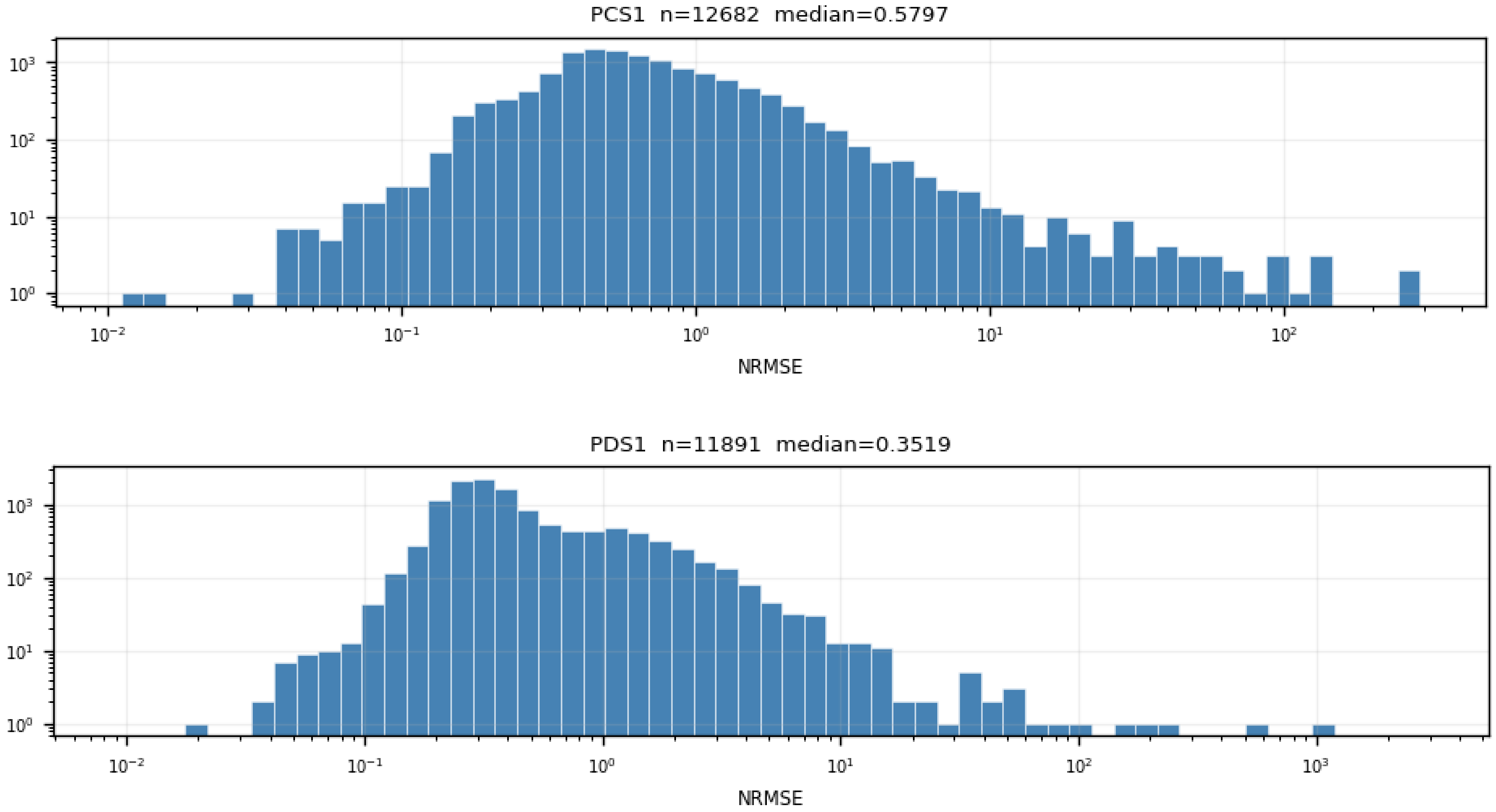




Z24 · nrmse — NRMSE distribution per channel (2/6)

figure: zip24_nrmse.png — channel panels 3–4 of 12, top to bottom

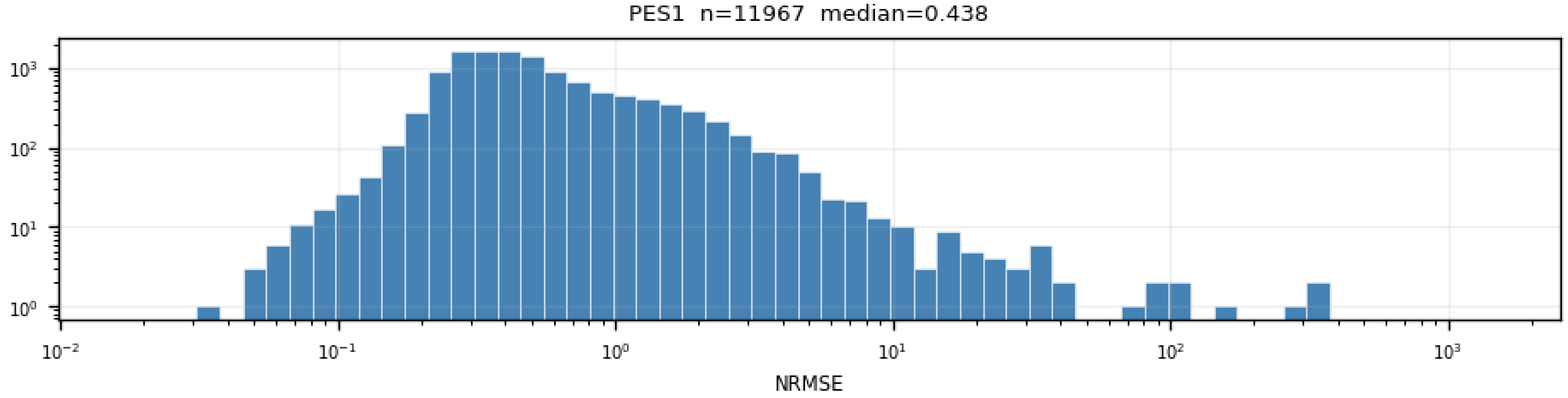
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits vs noise triggers; the valley at ≈0.4 sets quality cut 1.

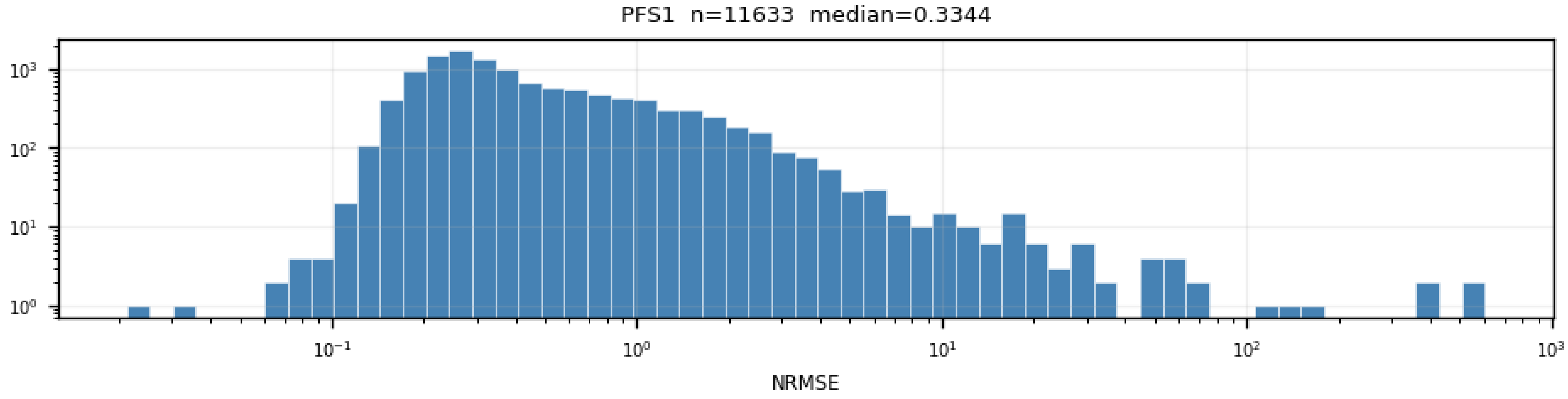


Z24 · nrmse — NRMSE distribution per channel (3/6)

figure: zip24_nrmse.png — channel panels 5–6 of 12, top to bottom

NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits vs noise triggers; the valley at ≈0.4 sets quality cut 1.

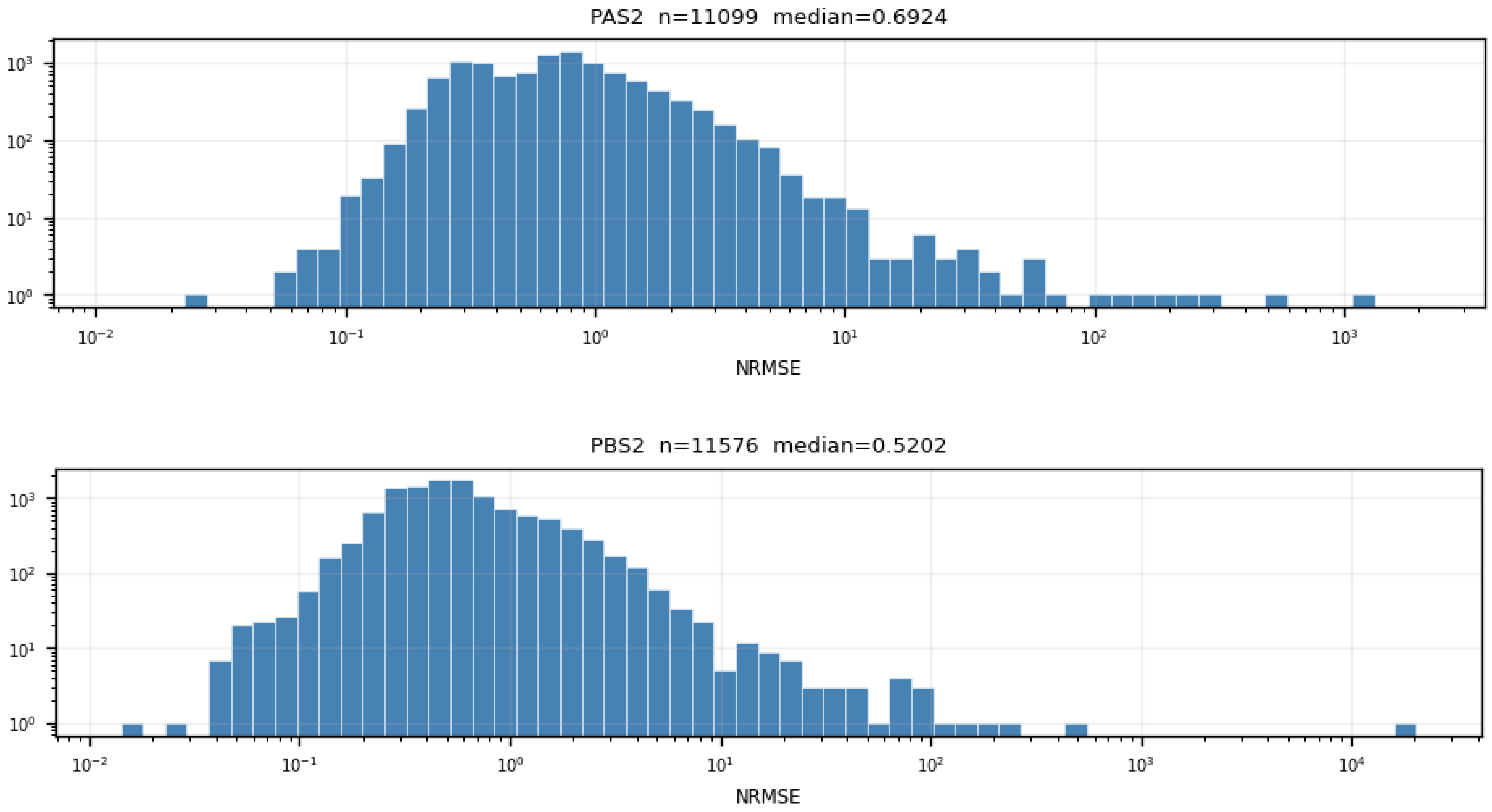




Z24 · nrmse — NRMSE distribution per channel (4/6)

figure: zip24_nrmse.png — channel panels 7–8 of 12, top to bottom

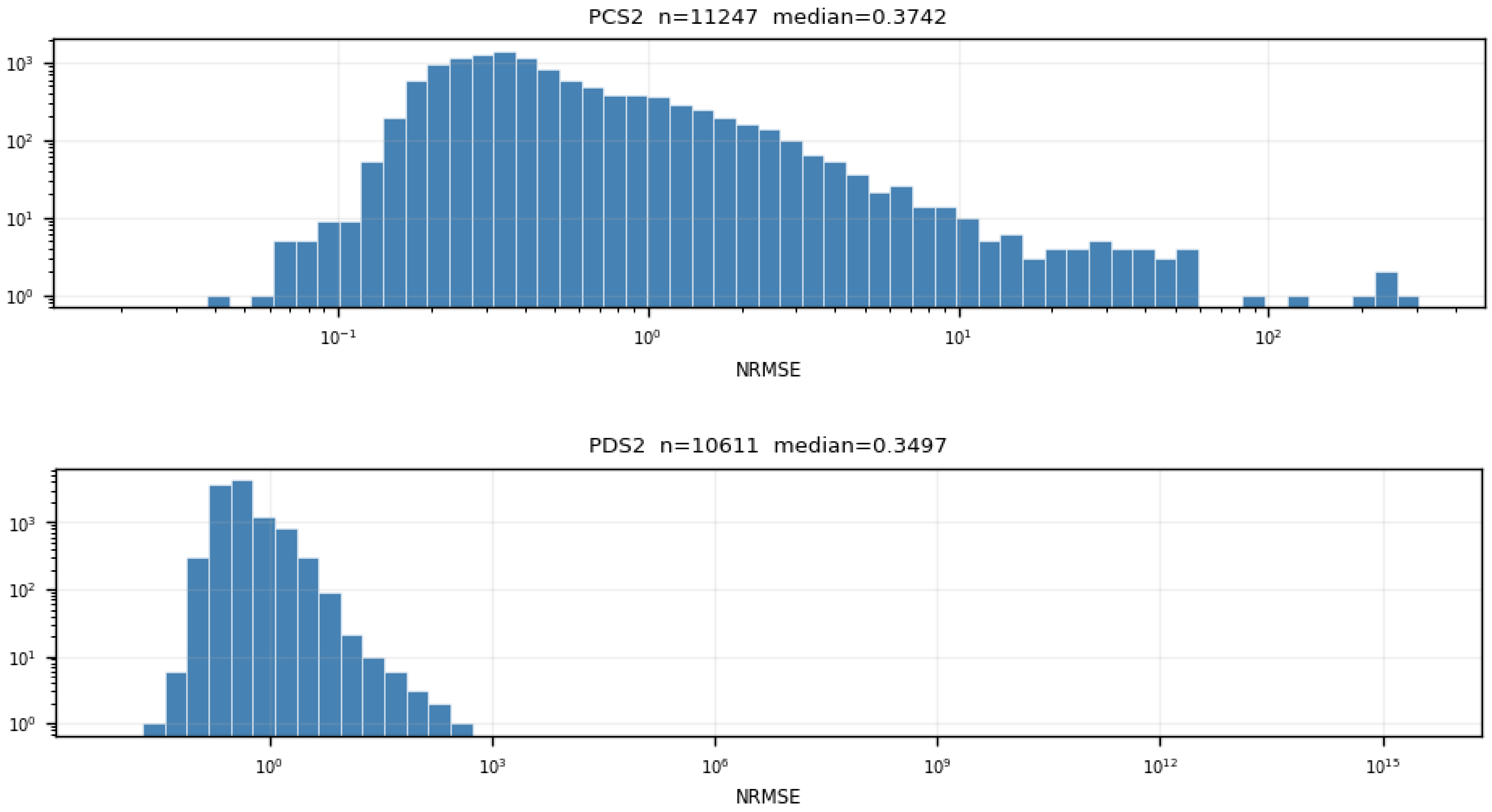
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits vs noise triggers; the valley at ≈0.4 sets quality cut 1.



Z24 · nrmse — NRMSE distribution per channel (5/6)

figure: zip24_nrmse.png — channel panels 9–10 of 12, top to bottom

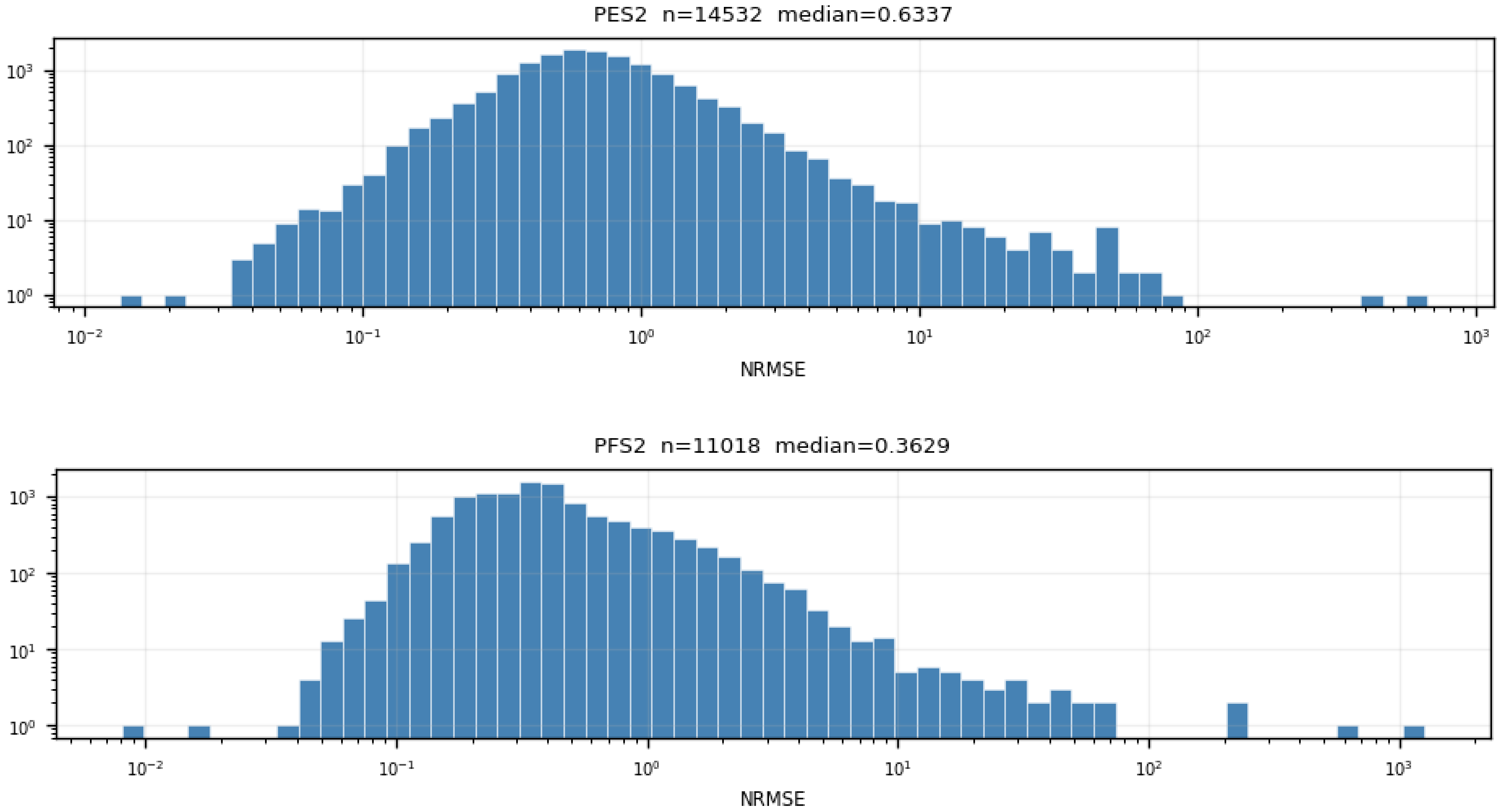
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits vs noise triggers; the valley at ≈0.4 sets quality cut 1.



Z24 · nrmse — NRMSE distribution per channel (6/6)

figure: zip24_nrmse.png — channel panels 11–12 of 12, top to bottom

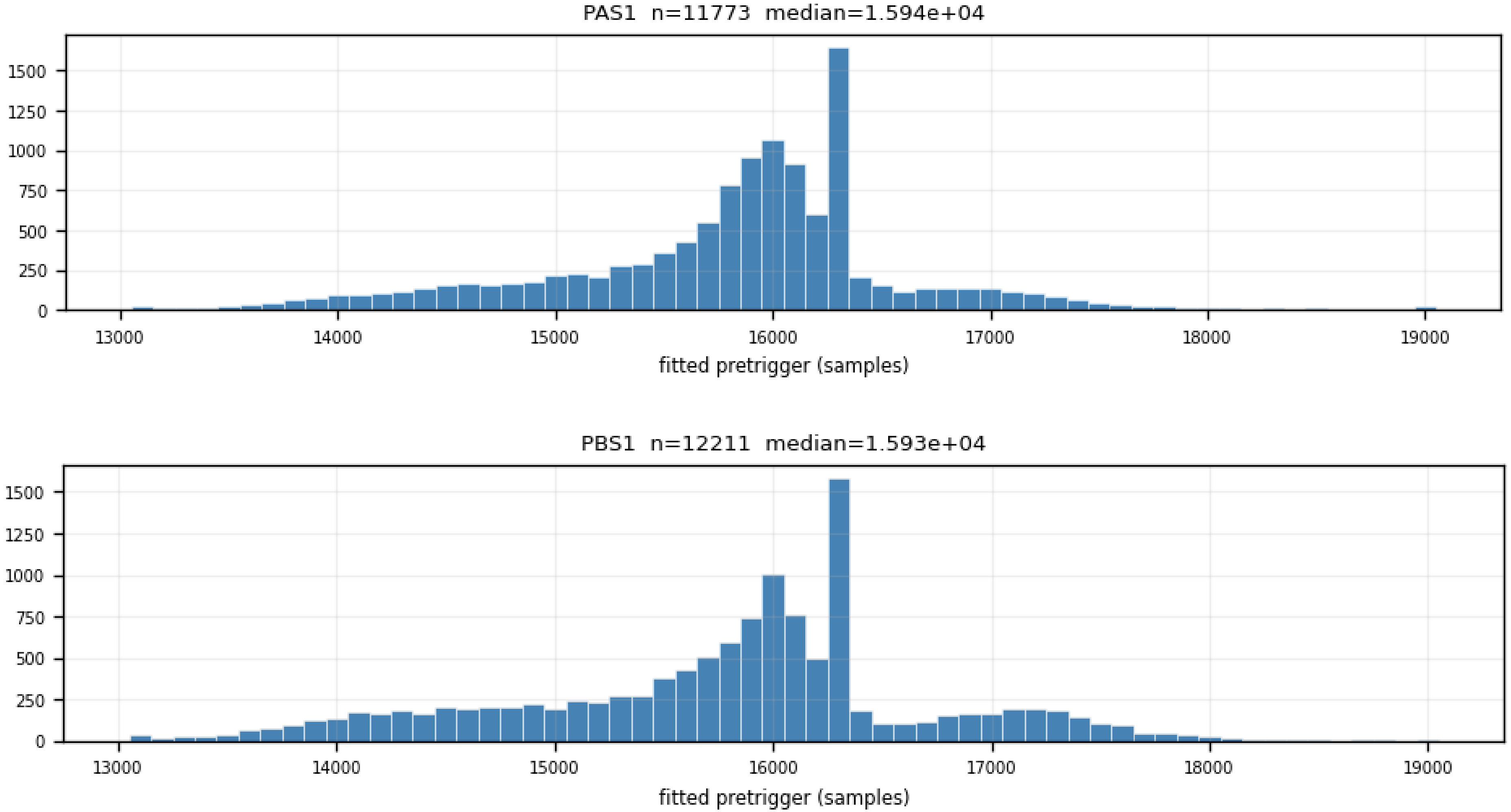
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits vs noise triggers; the valley at ≈0.4 sets quality cut 1.



Z24 · pretrigger — fitted onset per channel (1/6)

figure: zip24_pretrigger.png — channel panels 1–2 of 12, top to bottom

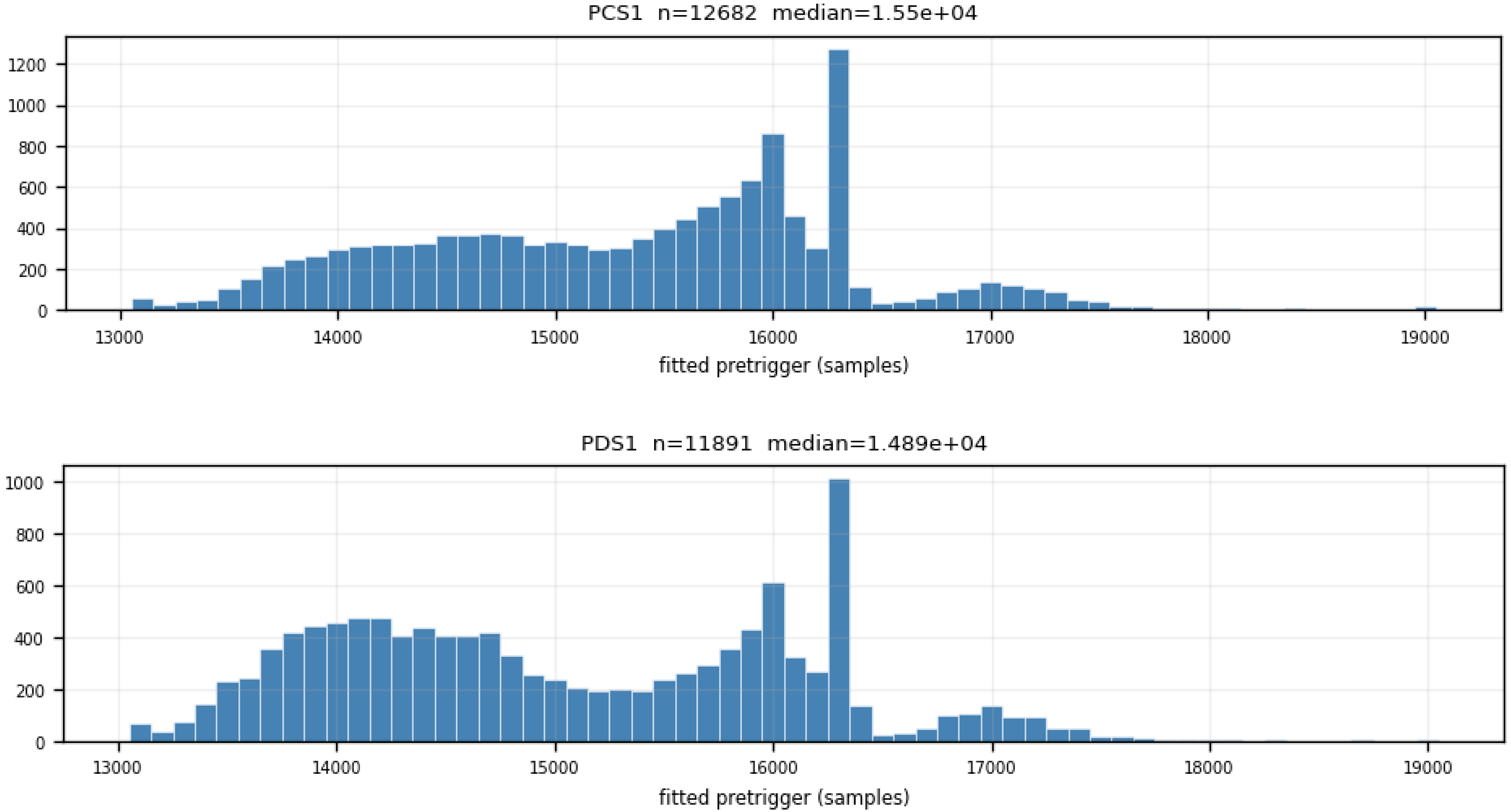
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z24 · pretrigger — fitted onset per channel (2/6)

figure: zip24_pretrigger.png — channel panels 3–4 of 12, top to bottom

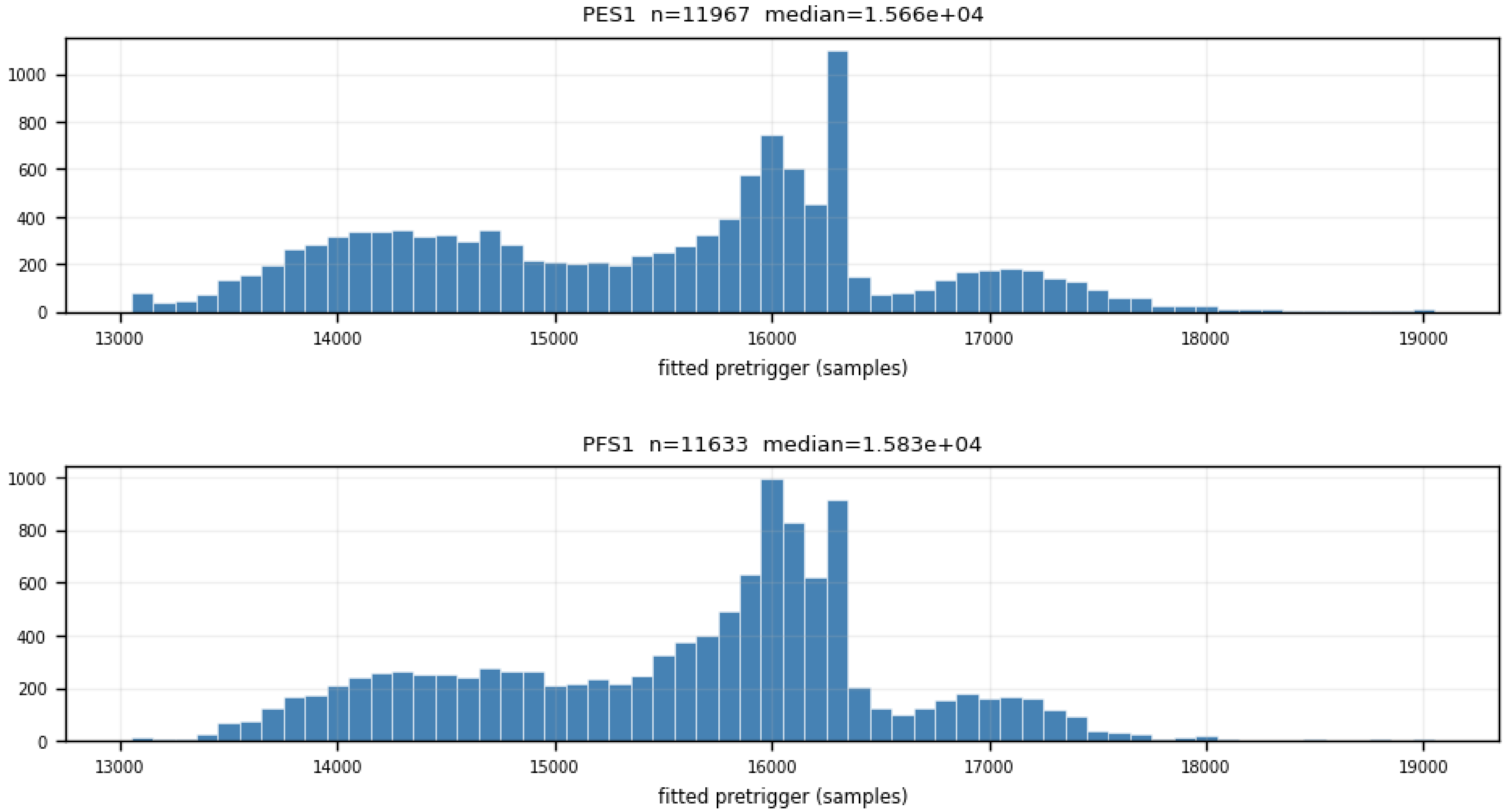
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z24 · pretrigger — fitted onset per channel (3/6)

figure: zip24_pretrigger.png — channel panels 5–6 of 12, top to bottom

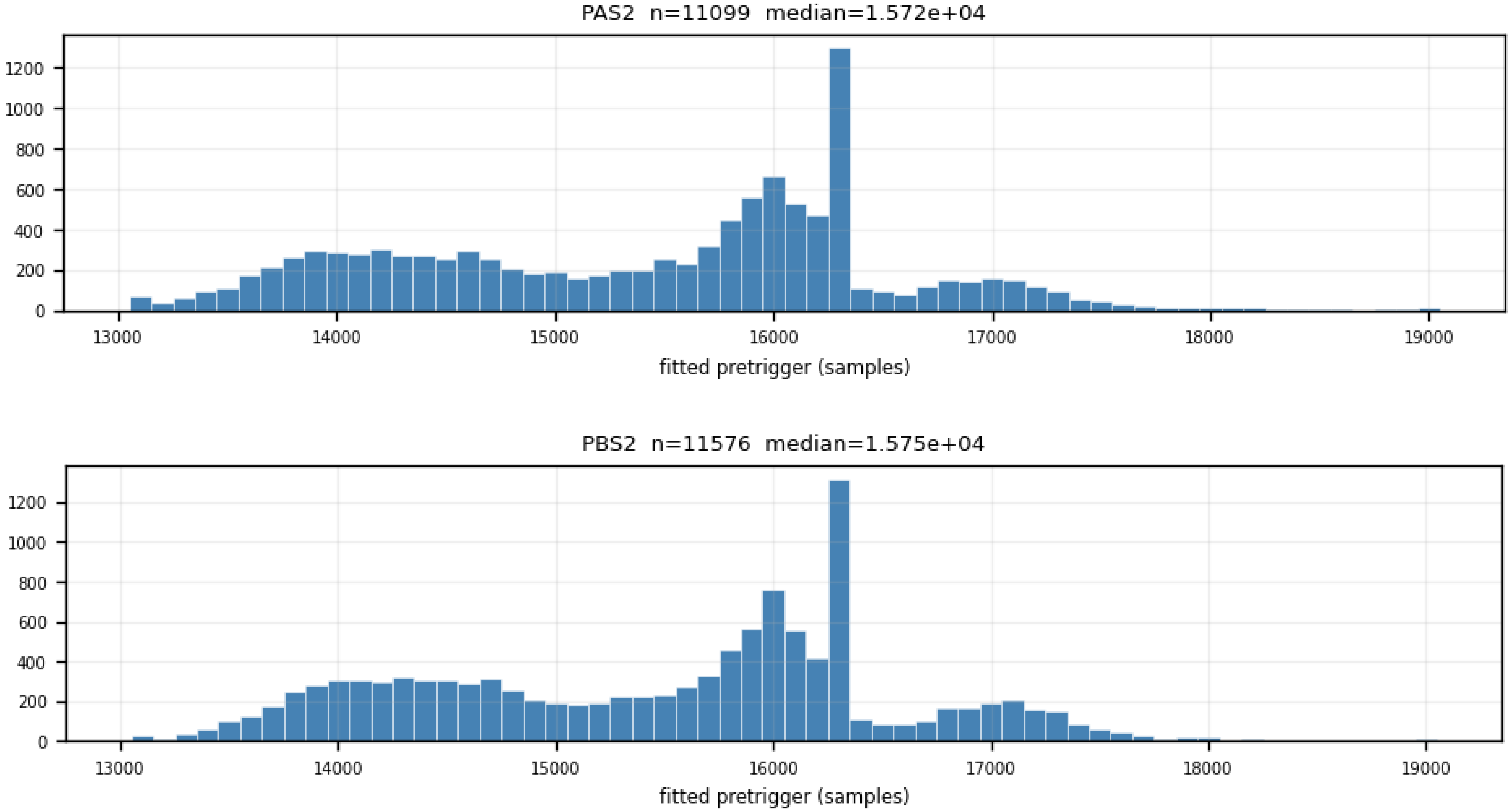
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z24 · pretrigger — fitted onset per channel (4/6)

figure: zip24_pretrigger.png — channel panels 7–8 of 12, top to bottom

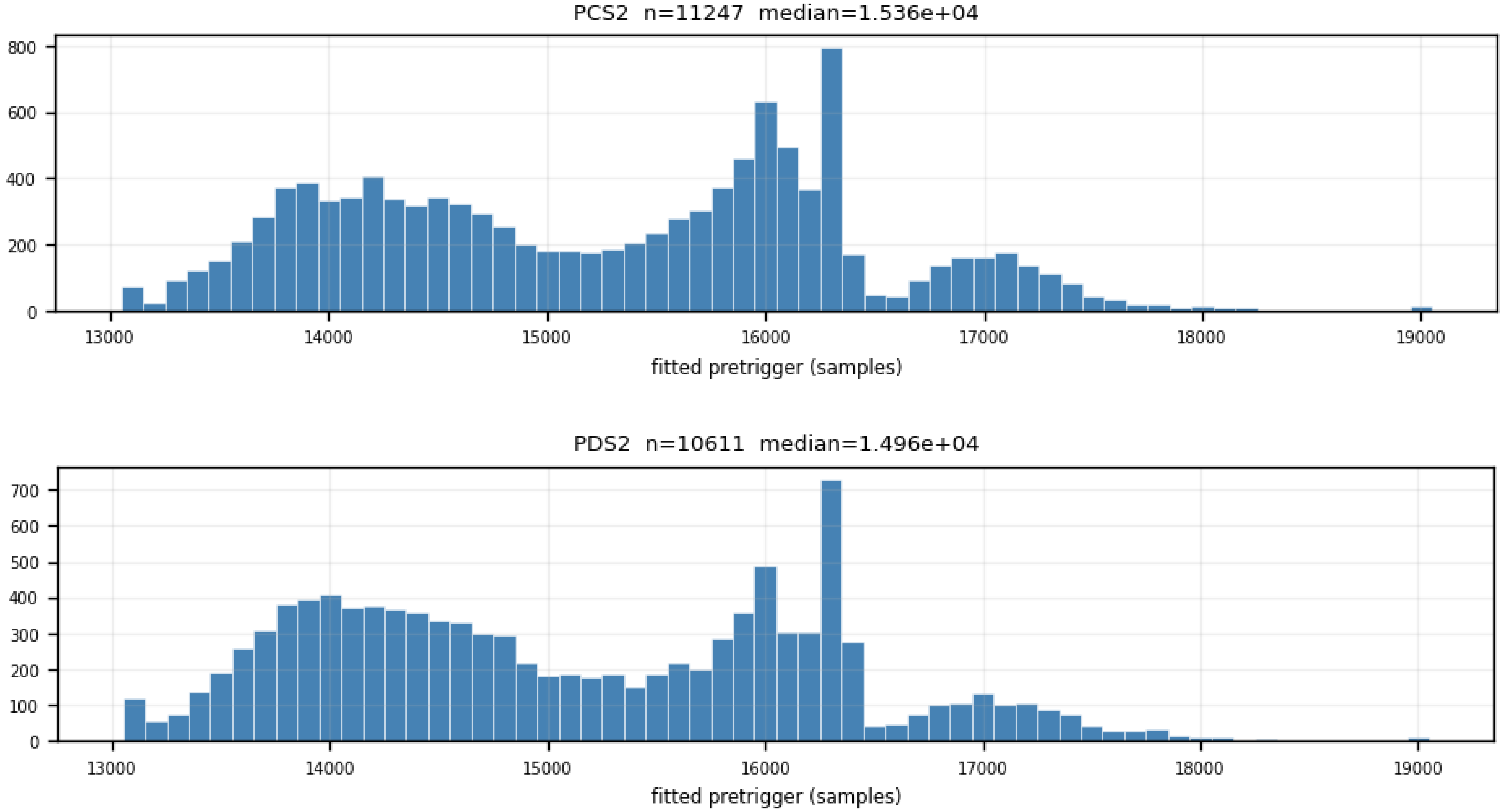
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z24 · pretrigger — fitted onset per channel (5/6)

figure: zip24_pretrigger.png — channel panels 9–10 of 12, top to bottom

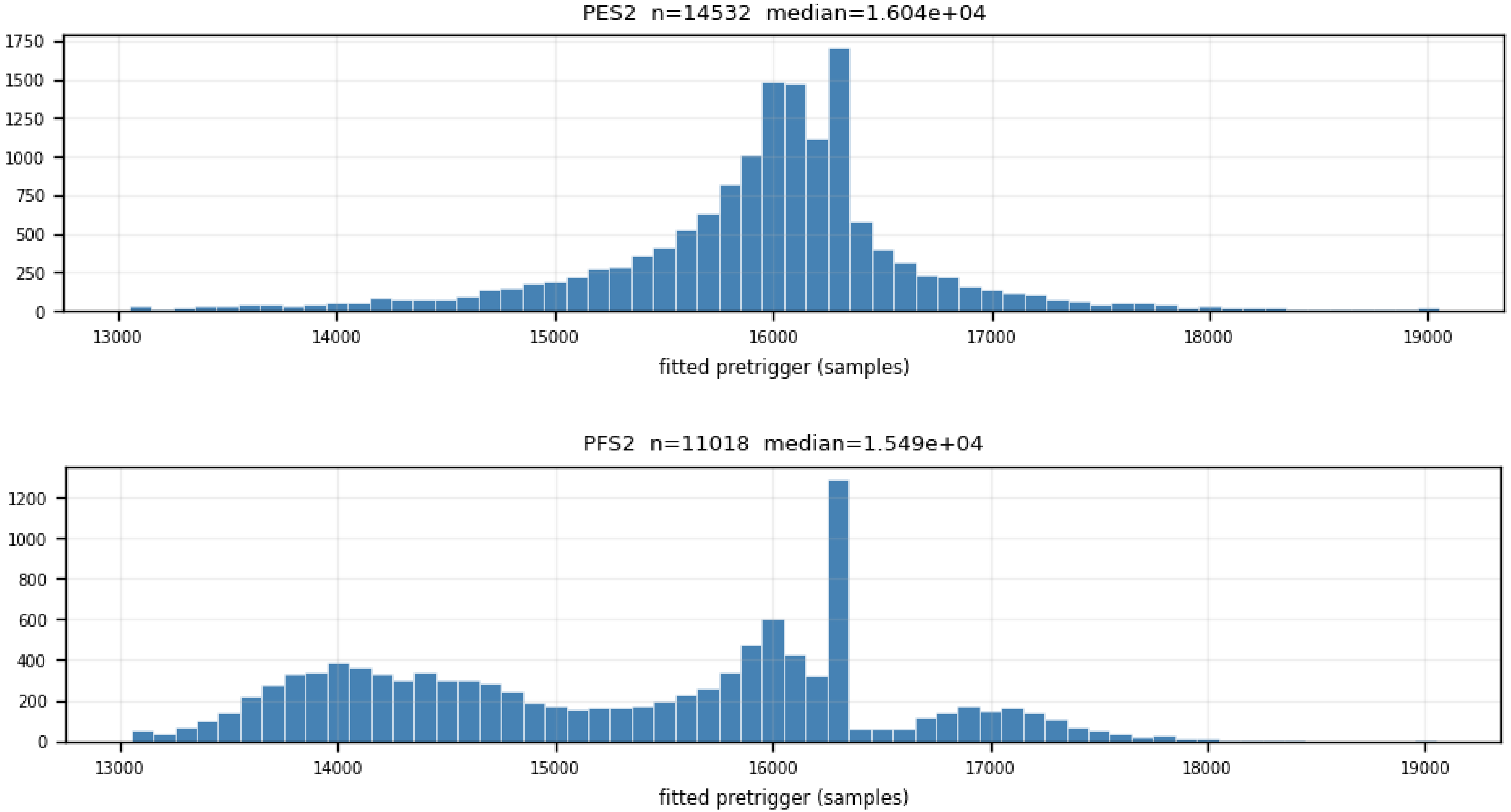
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z24 · pretrigger — fitted onset per channel (6/6)

figure: zip24_pretrigger.png — channel panels 11–12 of 12, top to bottom

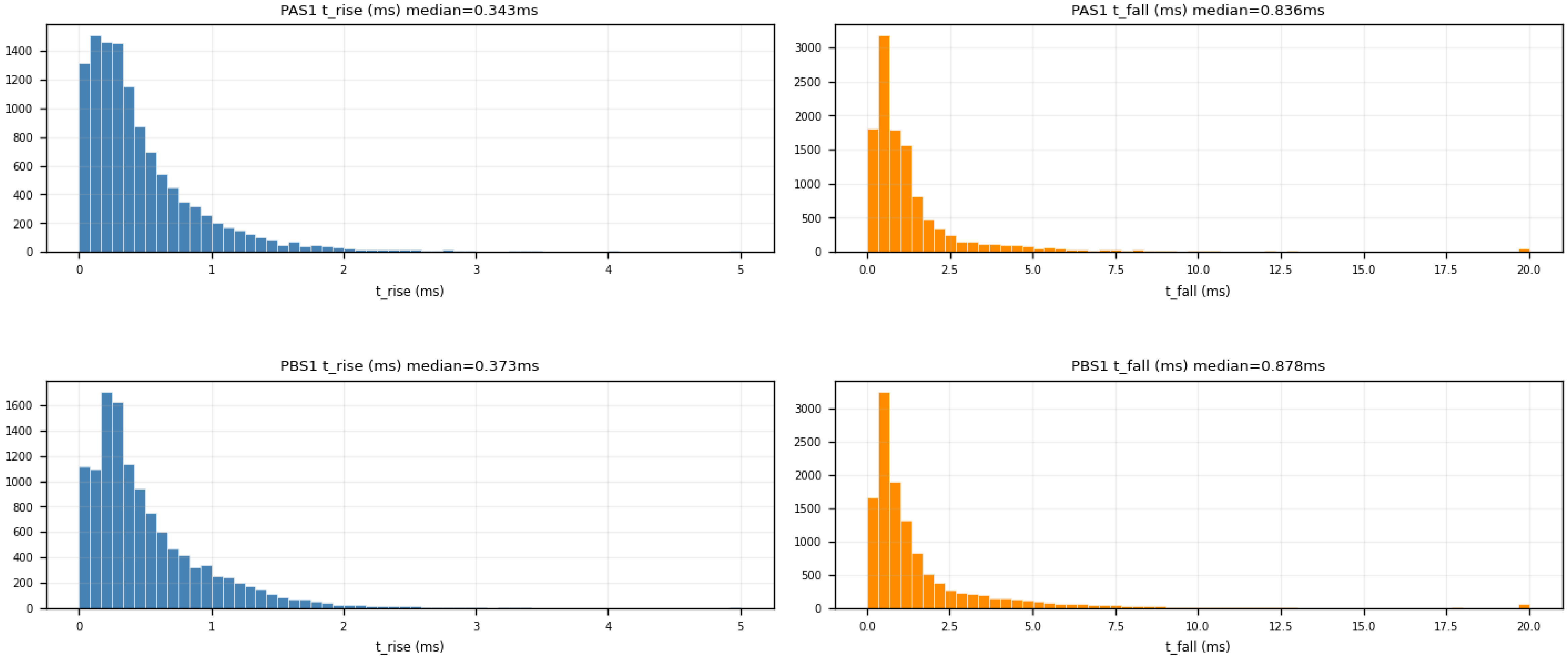
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z24 · time_constants — τ_rise / τ_fall per channel (1/6)

figure: zip24_time_constants.png — channel panels 1–2 of 12, top to bottom

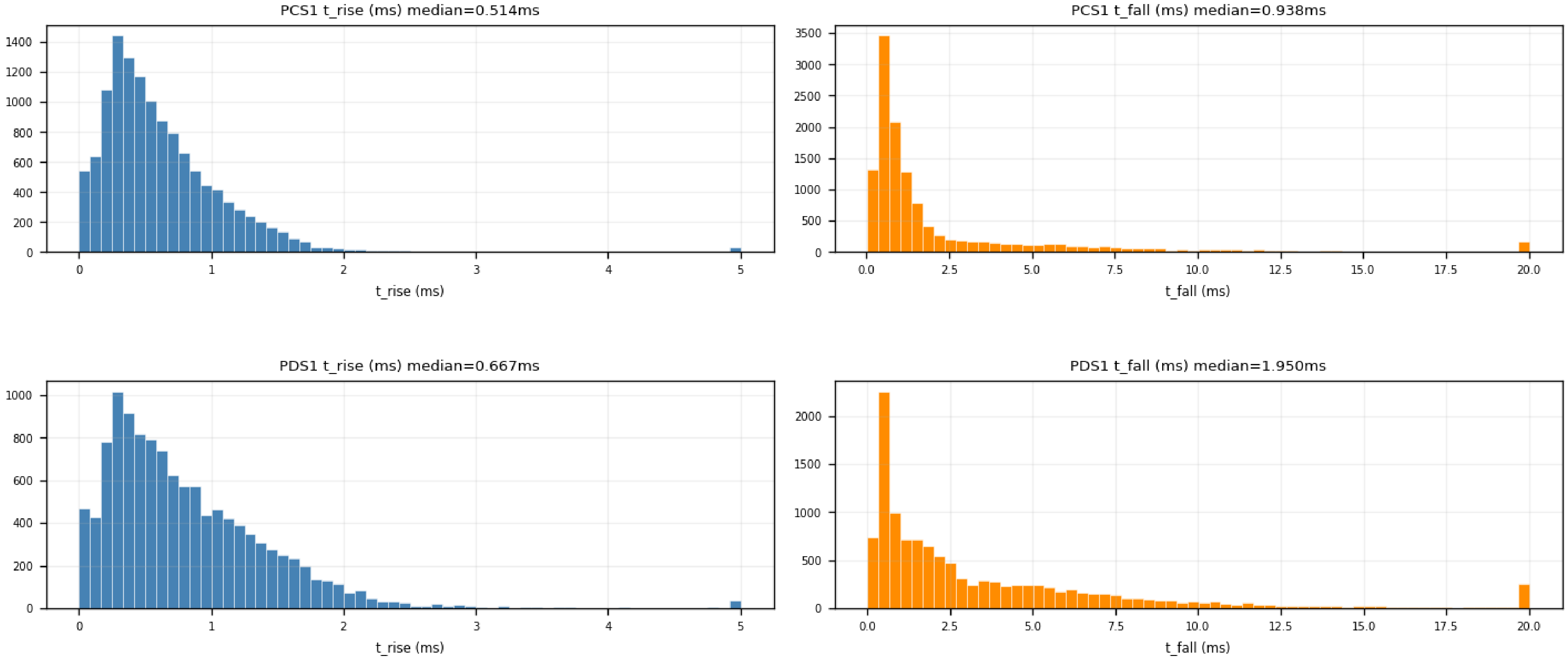
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z24 · time_constants — τ_rise / τ_fall per channel (2/6)

figure: zip24_time_constants.png — channel panels 3–4 of 12, top to bottom

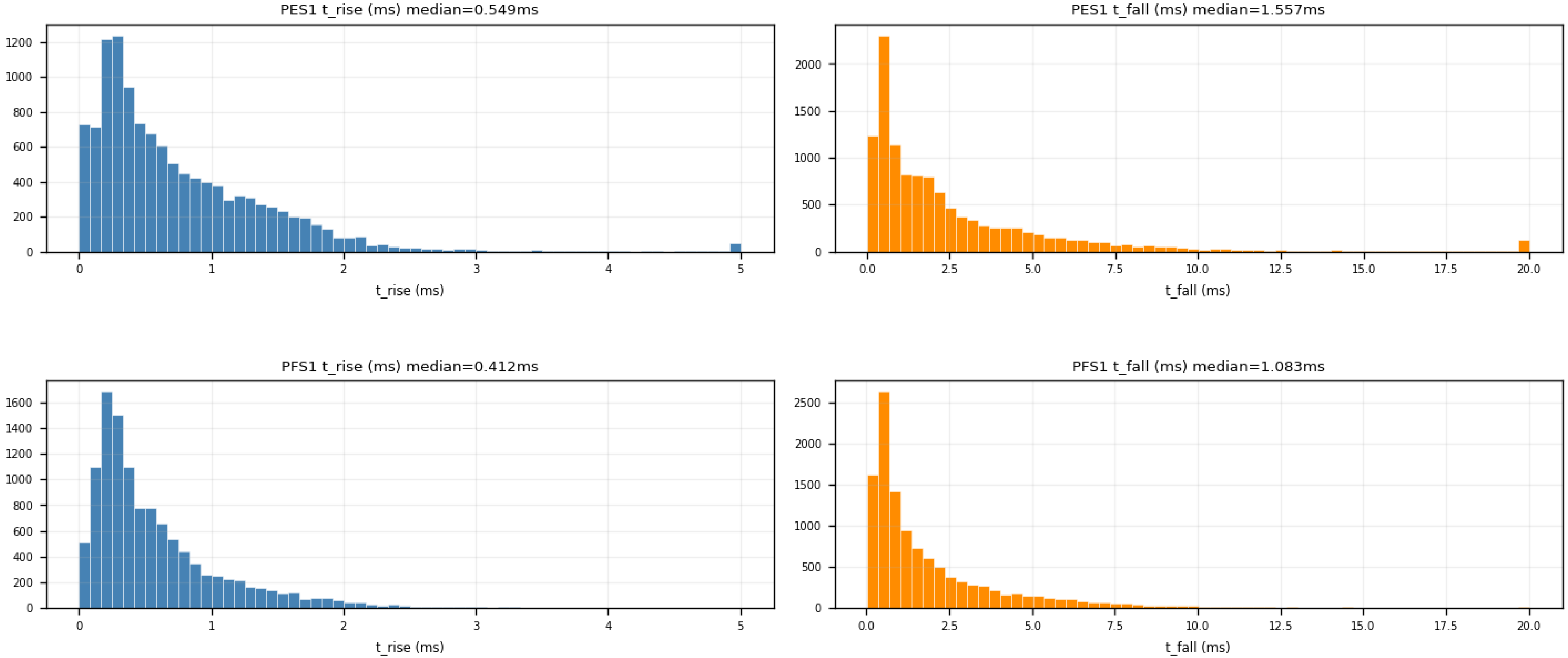
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z24 · time_constants — τ_rise / τ_fall per channel (3/6)

figure: zip24_time_constants.png — channel panels 5–6 of 12, top to bottom

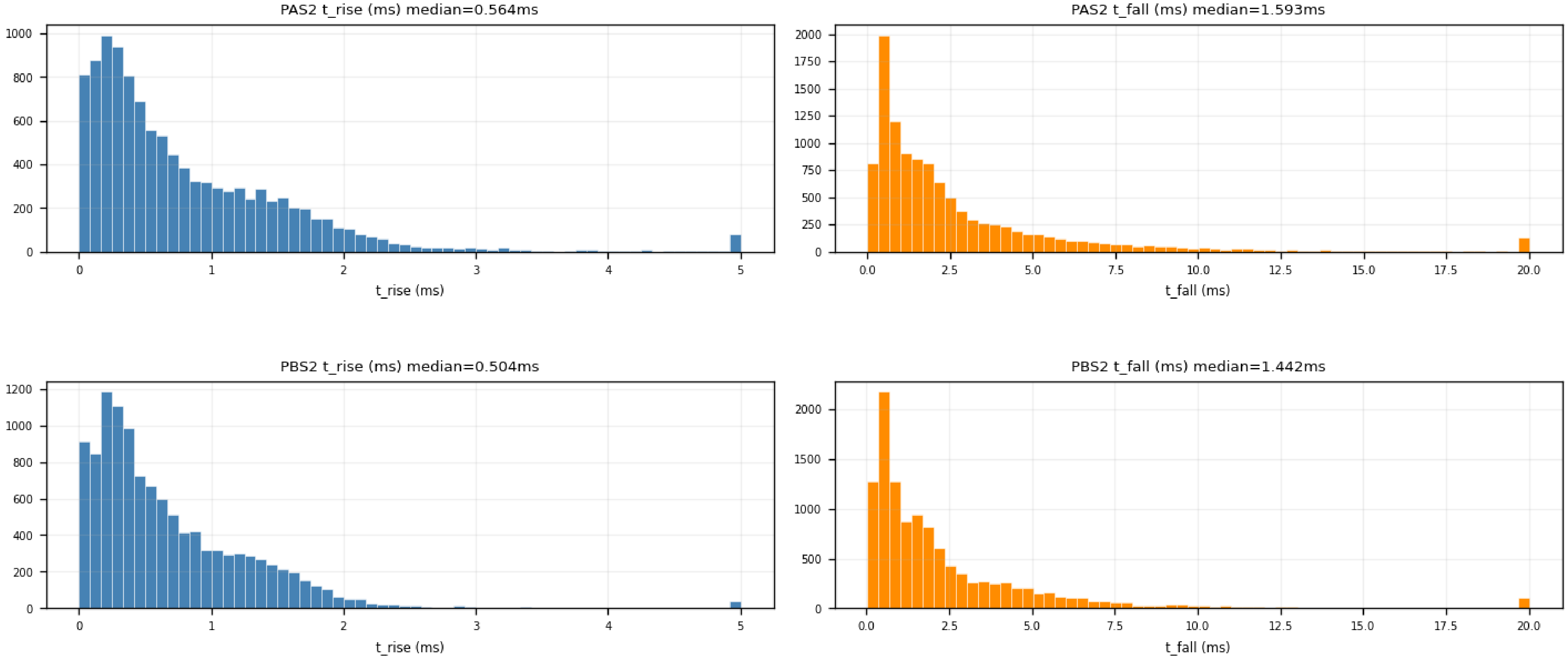
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z24 · time_constants — τ_rise / τ_fall per channel (4/6)

figure: zip24_time_constants.png — channel panels 7–8 of 12, top to bottom

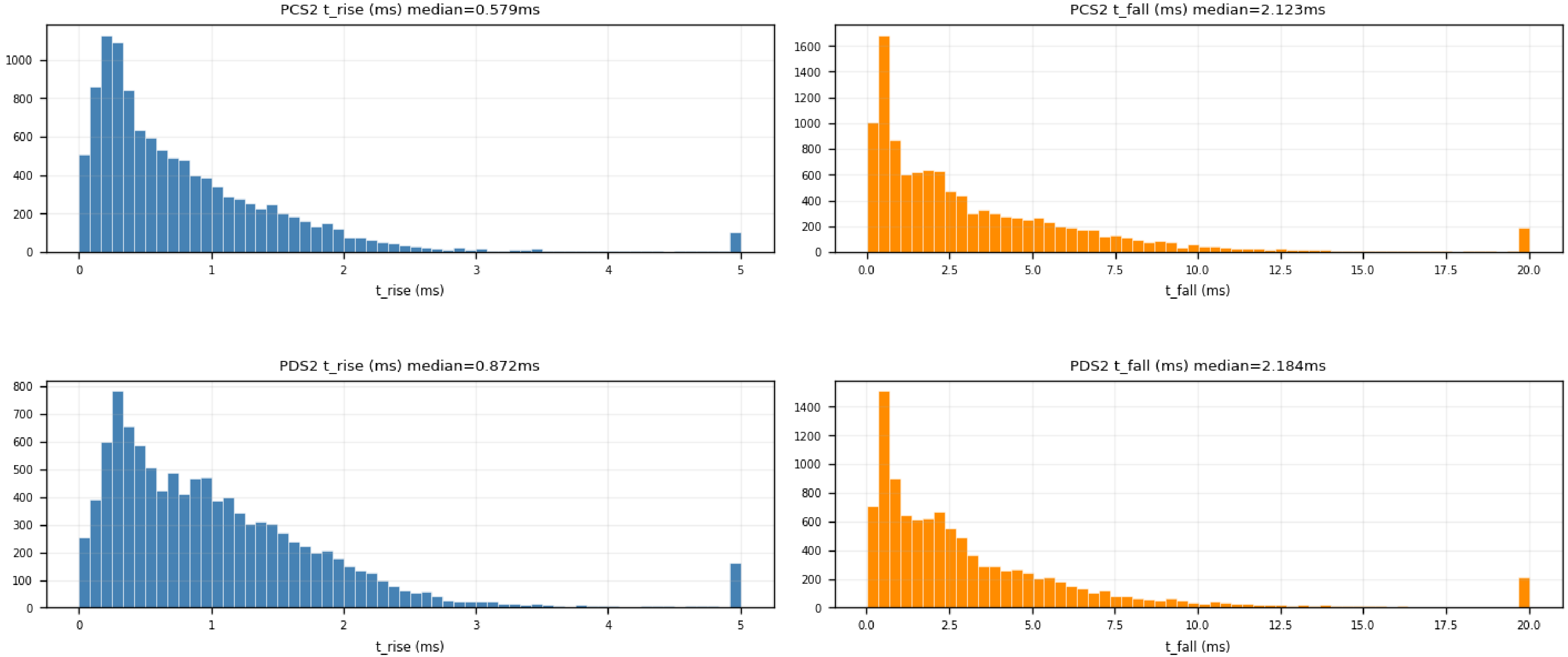
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z24 · time_constants — τ_rise / τ_fall per channel (5/6)

figure: zip24_time_constants.png — channel panels 9–10 of 12, top to bottom

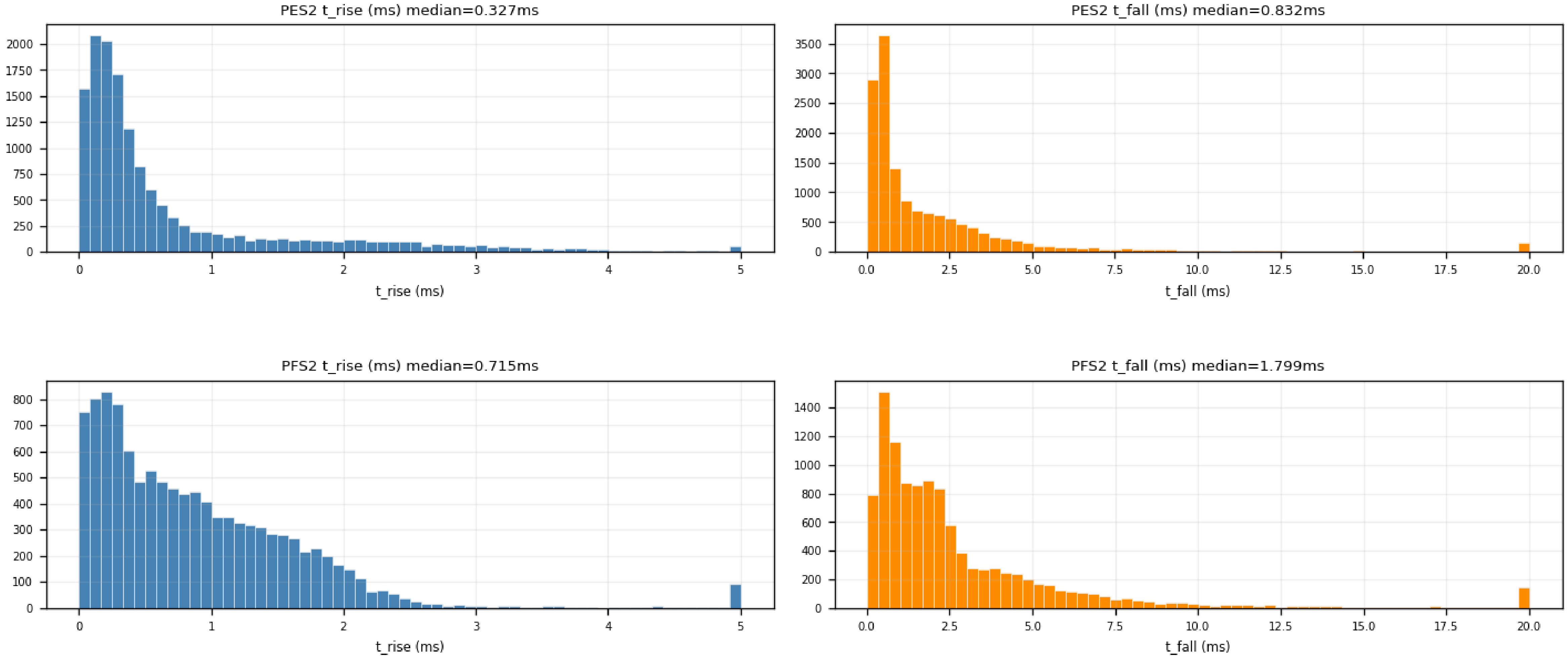
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z24 · time_constants — τ_rise / τ_fall per channel (6/6)

figure: zip24_time_constants.png — channel panels 11–12 of 12, top to bottom

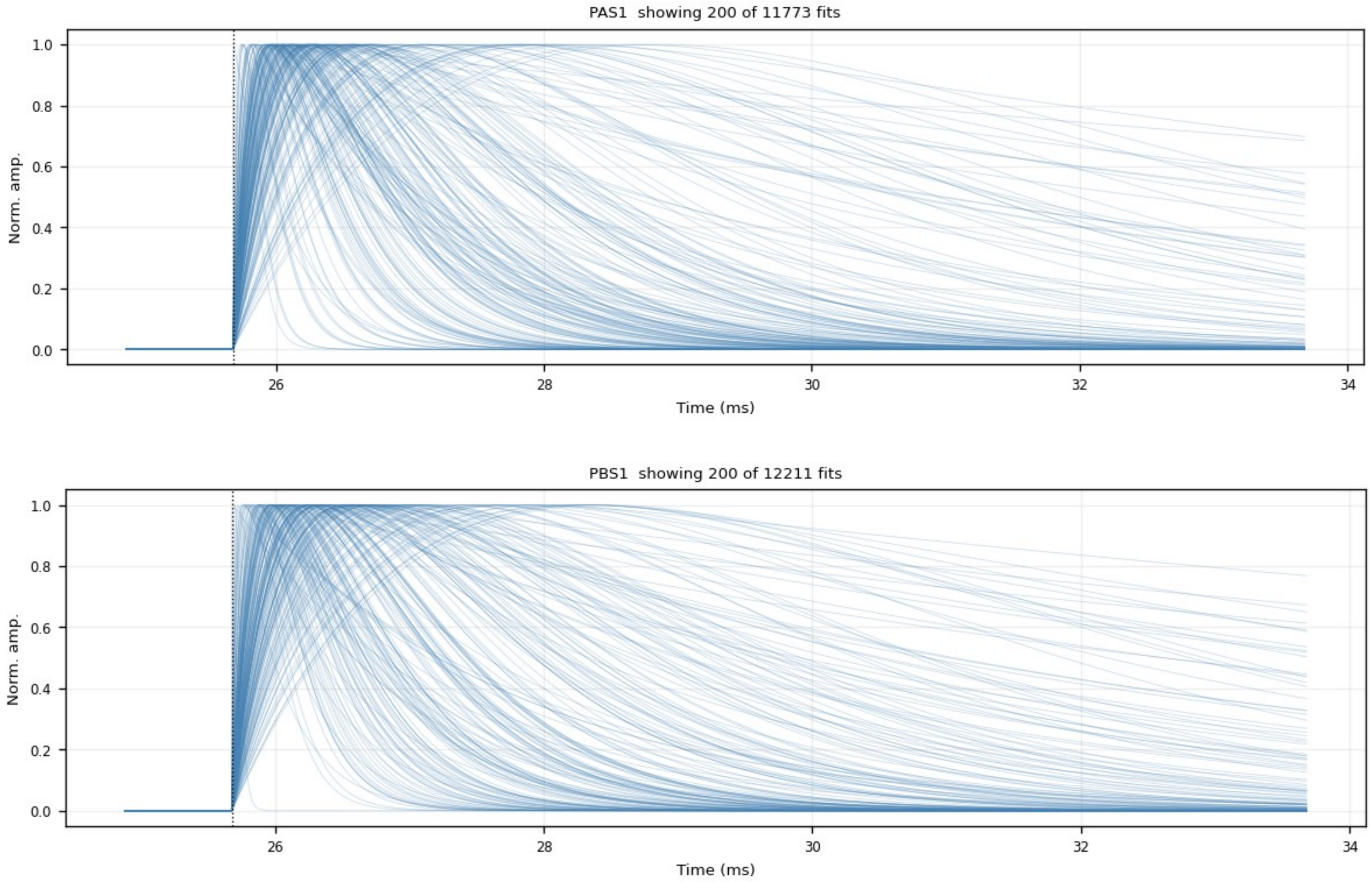
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z24 · fitted_curves_overlay — fan, no cut (1/6)

figure: zip24_fitted_curves_overlay.png — channel panels 1–2 of 12, top to bottom

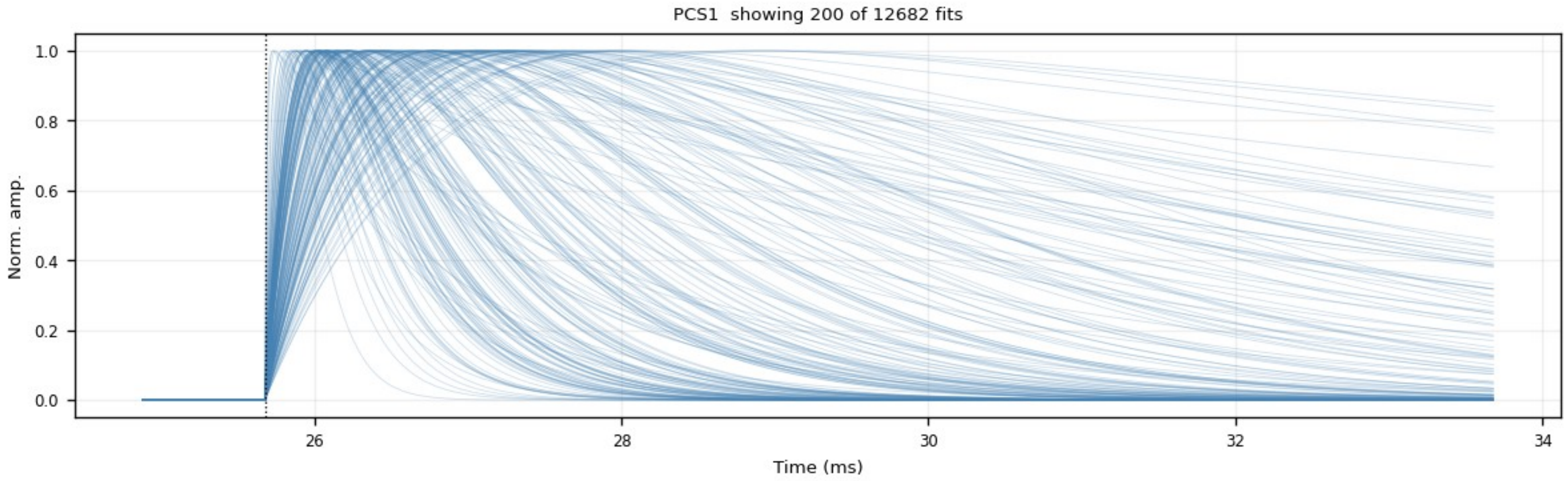
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

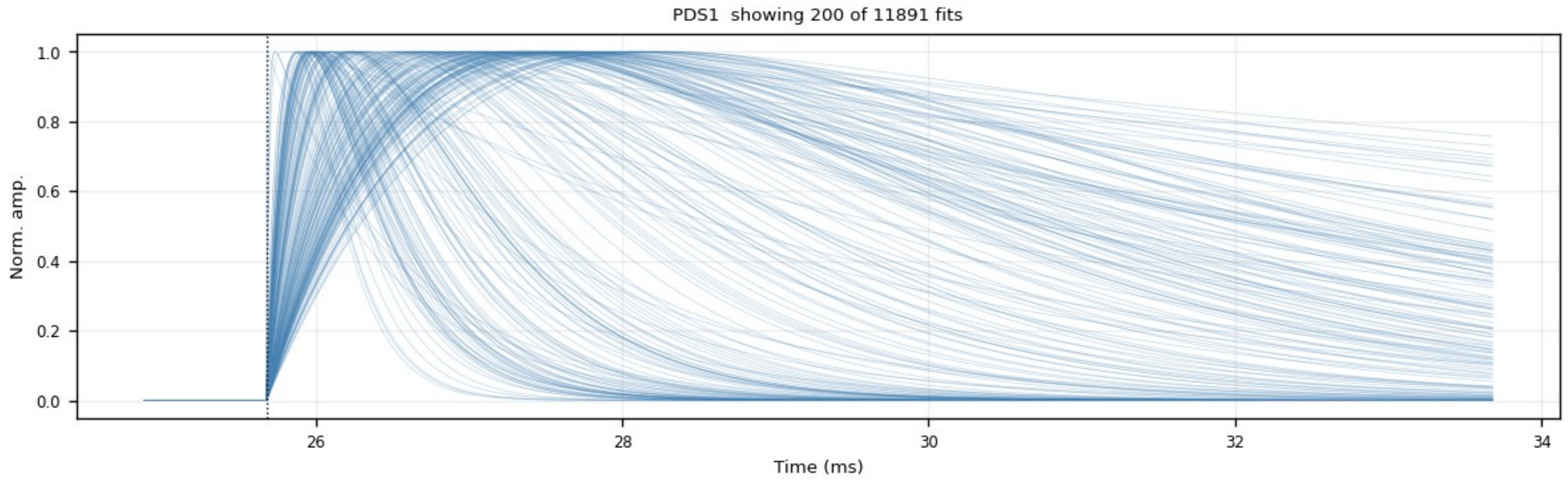


Z24 · fitted_curves_overlay — fan, no cut (2/6)

figure: zip24_fitted_curves_overlay.png — channel panels 3–4 of 12, top to bottom

Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

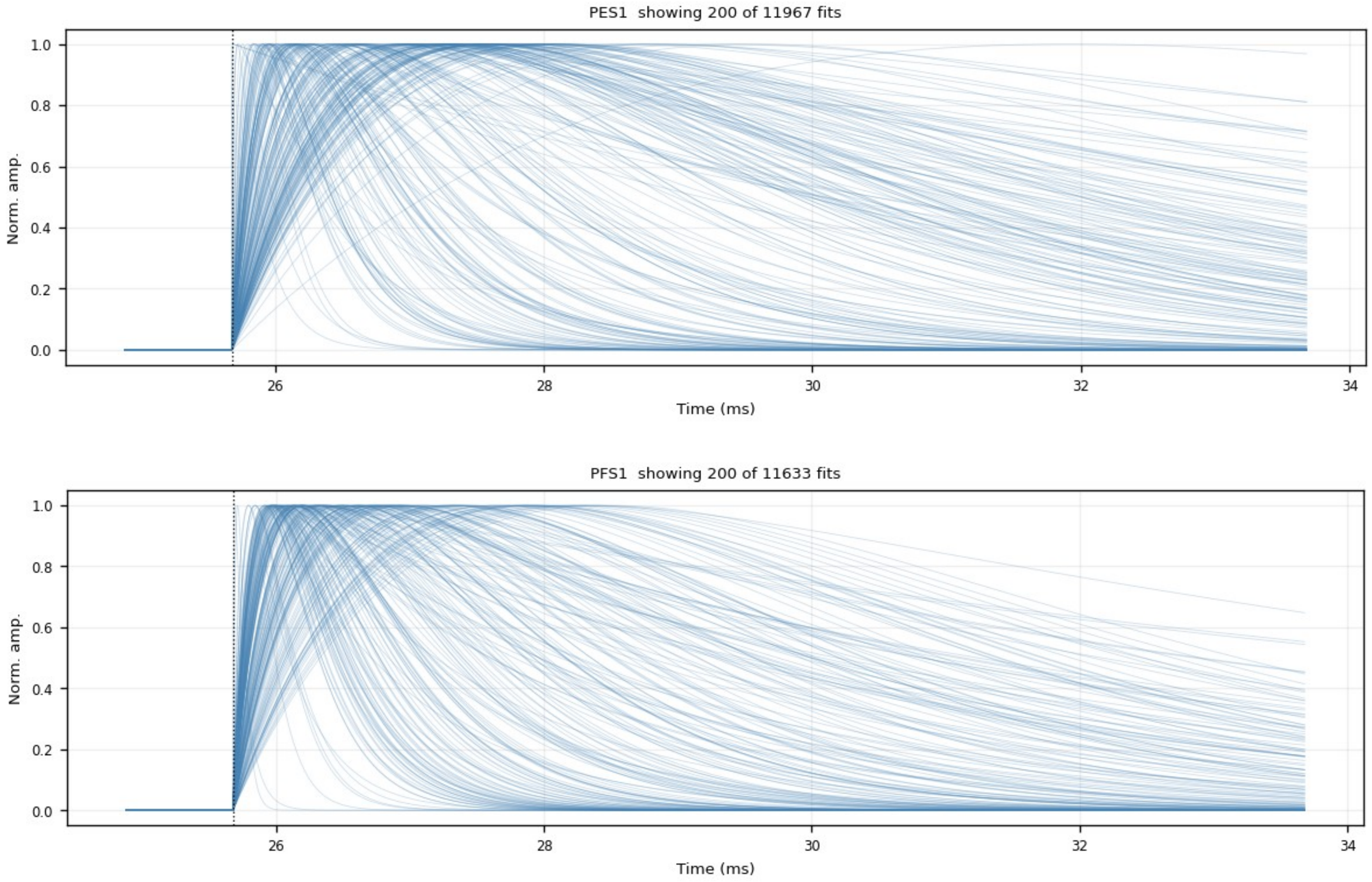




Z24 · fitted_curves_overlay — fan, no cut (3/6)

figure: zip24_fitted_curves_overlay.png — channel panels 5–6 of 12, top to bottom

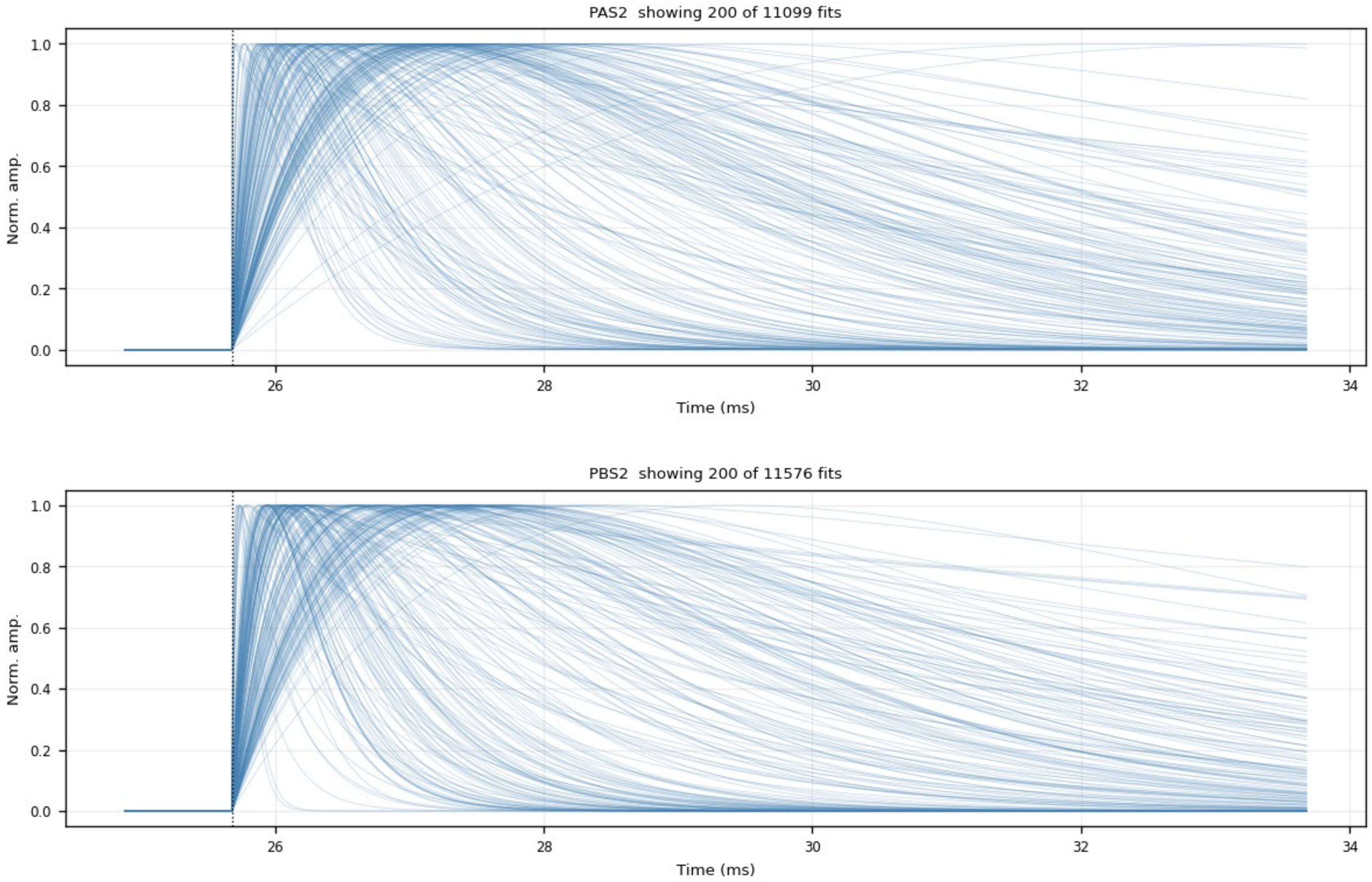
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Z24 · fitted_curves_overlay — fan, no cut (4/6)

figure: zip24_fitted_curves_overlay.png — channel panels 7–8 of 12, top to bottom

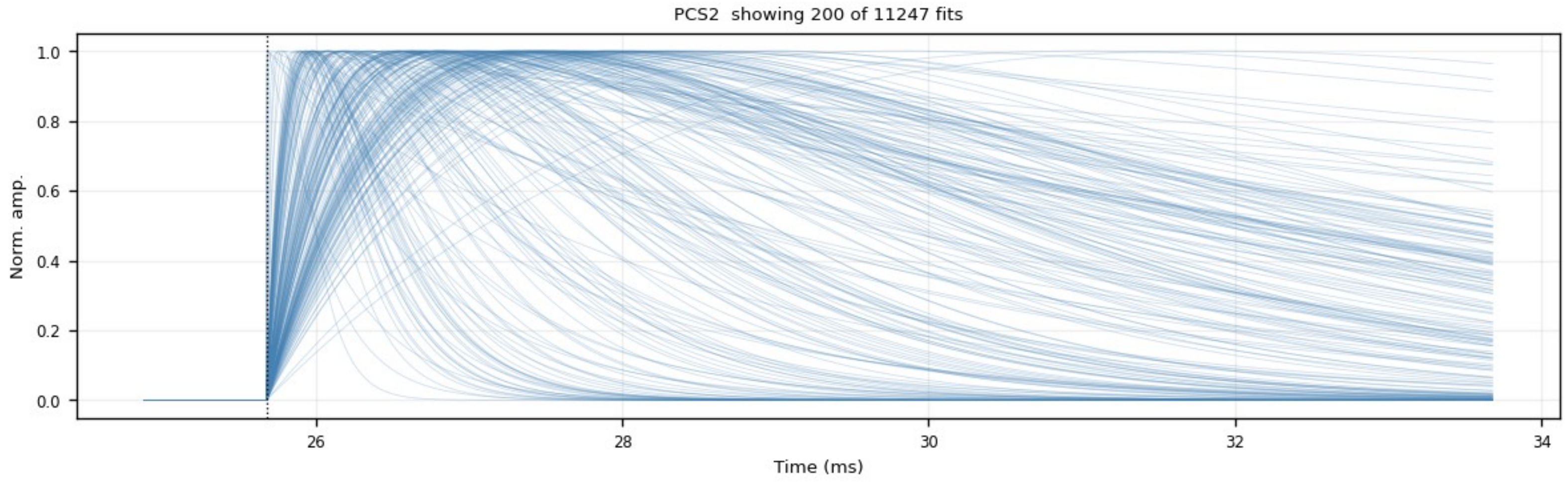
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

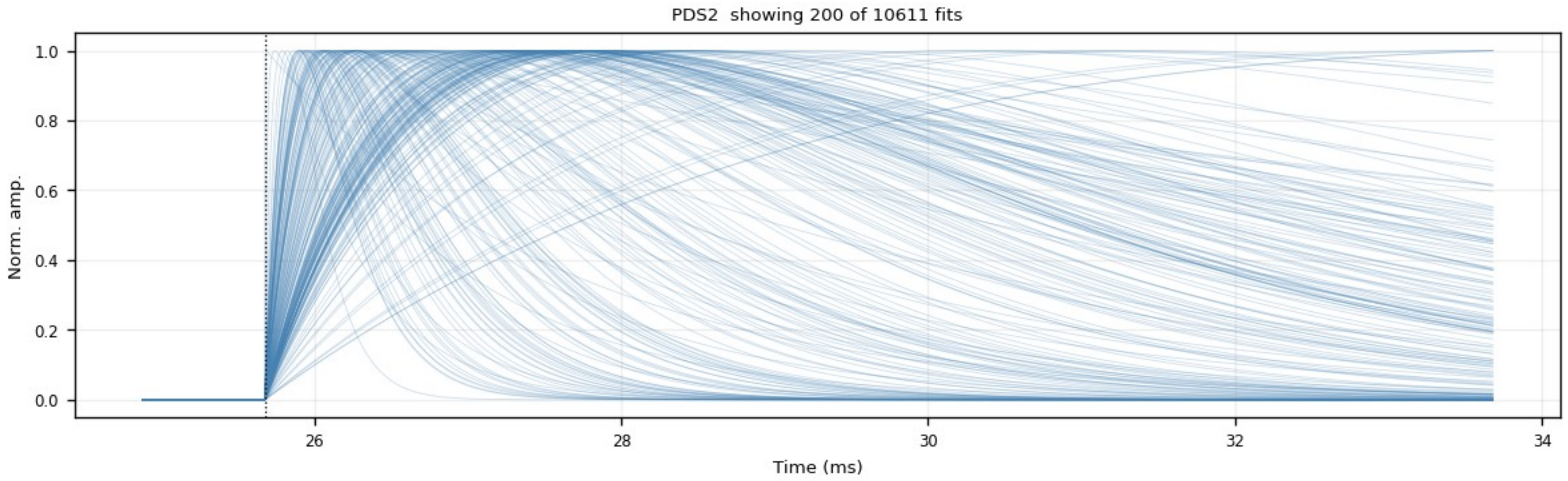


Z24 · fitted_curves_overlay — fan, no cut (5/6)

figure: zip24_fitted_curves_overlay.png — channel panels 9–10 of 12, top to bottom

Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

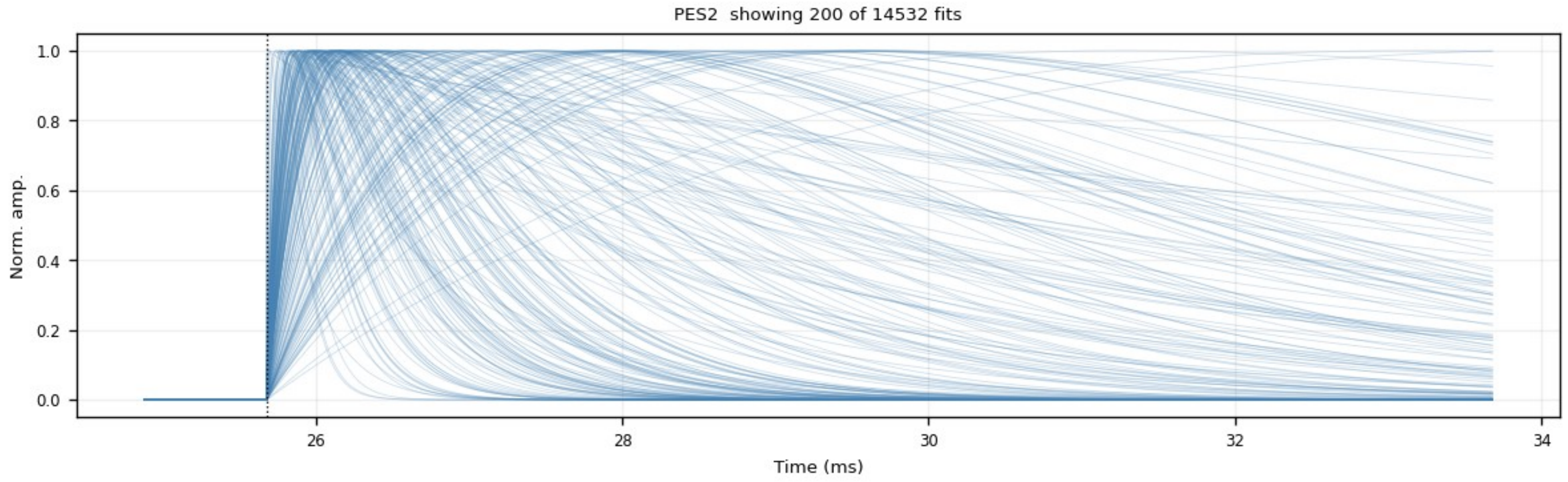


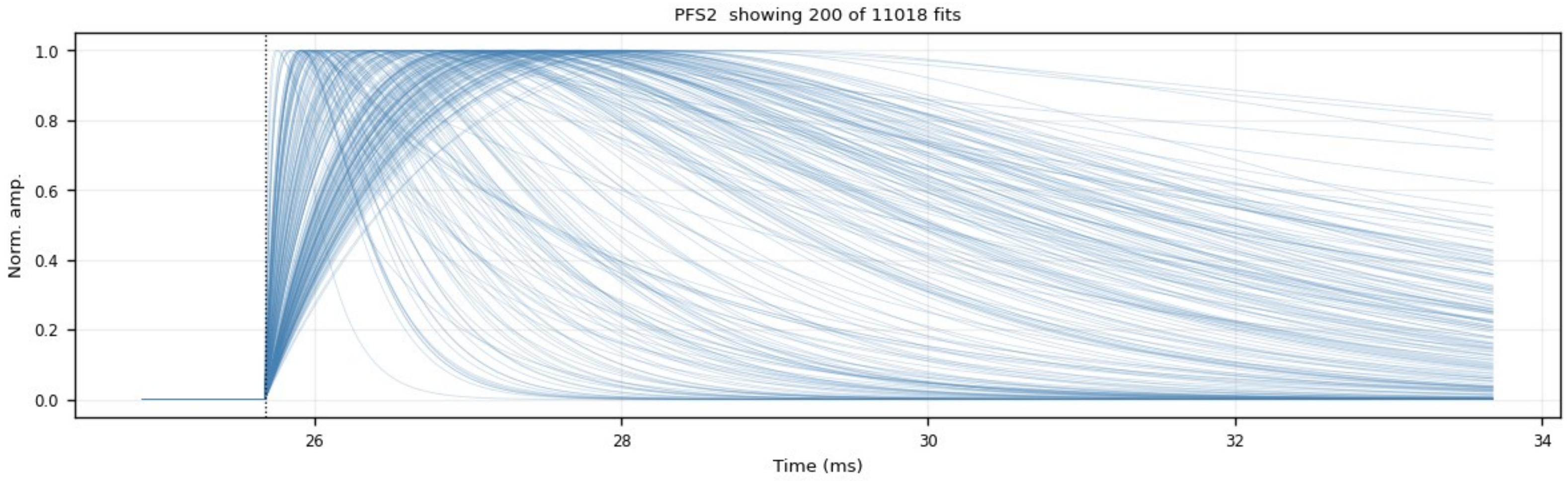


Z24 · fitted_curves_overlay — fan, no cut (6/6)

figure: zip24_fitted_curves_overlay.png — channel panels 11–12 of 12, top to bottom

Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

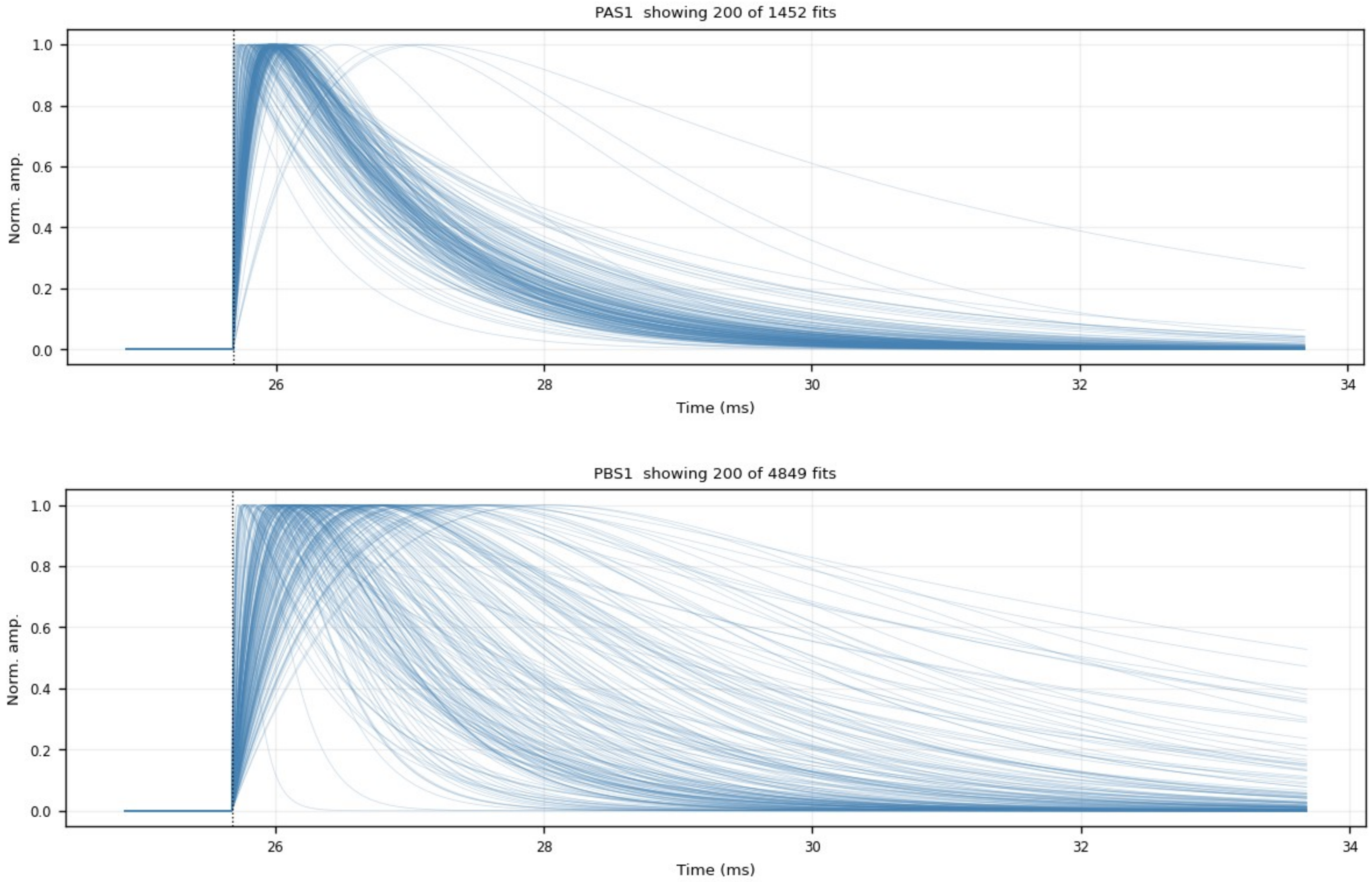




Z24 · fitted_curves_overlay — after NRMSE ≤ 0.4 (1/6)

figure: zip24_fitted_curves_overlay_nrmse0.4.png — channel panels 1–2 of 12, top to bottom

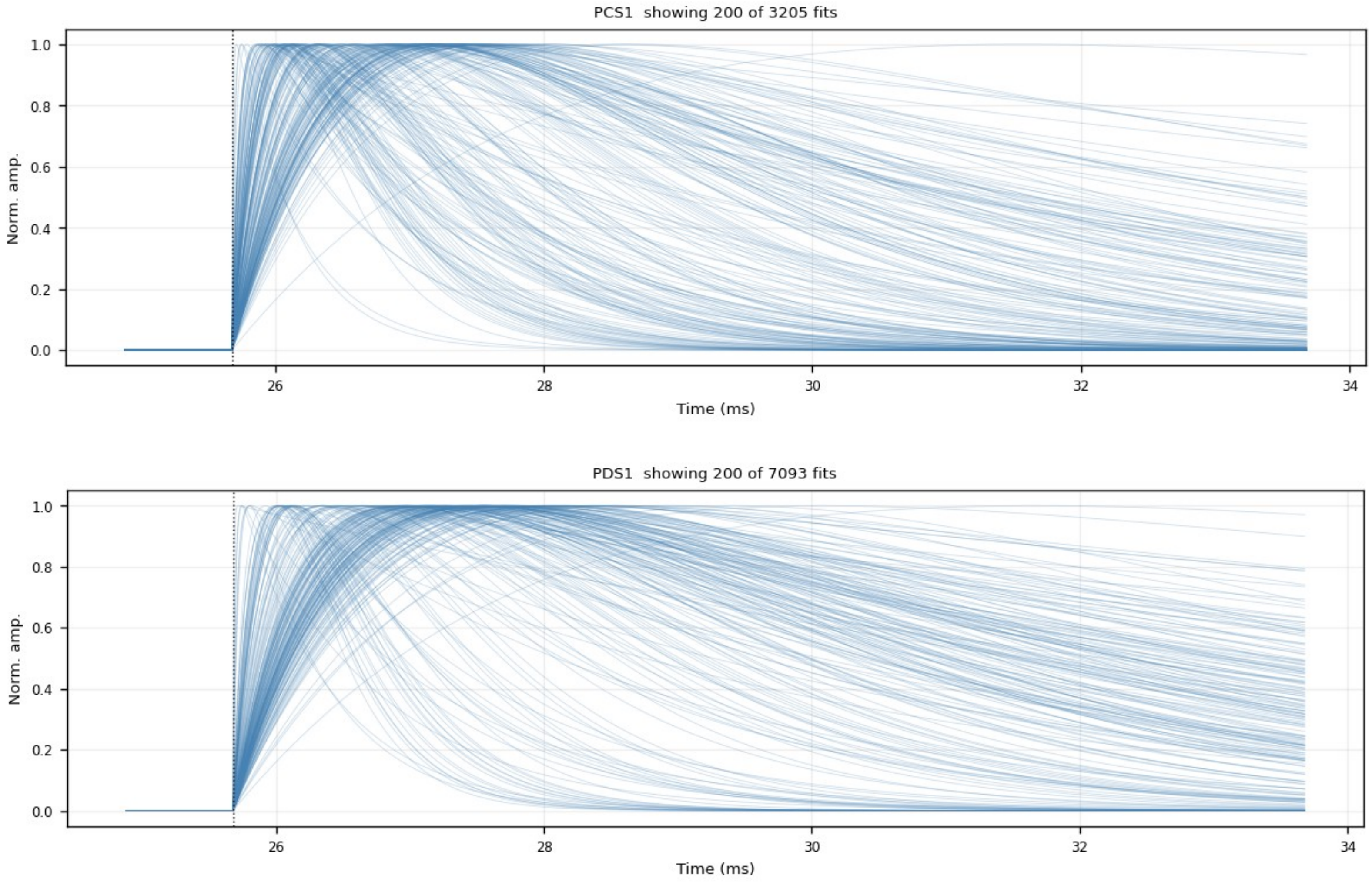
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z24 · fitted_curves_overlay — after NRMSE ≤ 0.4 (2/6)

figure: zip24_fitted_curves_overlay_nrmse0.4.png — channel panels 3–4 of 12, top to bottom

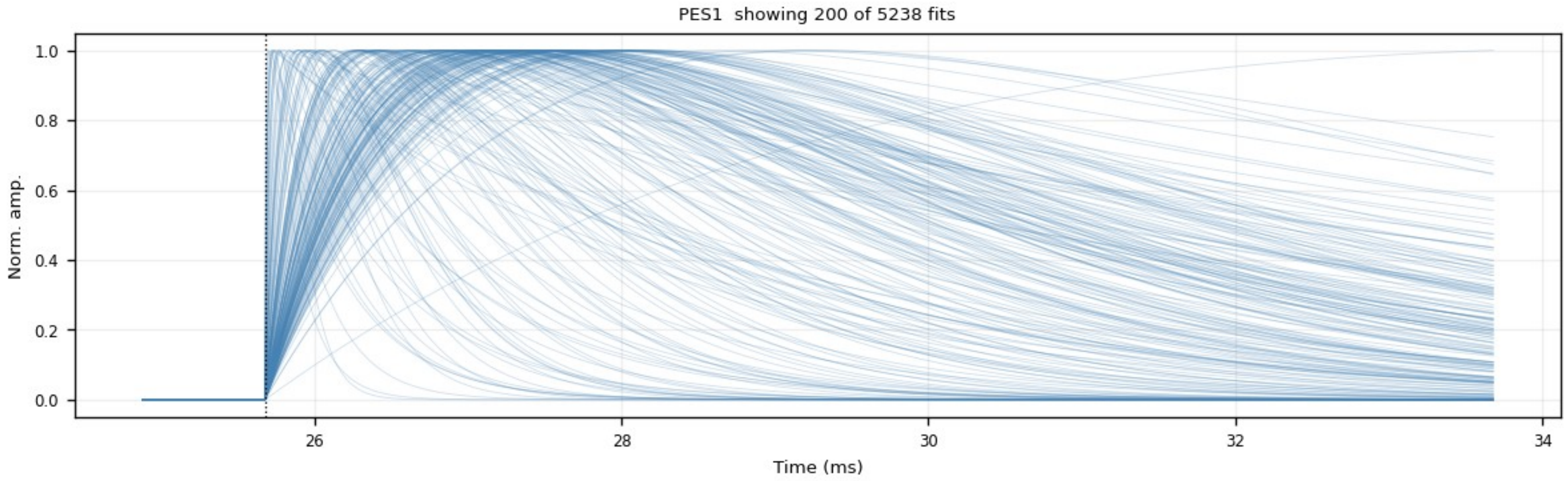
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.

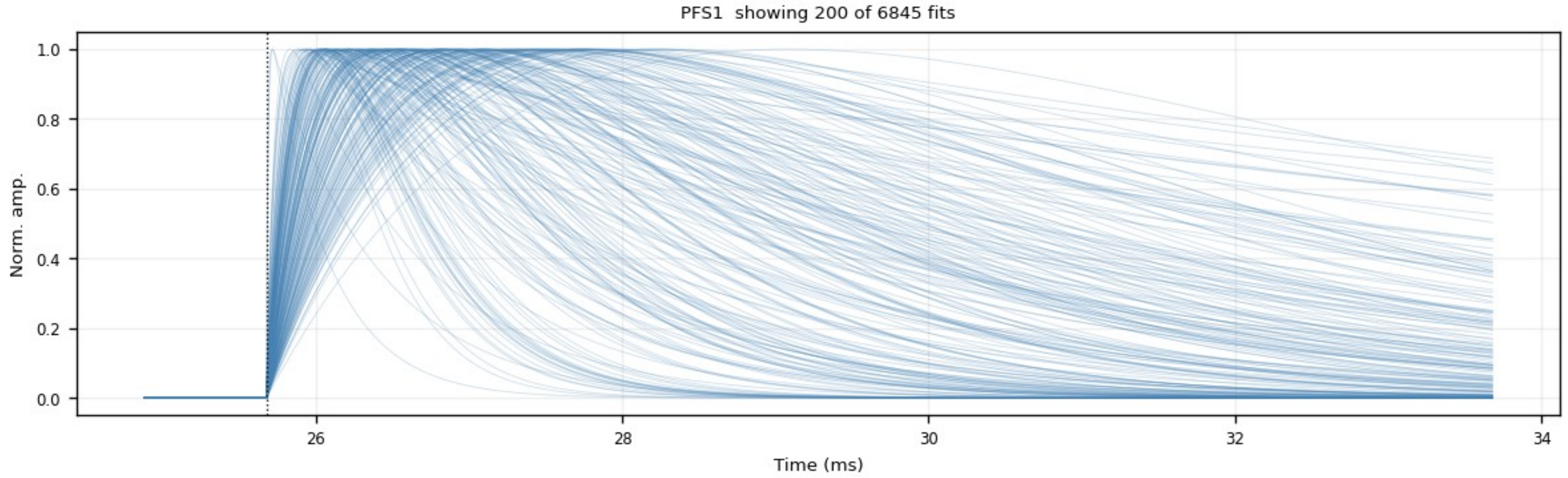


Z24 · fitted_curves_overlay — after NRMSE ≤ 0.4 (3/6)

figure: zip24_fitted_curves_overlay_nrmse0.4.png — channel panels 5–6 of 12, top to bottom

Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.

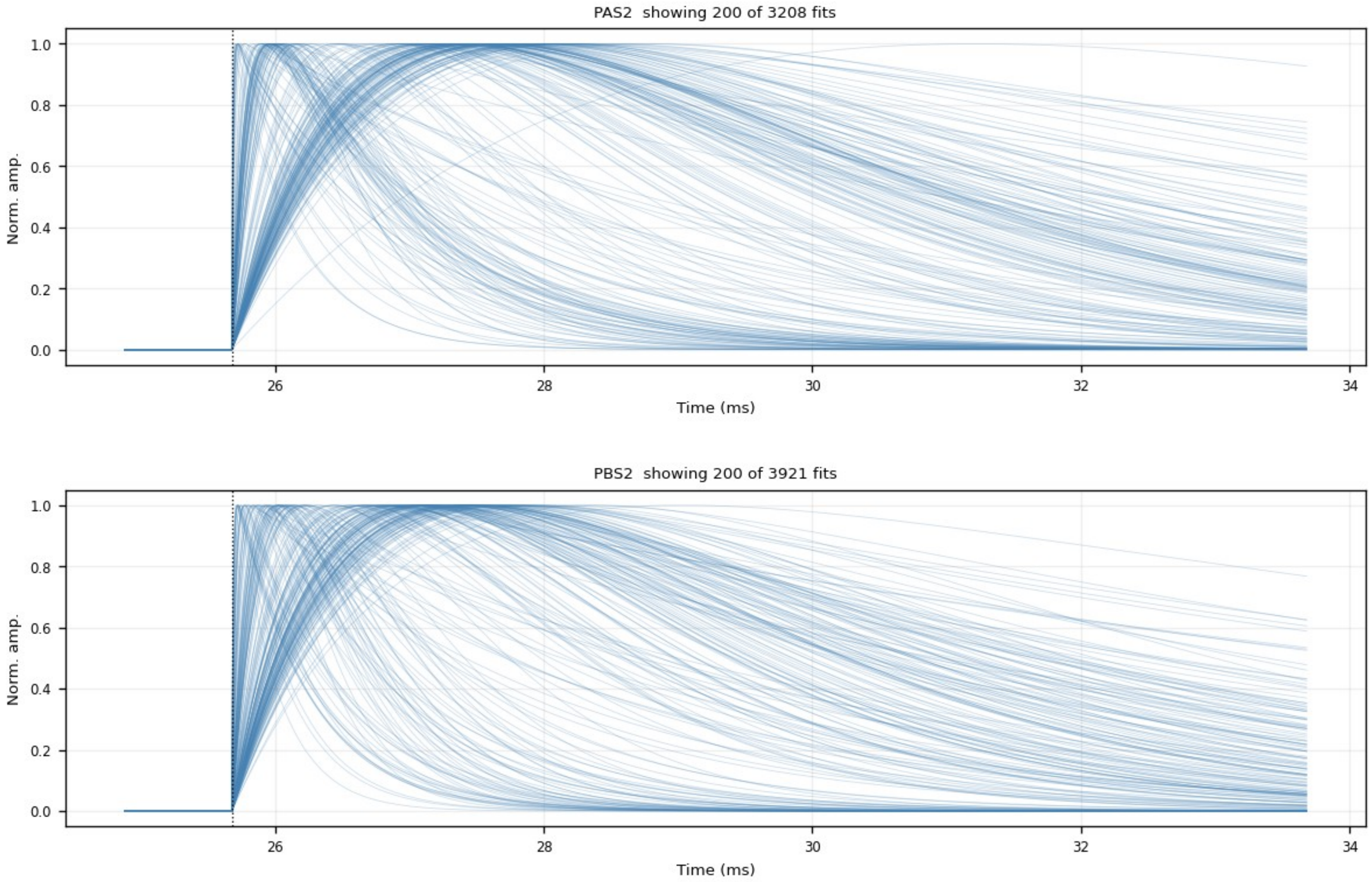




Z24 · fitted_curves_overlay — after NRMSE ≤ 0.4 (4/6)

figure: zip24_fitted_curves_overlay_nrmse0.4.png — channel panels 7–8 of 12, top to bottom

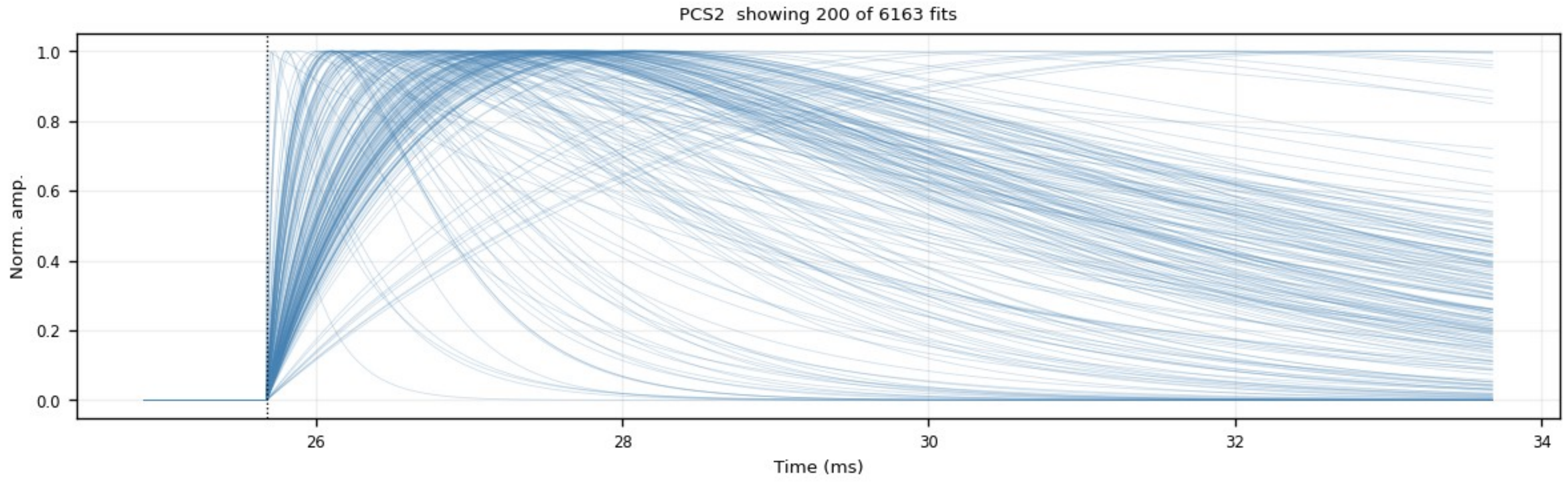
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.

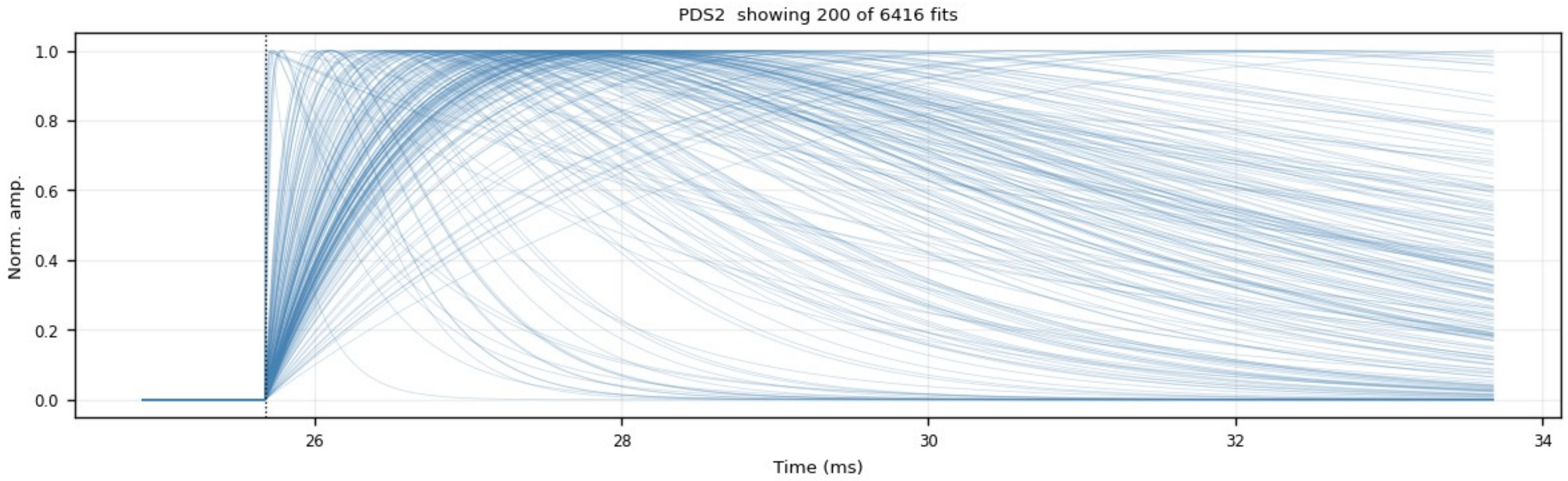


Z24 · fitted_curves_overlay — after NRMSE ≤ 0.4 (5/6)

figure: zip24_fitted_curves_overlay_nrmse0.4.png — channel panels 9–10 of 12, top to bottom

Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.

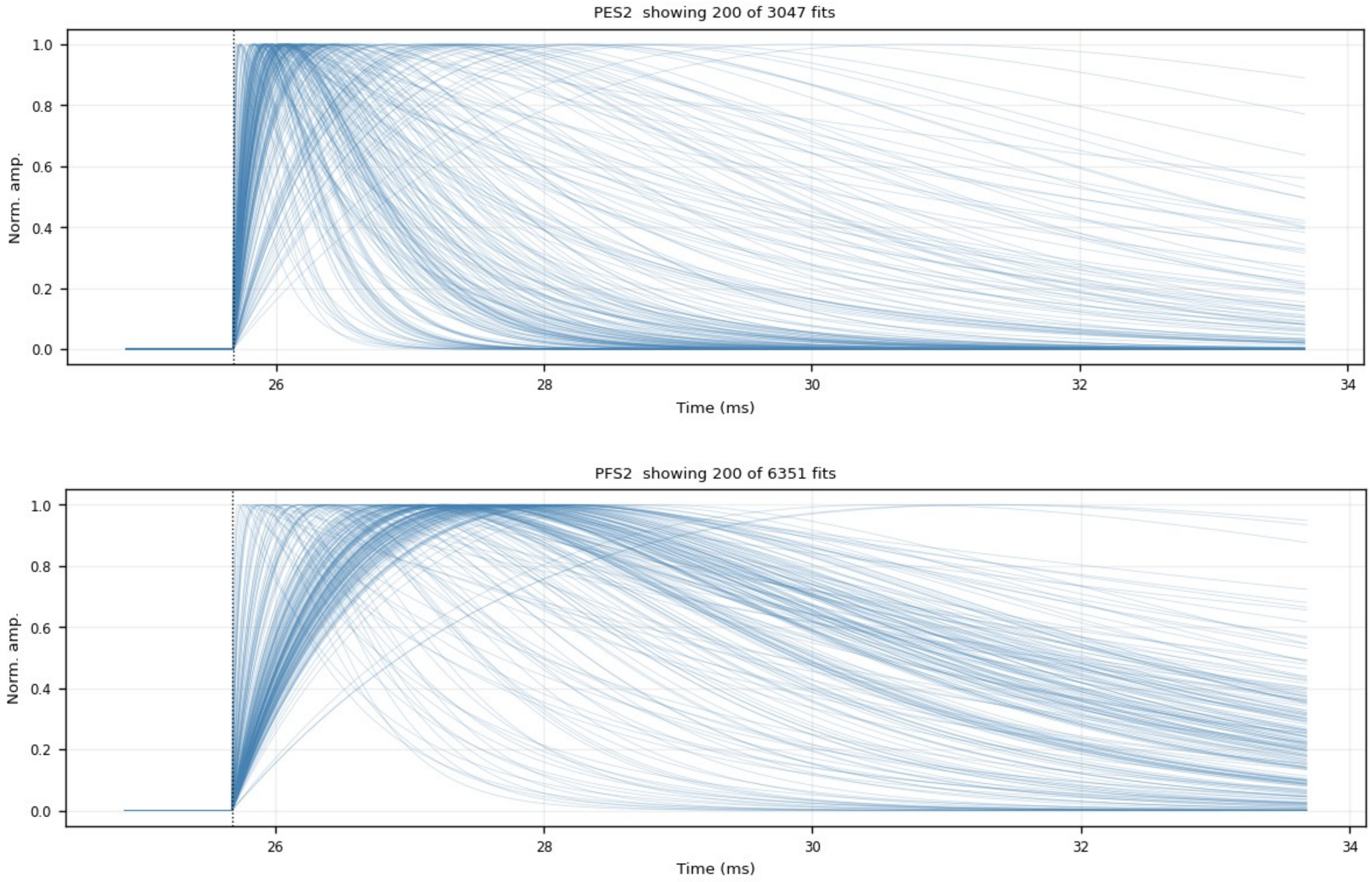




Z24 · fitted_curves_overlay — after NRMSE ≤ 0.4 (6/6)

figure: zip24_fitted_curves_overlay_nrmse0.4.png — channel panels 11–12 of 12, top to bottom

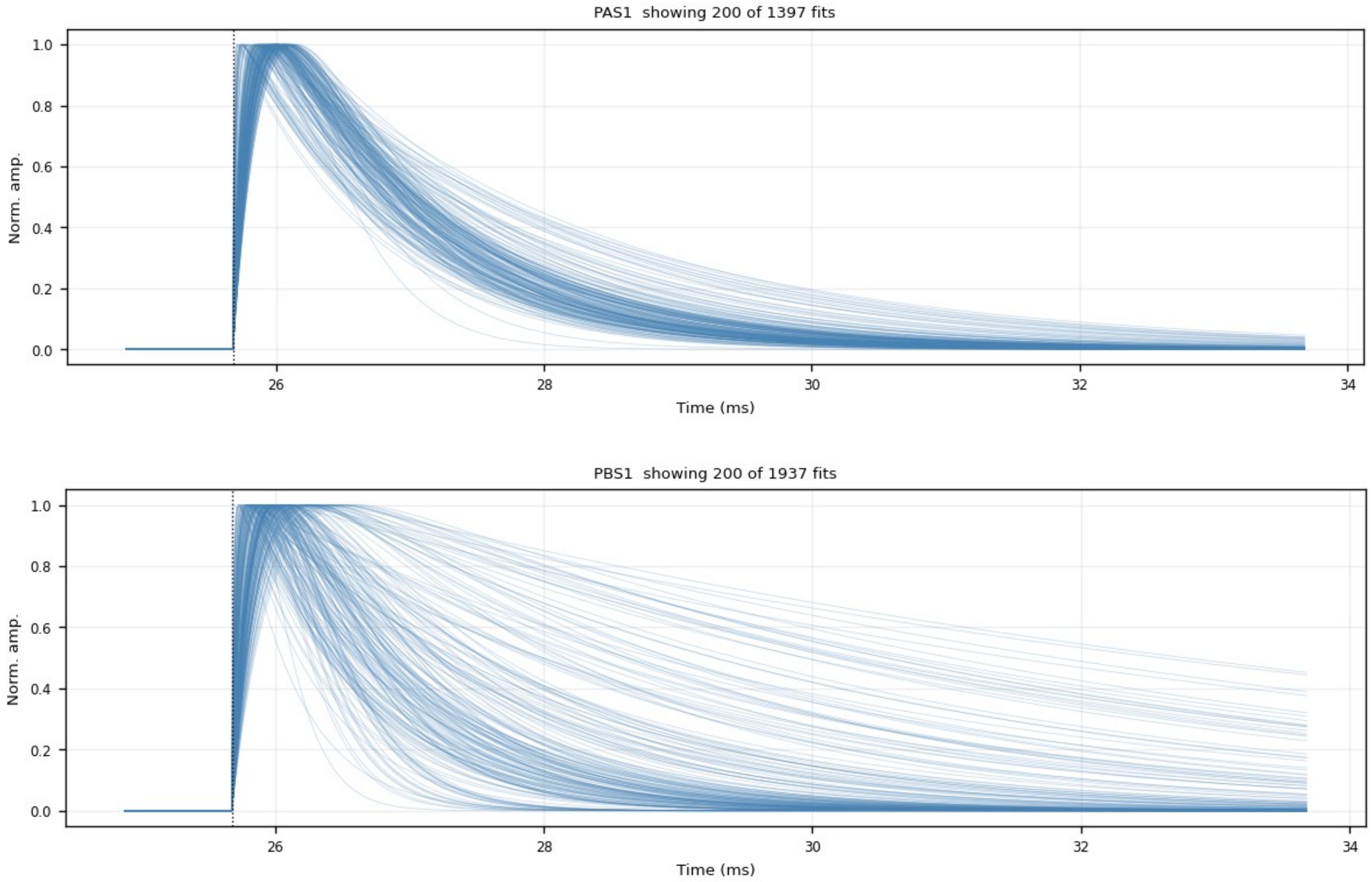
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z24 · fitted_curves_overlay — after both cuts (1/6)

figure: zip24_fitted_curves_overlay_nrmse0.4_trise0.30ms.png — channel panels 1–2 of 12, top to bottom

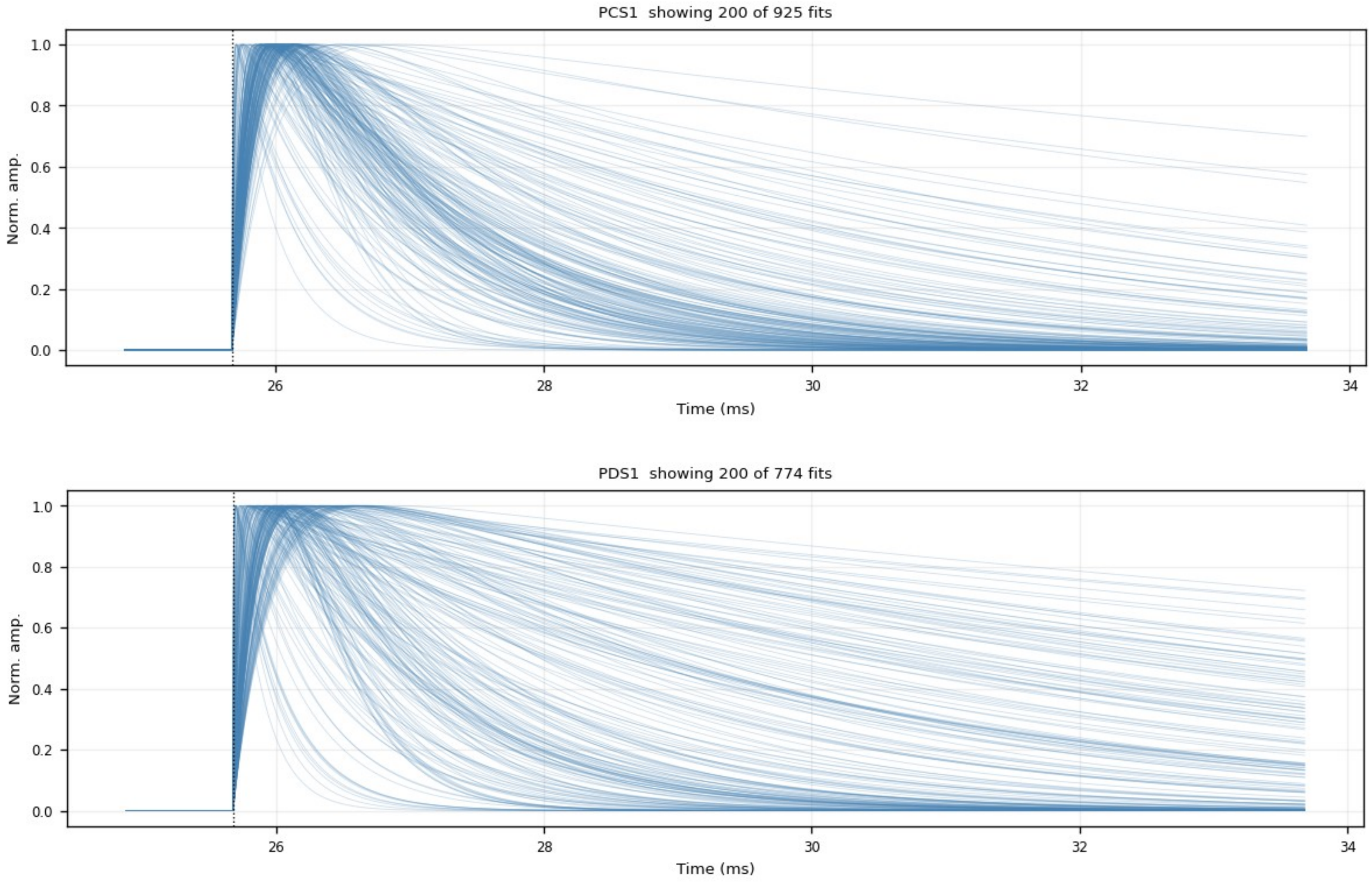
After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms — the exact PCA input population: the clean fast-pulse bundle only.



Z24 · fitted_curves_overlay — after both cuts (2/6)

figure: zip24_fitted_curves_overlay_nrmse0.4_trise0.30ms.png — channel panels 3–4 of 12, top to bottom

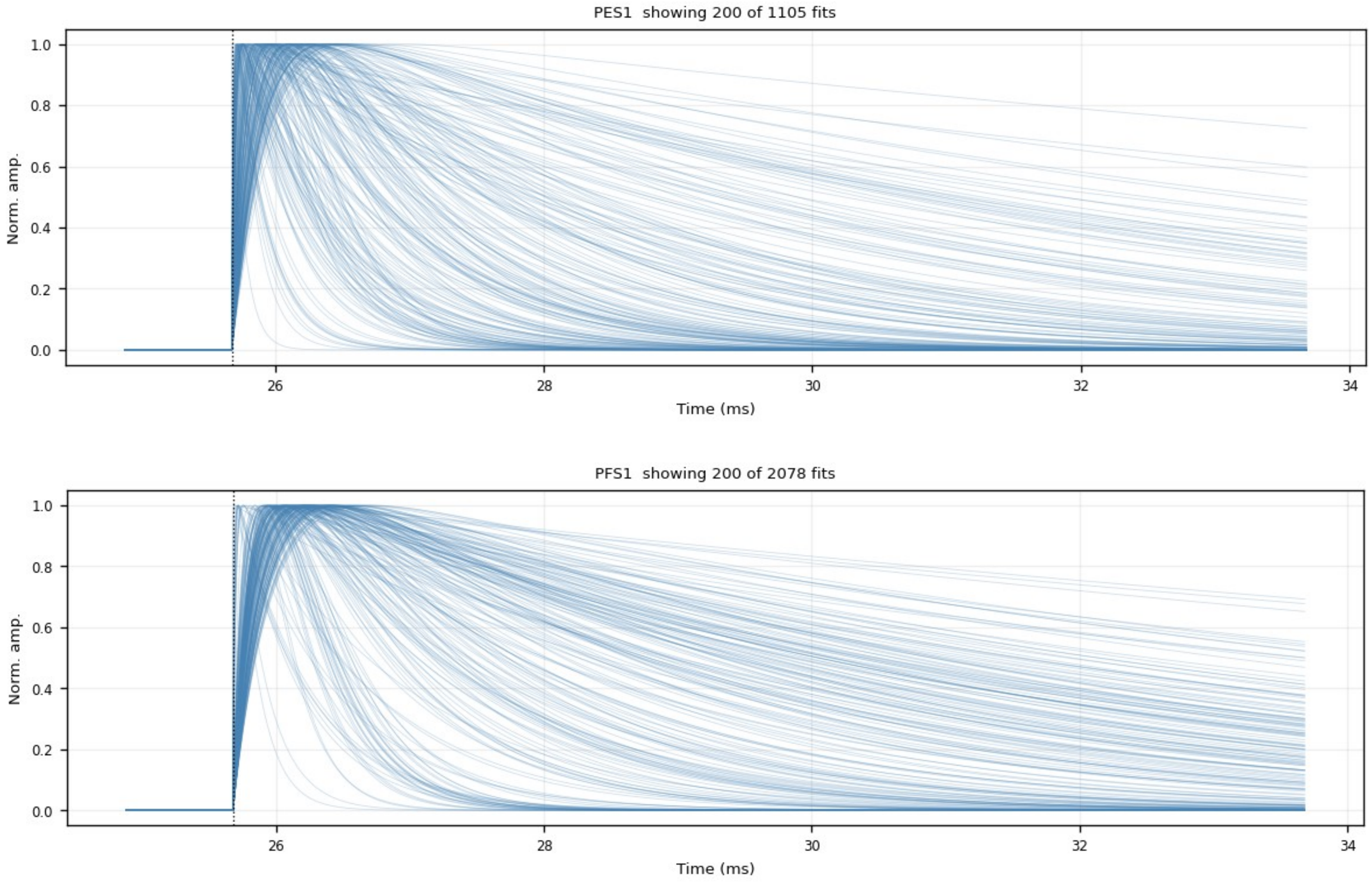
After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms — the exact PCA input population: the clean fast-pulse bundle only.



Z24 · fitted_curves_overlay — after both cuts (3/6)

figure: zip24_fitted_curves_overlay_nrmse0.4_trise0.30ms.png — channel panels 5–6 of 12, top to bottom

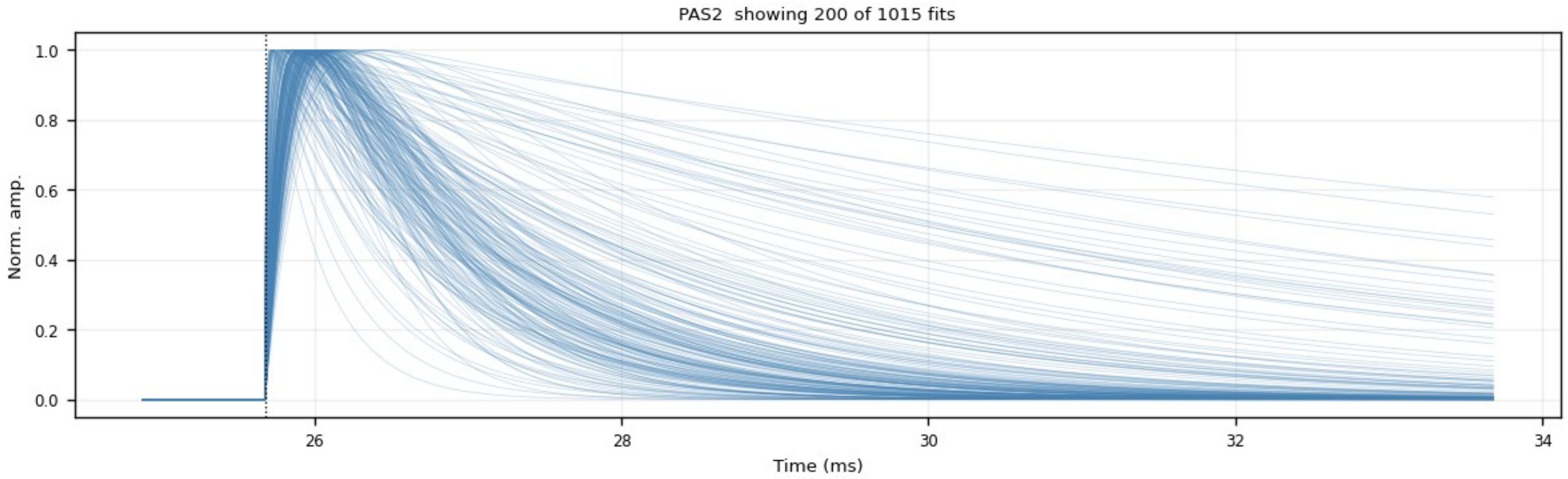
After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms — the exact PCA input population: the clean fast-pulse bundle only.

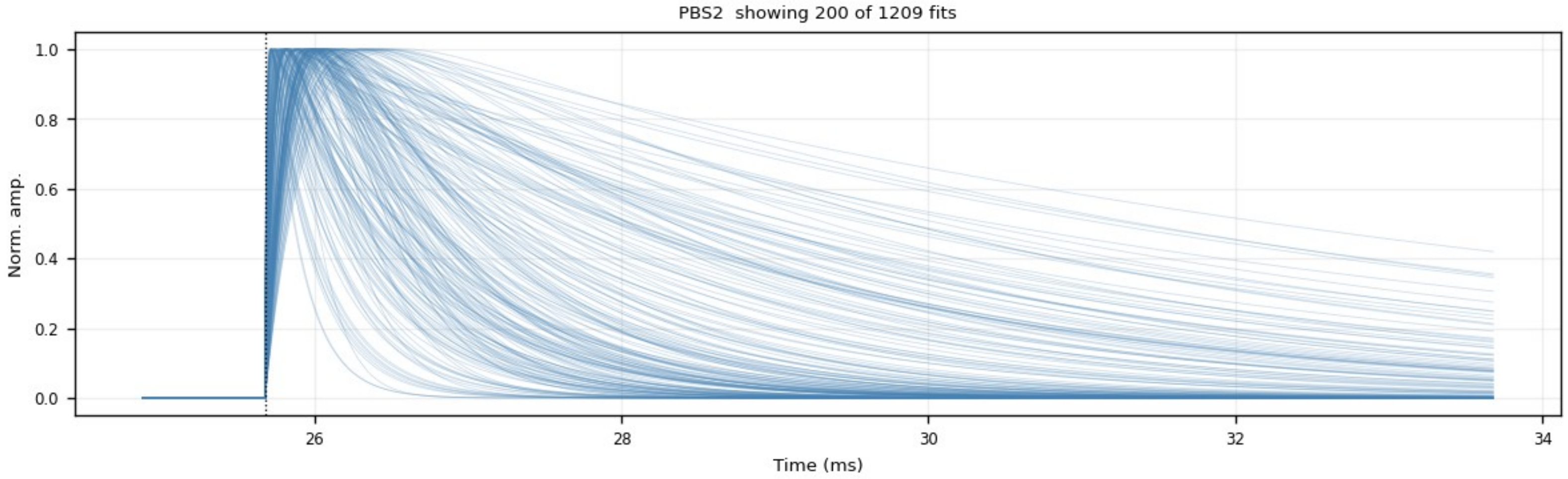


Z24 · fitted_curves_overlay — after both cuts (4/6)

figure: zip24_fitted_curves_overlay_nrmse0.4_trise0.30ms.png — channel panels 7–8 of 12, top to bottom

After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms — the exact PCA input population: the clean fast-pulse bundle only.

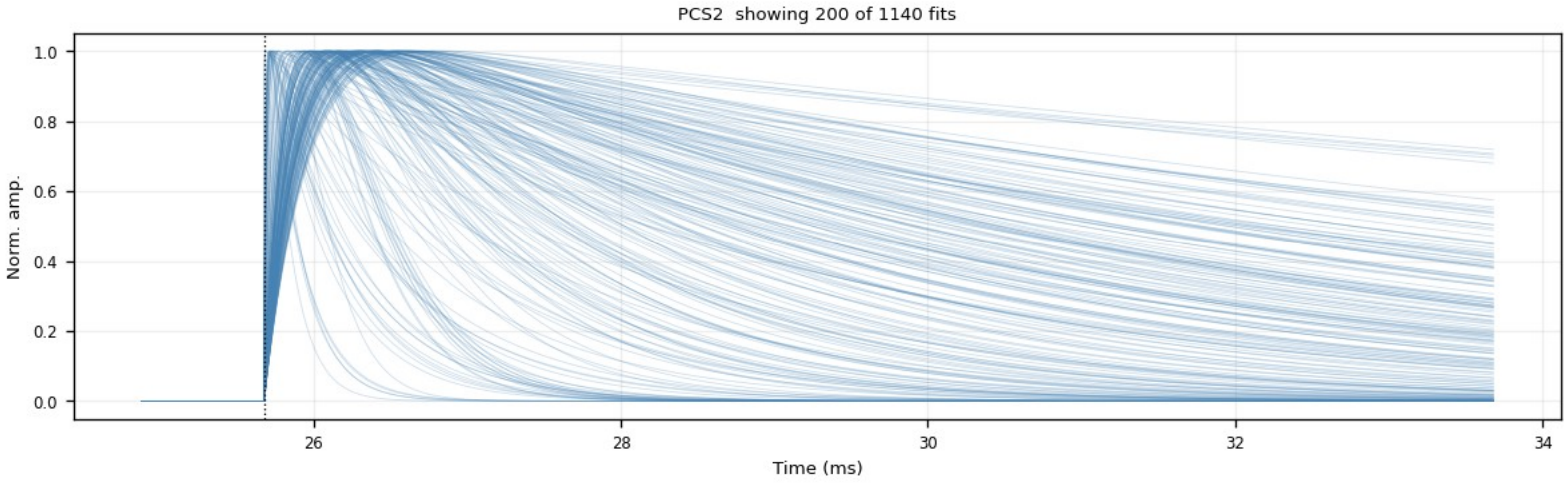


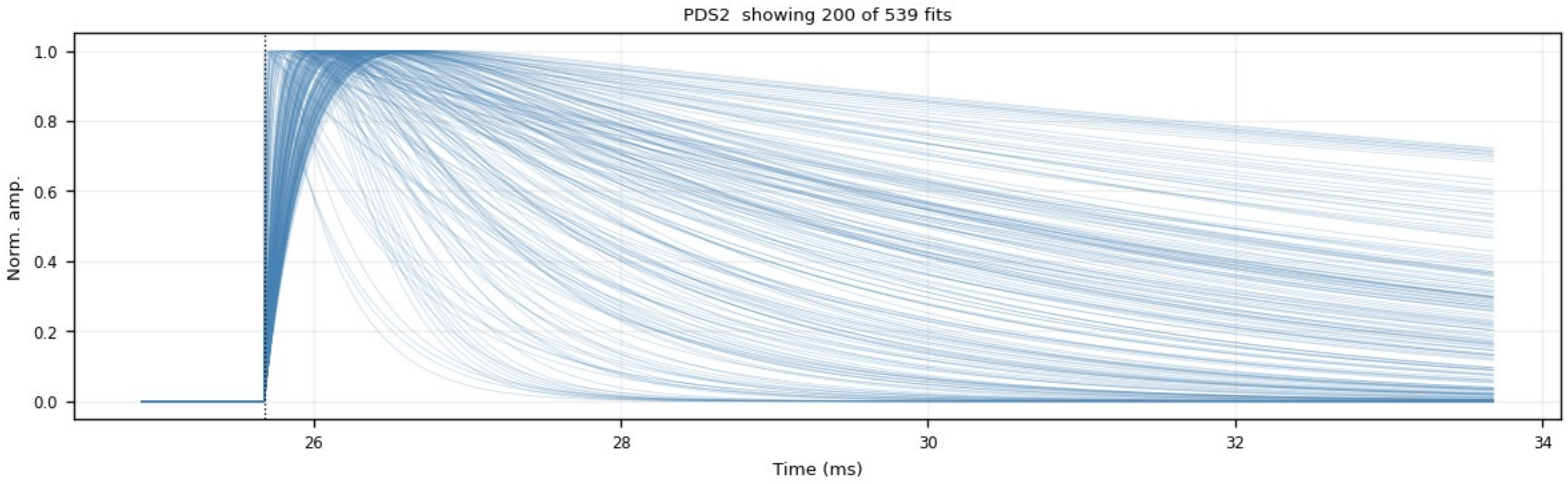


Z24 · fitted_curves_overlay — after both cuts (5/6)

figure: zip24_fitted_curves_overlay_nrmse0.4_trise0.30ms.png — channel panels 9–10 of 12, top to bottom

After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms — the exact PCA input population: the clean fast-pulse bundle only.

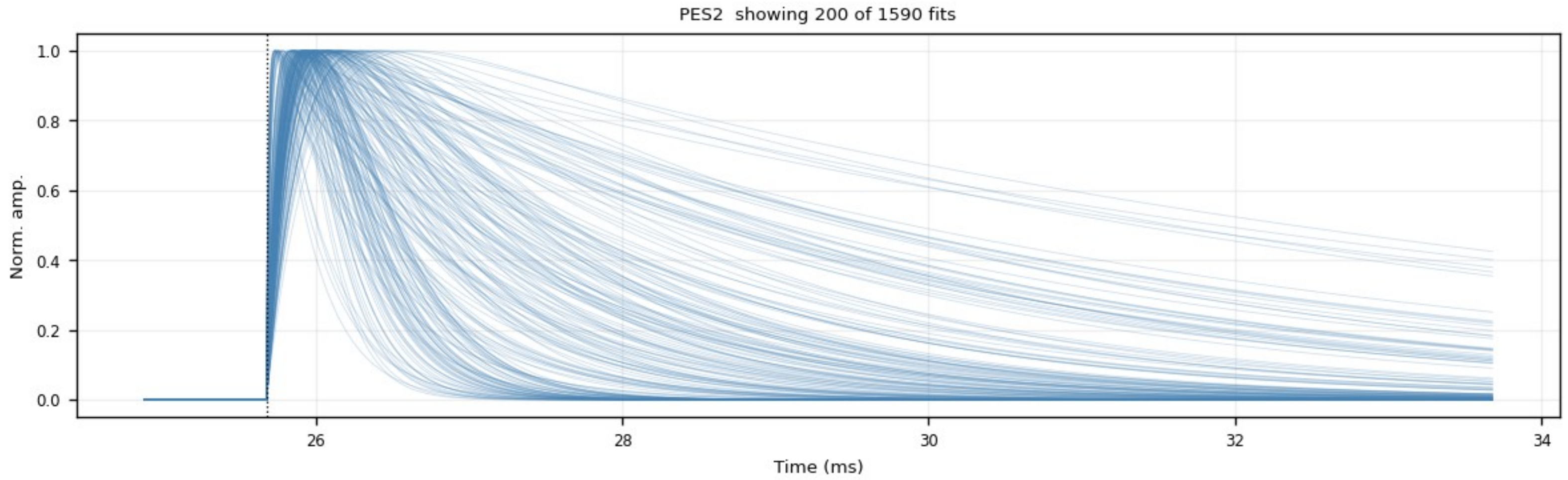


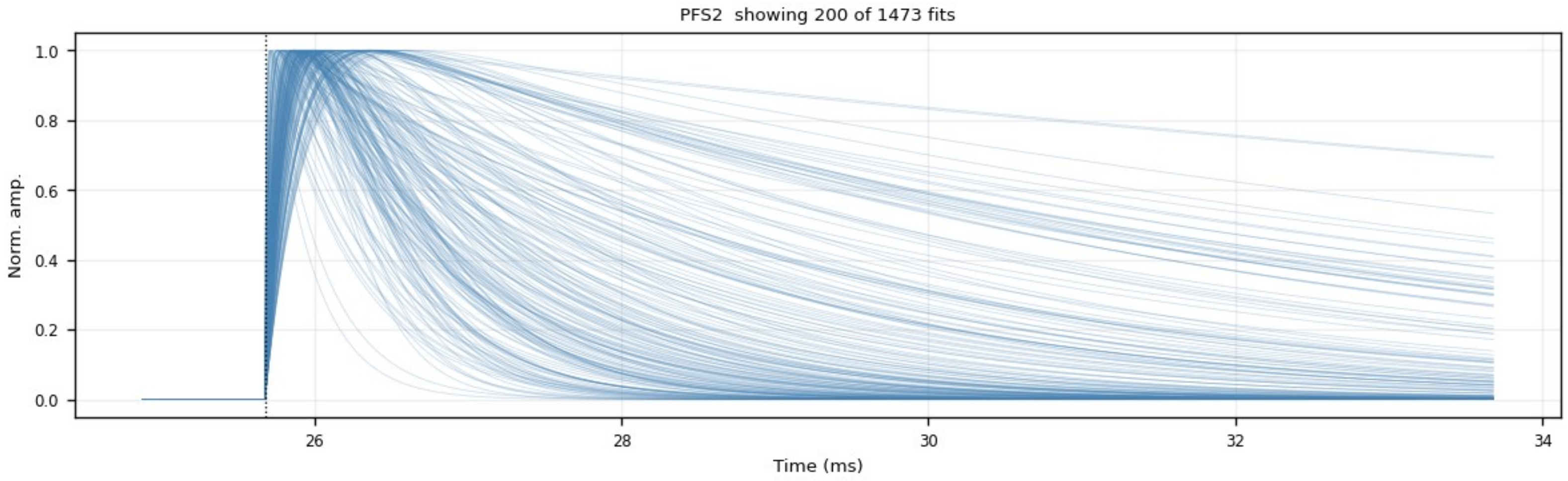


Z24 · fitted_curves_overlay — after both cuts (6/6)

figure: zip24_fitted_curves_overlay_nrmse0.4_trise0.30ms.png — channel panels 11–12 of 12, top to bottom

After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms — the exact PCA input population: the clean fast-pulse bundle only.

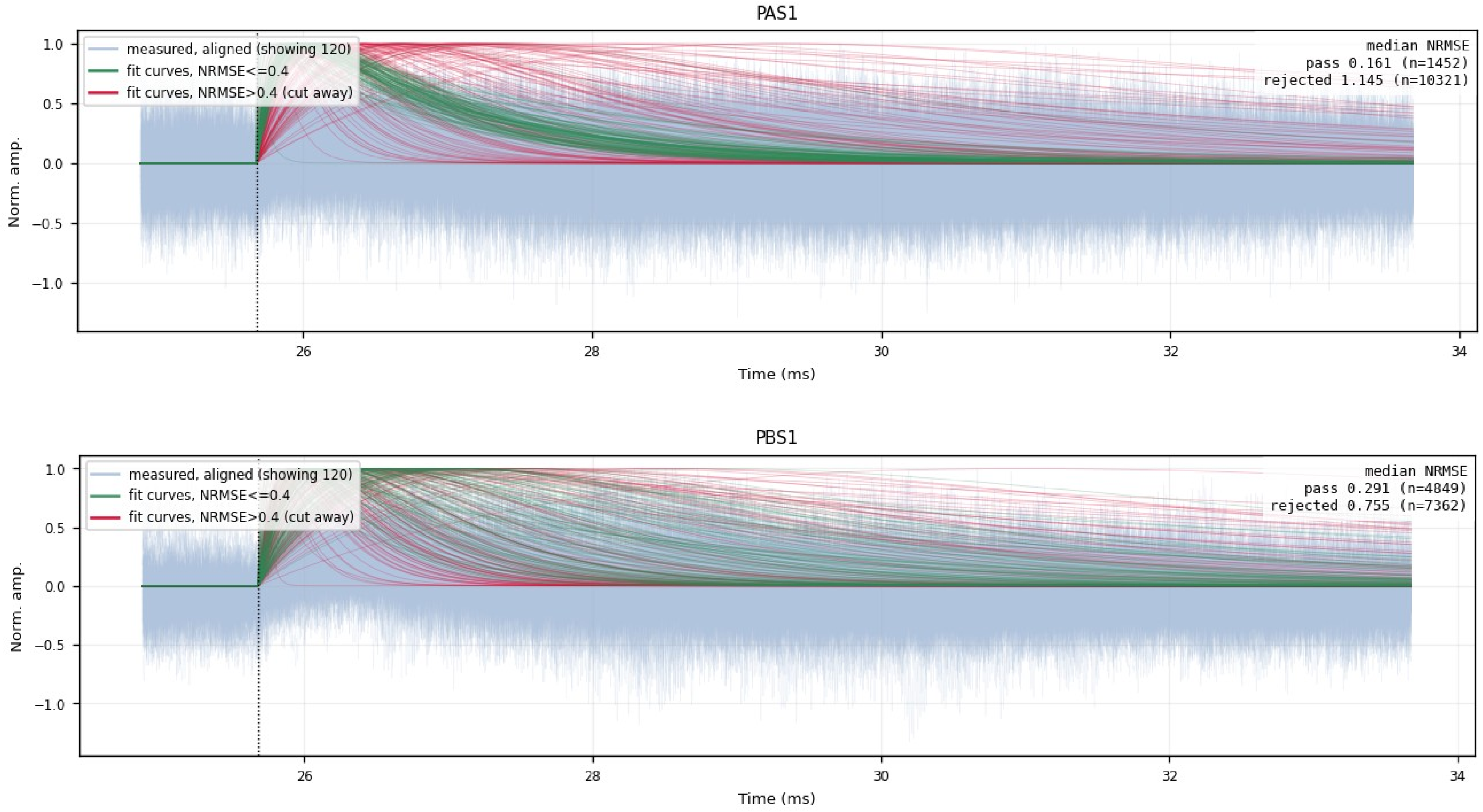




Z24 · overlay_fan_cut — kept vs rejected fits (1/6)

figure: zip24_overlay_fan_cut_nrmse0.4.png — channel panels 1–2 of 12, top to bottom

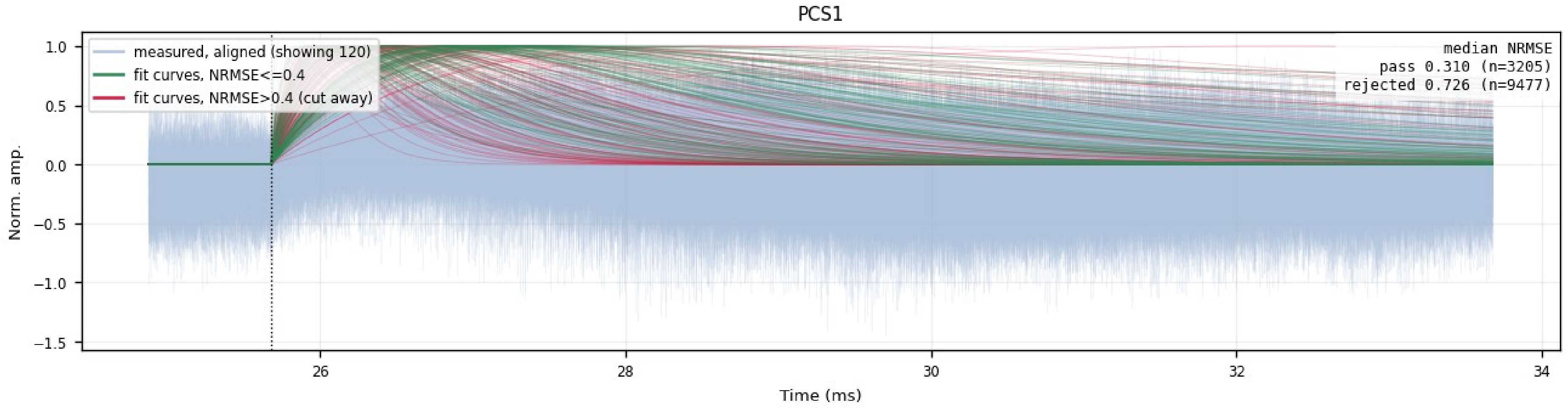
Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

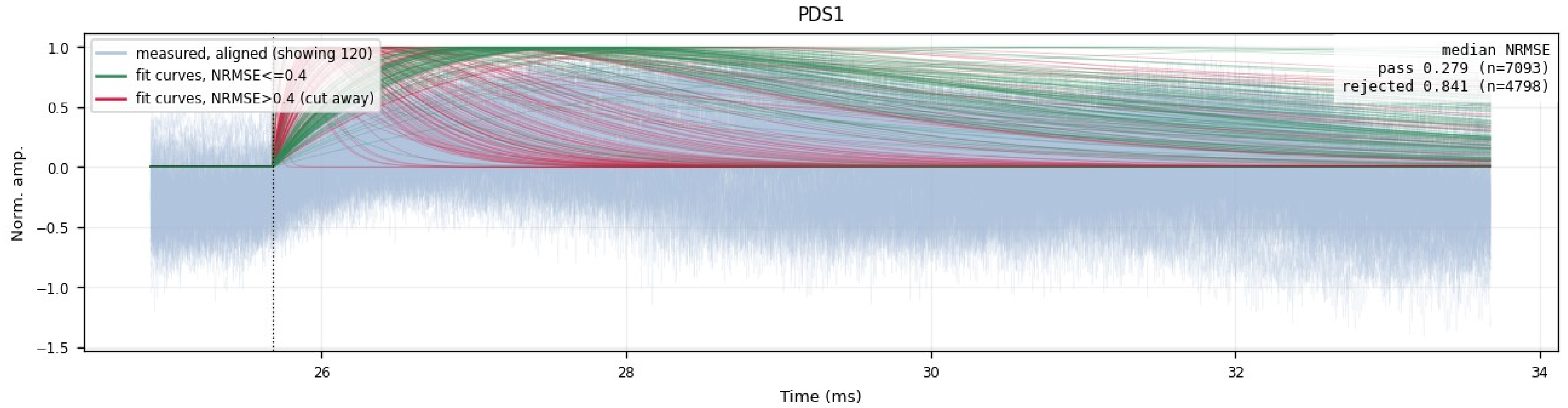


Z24 · overlay_fan_cut — kept vs rejected fits (2/6)

figure: zip24_overlay_fan_cut_nrmse0.4.png — channel panels 3–4 of 12, top to bottom

Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

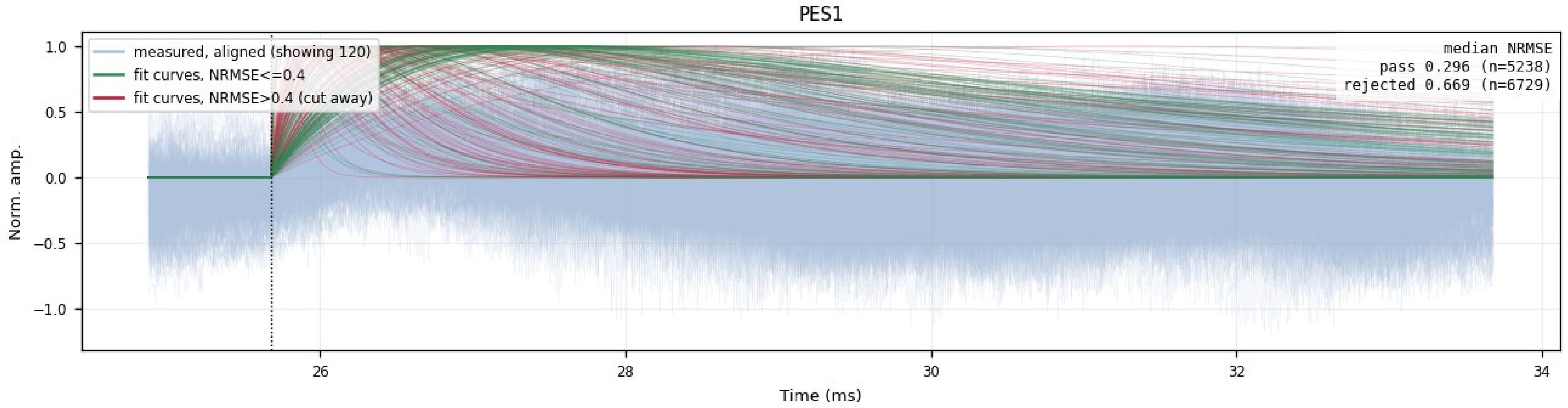


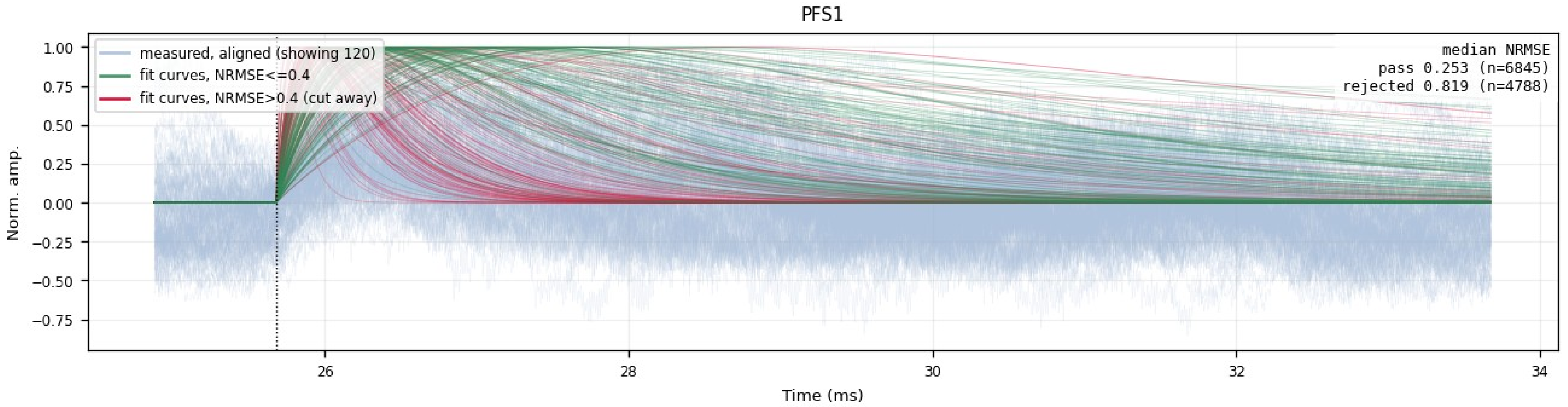


Z24 · overlay_fan_cut — kept vs rejected fits (3/6)

figure: zip24_overlay_fan_cut_nrmse0.4.png — channel panels 5–6 of 12, top to bottom

Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

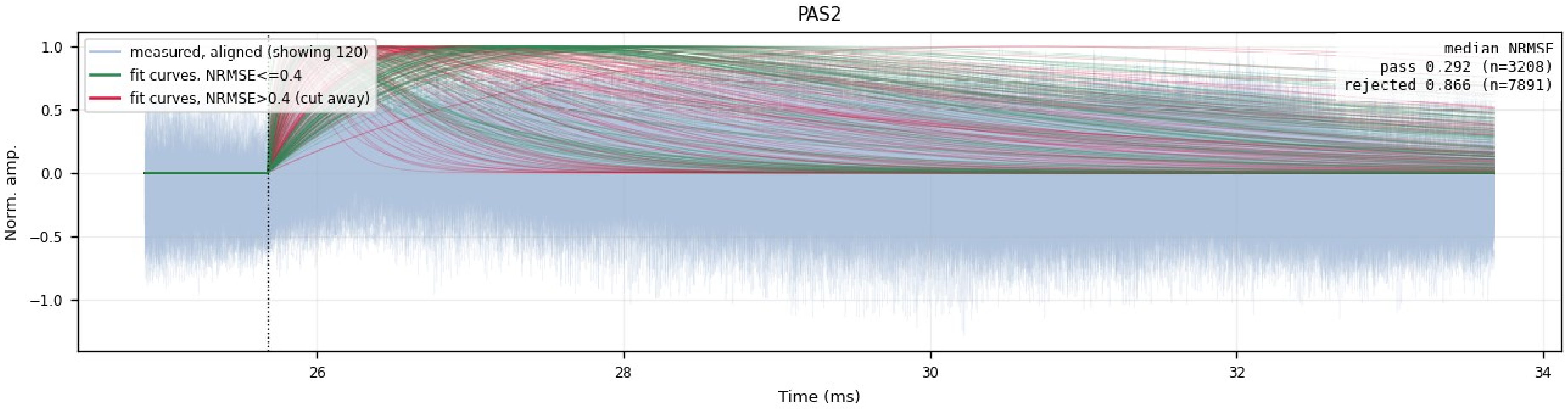


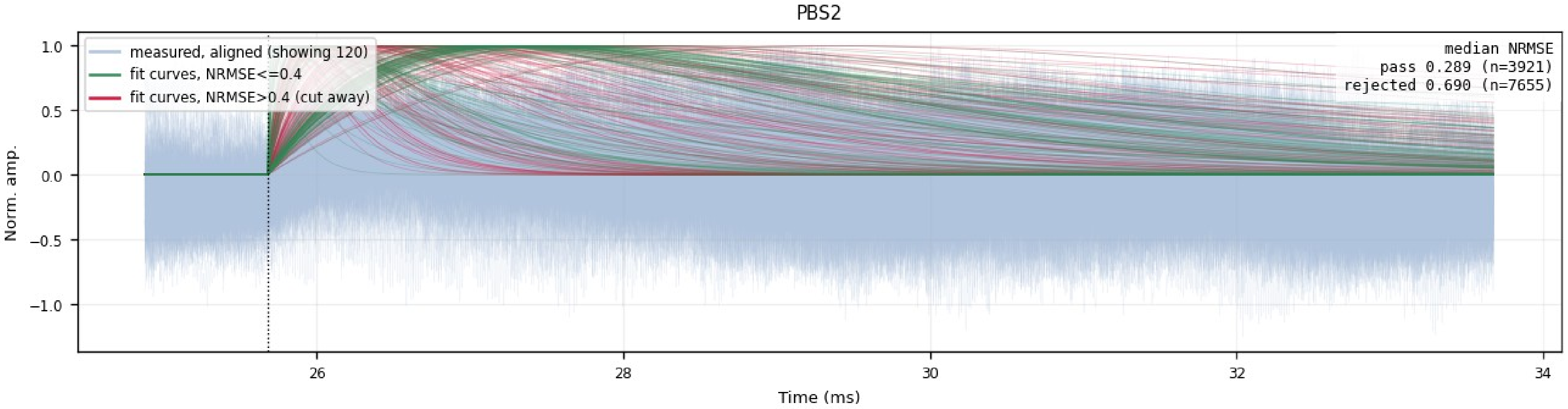


Z24 · overlay_fan_cut — kept vs rejected fits (4/6)

figure: zip24_overlay_fan_cut_nrmse0.4.png — channel panels 7–8 of 12, top to bottom

Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

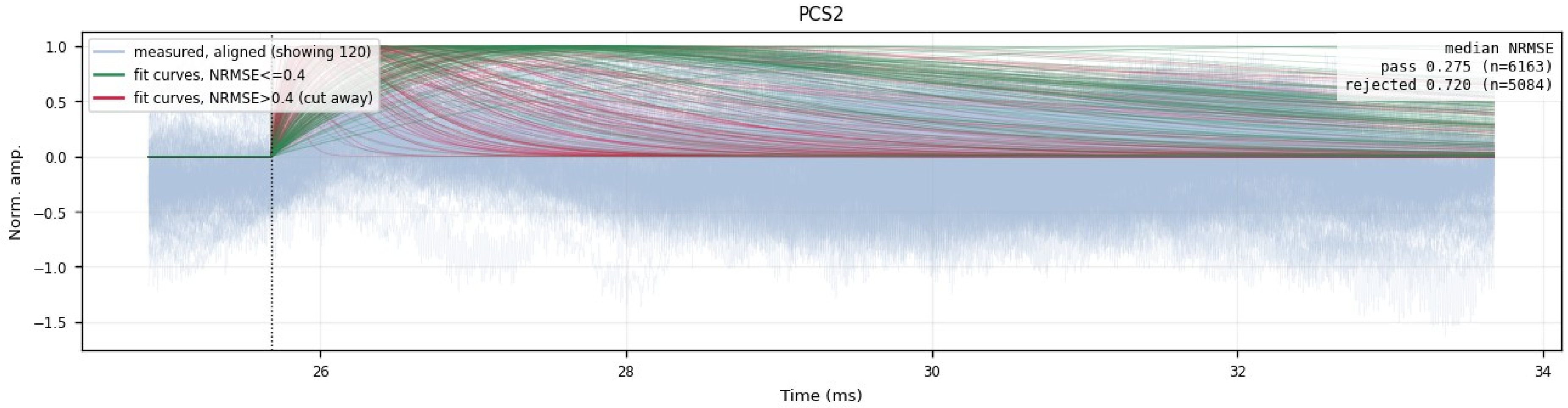


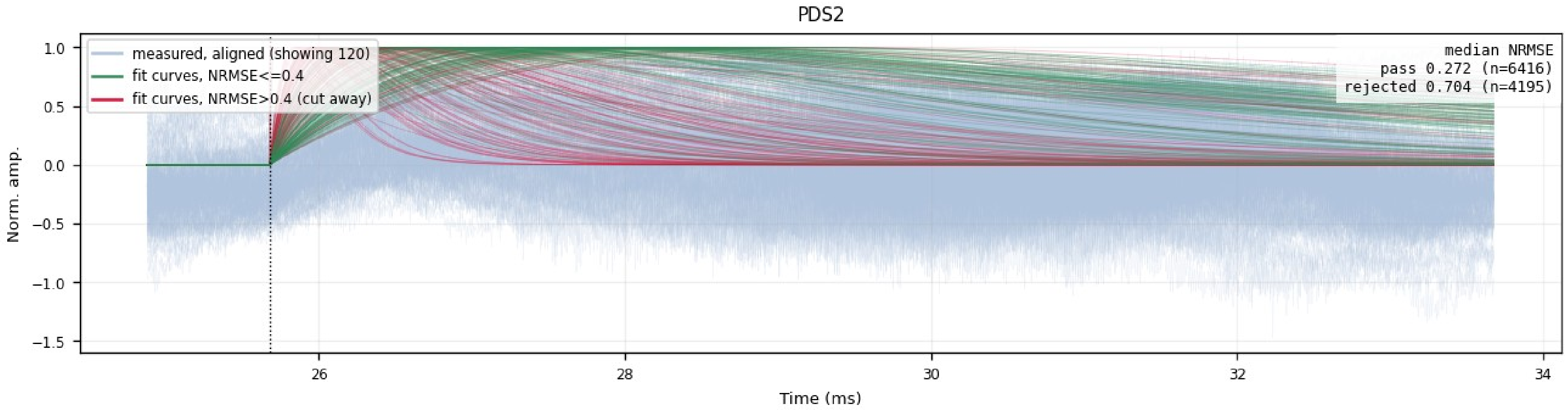


Z24 · overlay_fan_cut — kept vs rejected fits (5/6)

figure: zip24_overlay_fan_cut_nrmse0.4.png — channel panels 9–10 of 12, top to bottom

Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

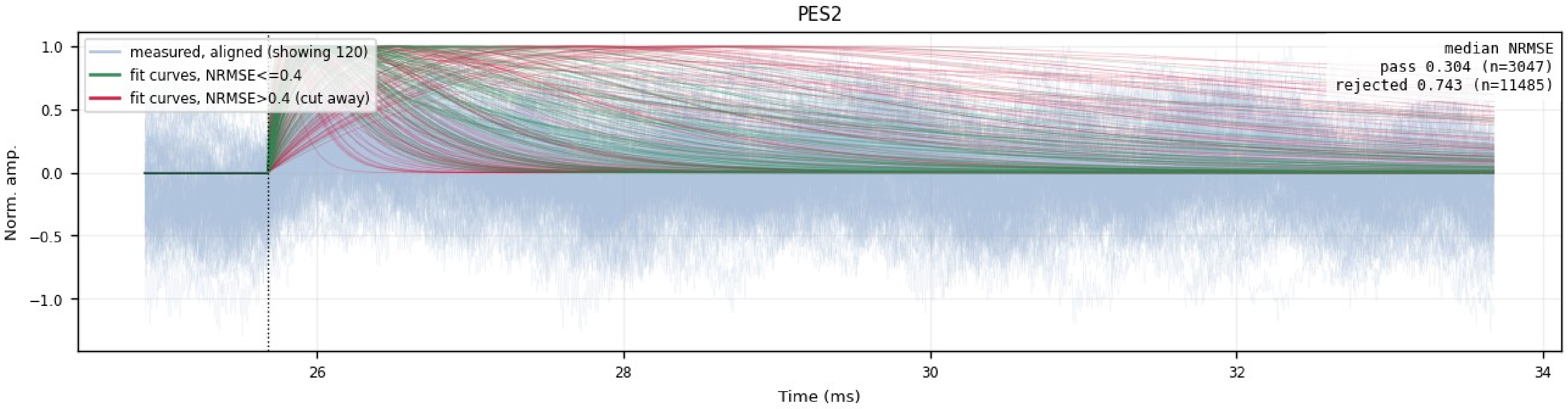


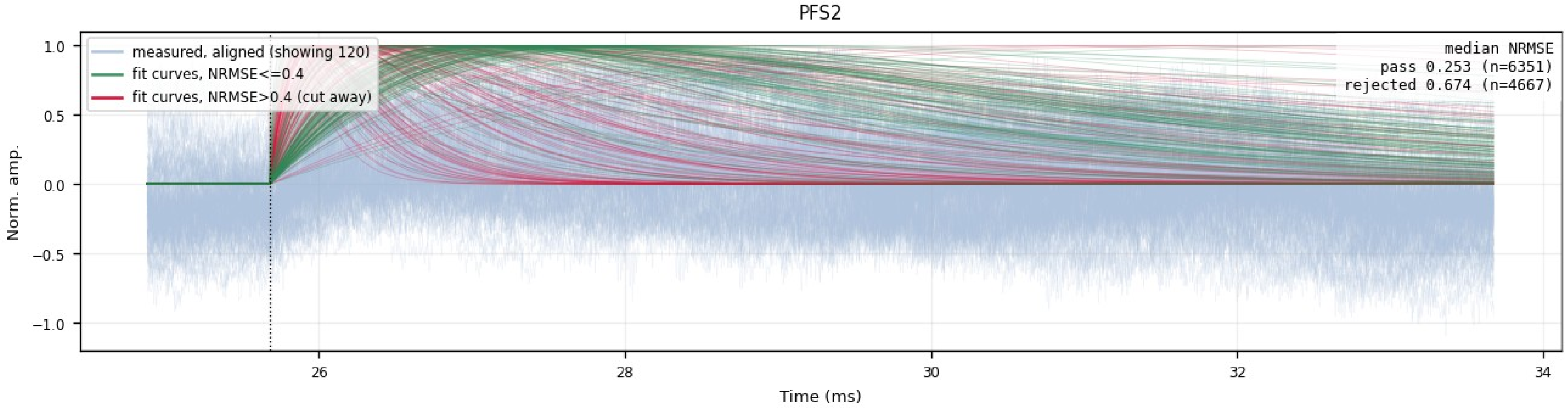


Z24 · overlay_fan_cut — kept vs rejected fits (6/6)

figure: zip24_overlay_fan_cut_nrmse0.4.png — channel panels 11–12 of 12, top to bottom

Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

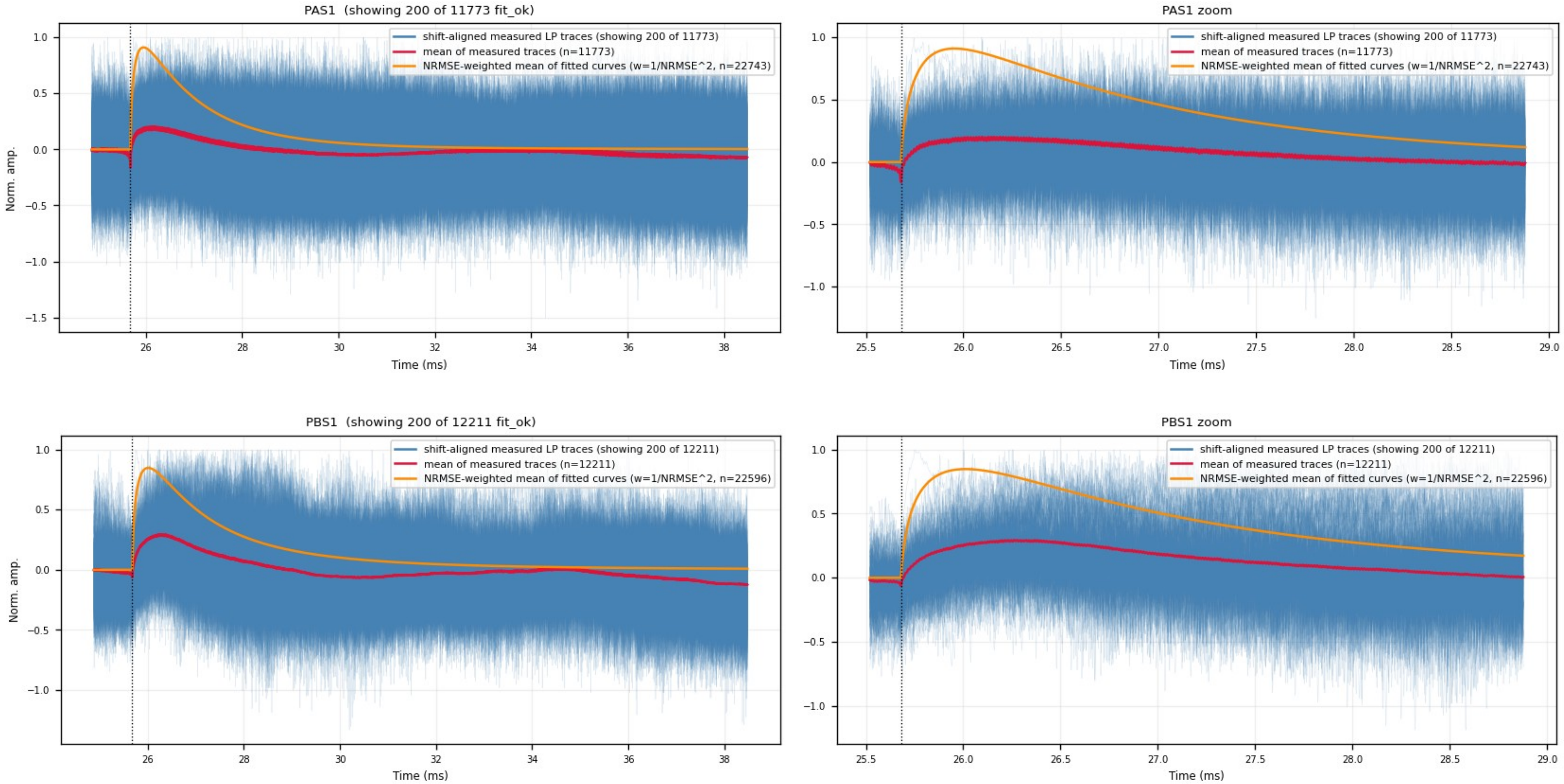




Z24 · aligned_overlay — aligned measured traces (1/6)

figure: zip24_lp_aligned_overlay.png — channel panels 1–2 of 12, top to bottom

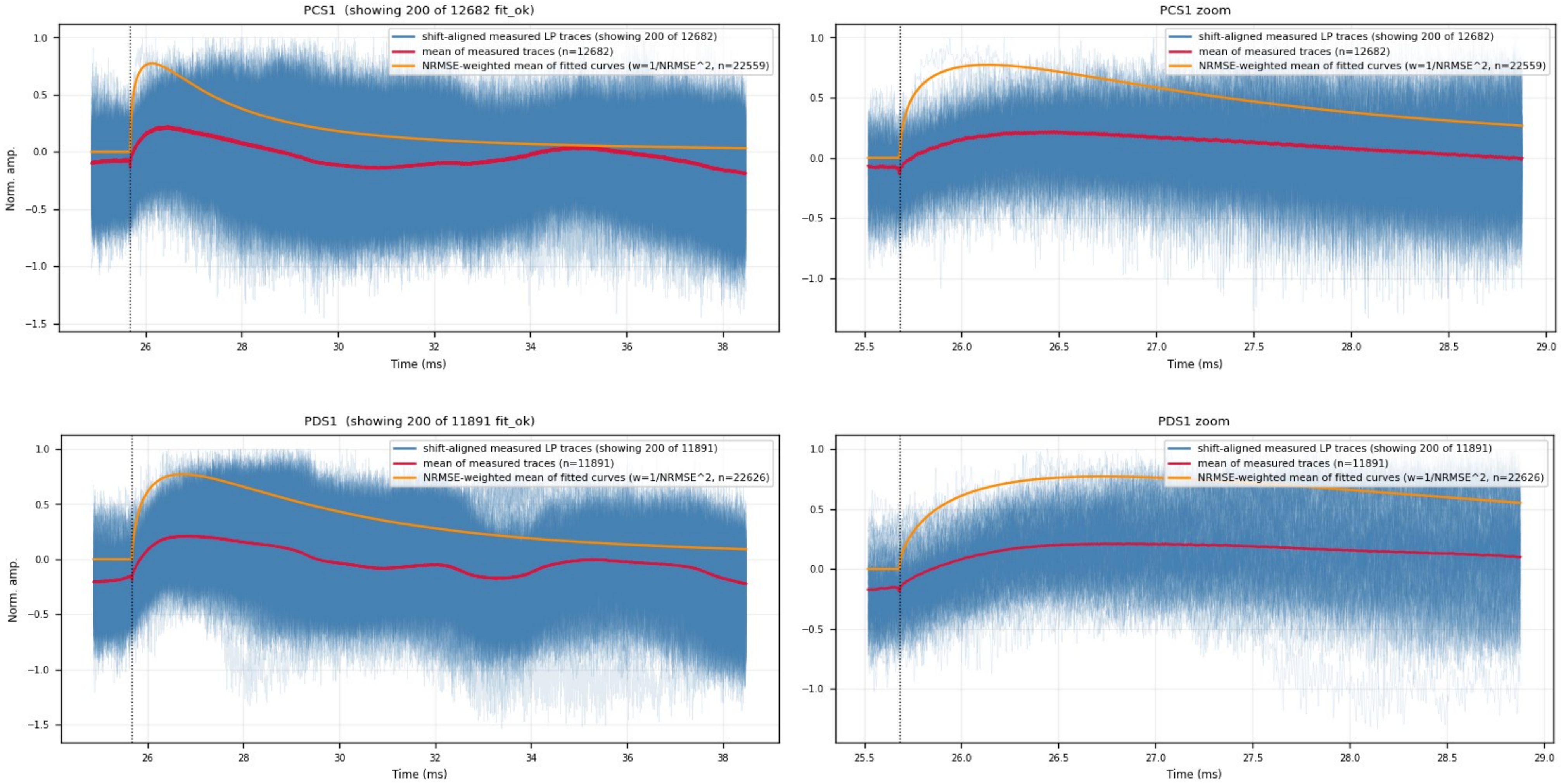
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z24 · aligned_overlay — aligned measured traces (2/6)

figure: zip24_lp_aligned_overlay.png — channel panels 3–4 of 12, top to bottom

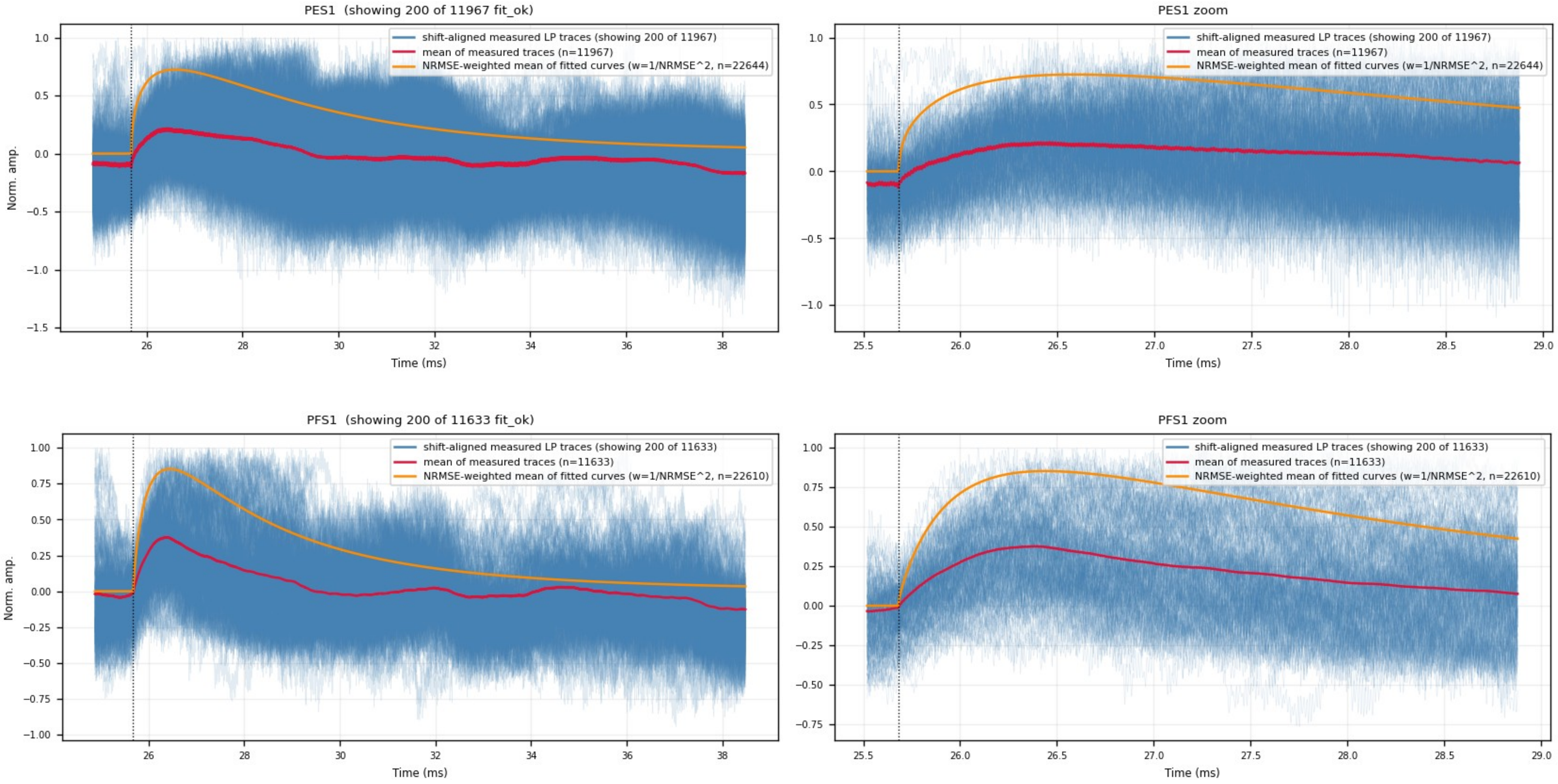
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z24 · aligned_overlay — aligned measured traces (3/6)

figure: zip24_lp_aligned_overlay.png — channel panels 5–6 of 12, top to bottom

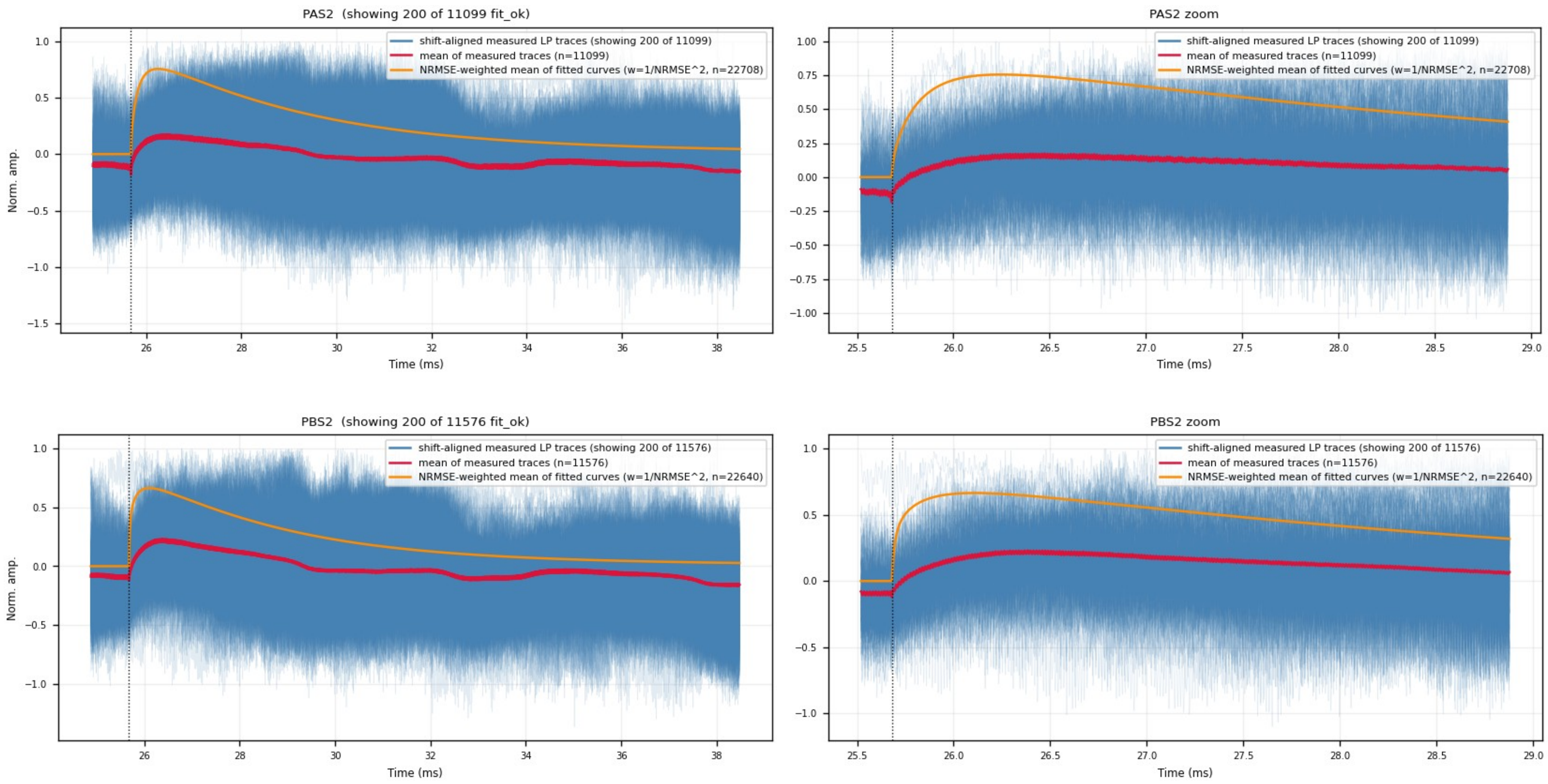
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z24 · aligned_overlay — aligned measured traces (4/6)

figure: zip24_lp_aligned_overlay.png — channel panels 7–8 of 12, top to bottom

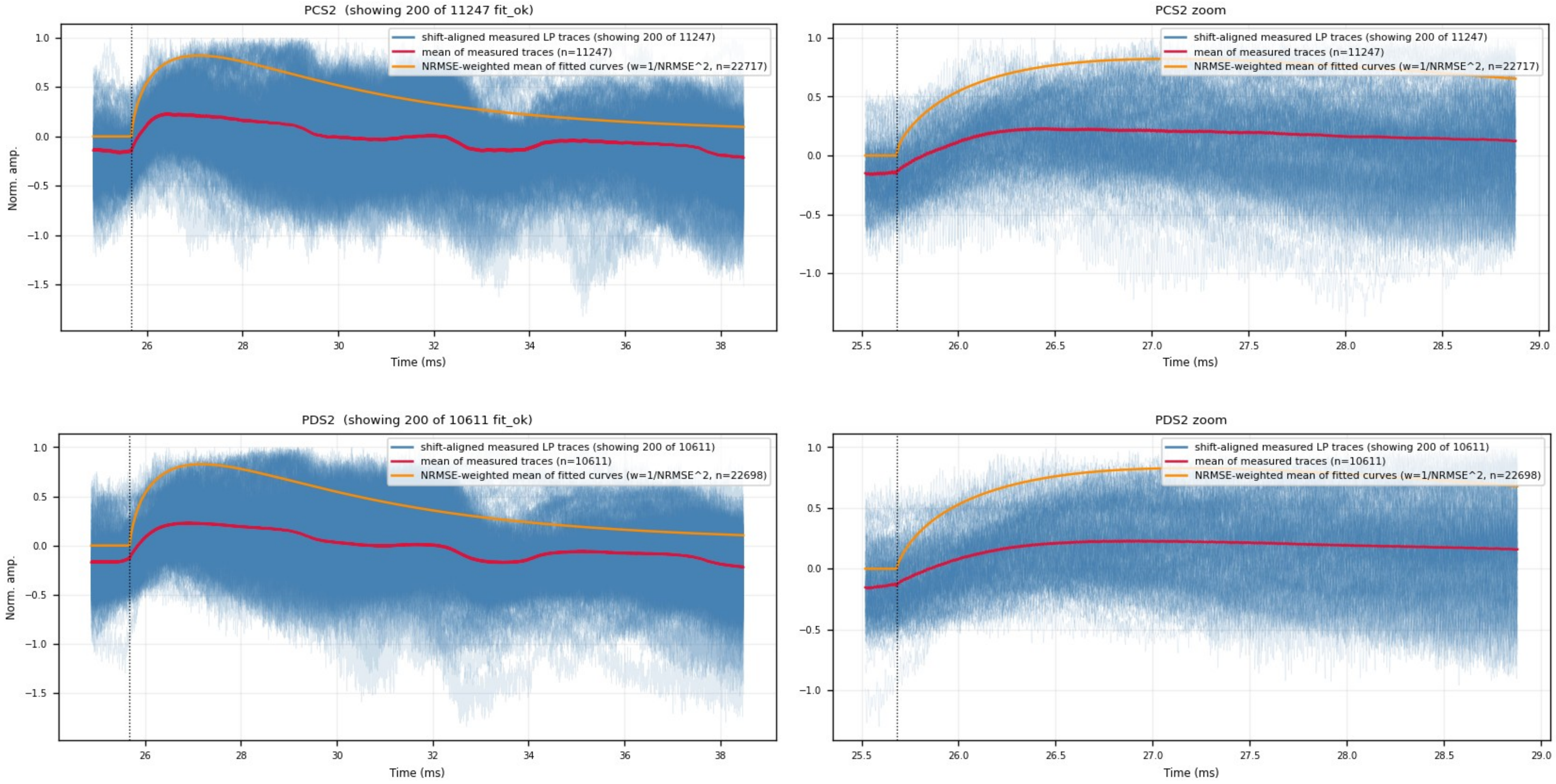
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z24 · aligned_overlay — aligned measured traces (5/6)

figure: zip24_lp_aligned_overlay.png — channel panels 9–10 of 12, top to bottom

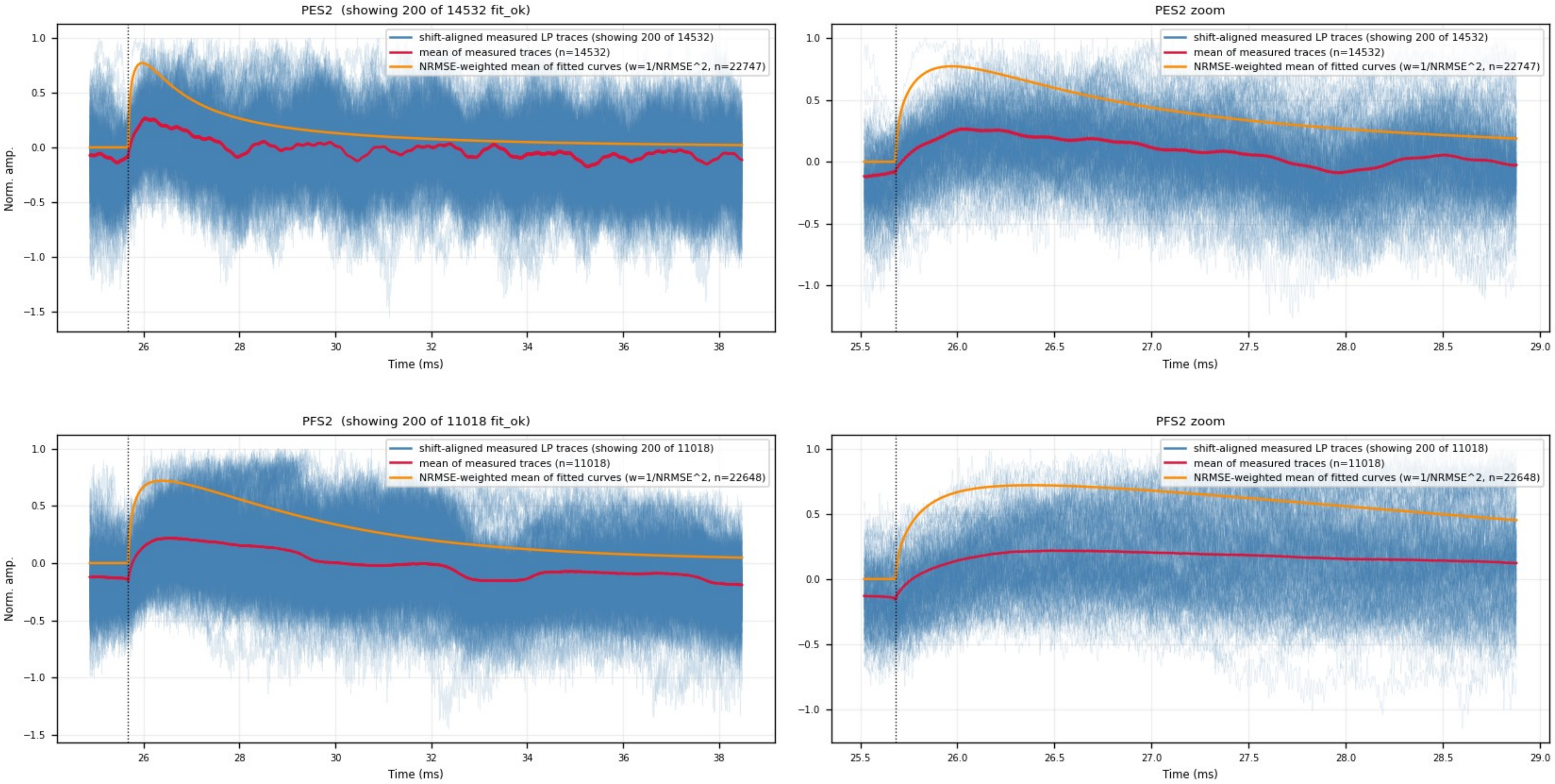
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z24 · aligned_overlay — aligned measured traces (6/6)

figure: zip24_lp_aligned_overlay.png — channel panels 11–12 of 12, top to bottom

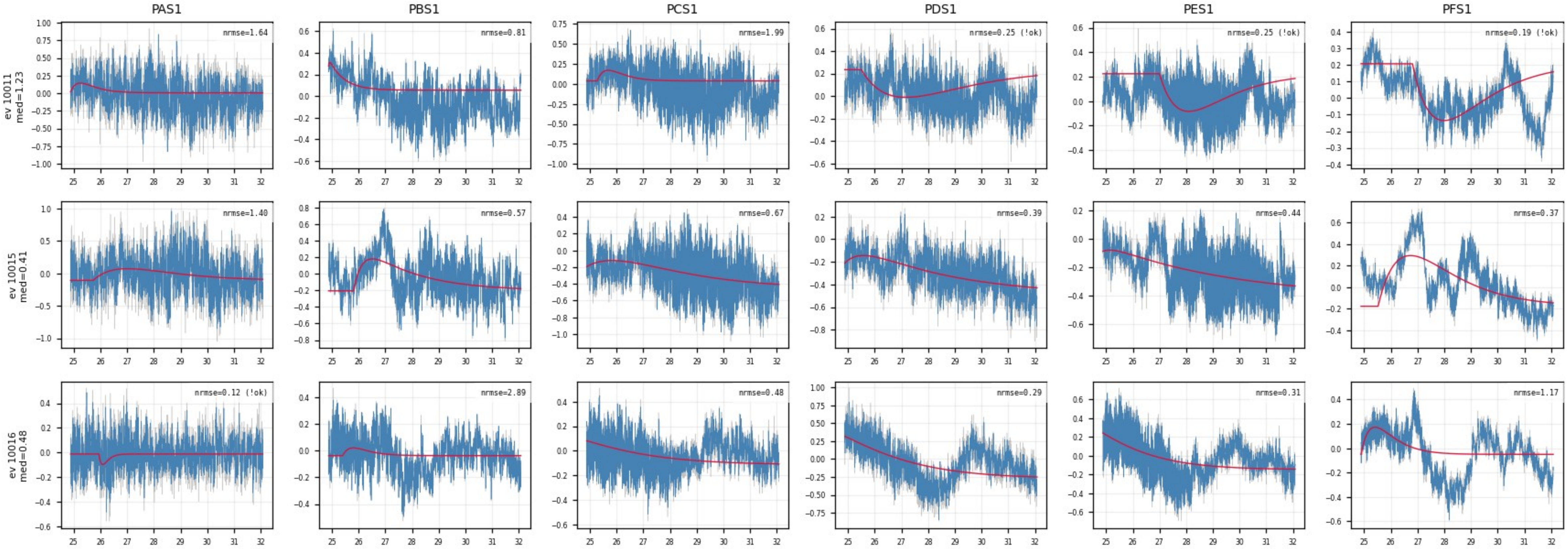
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z24 · slow_rise_events — NRMSE-rejected events (1/10)

figure: zip24_slow_rise_events.png — event rows 1–3 of 15, left half of the channels

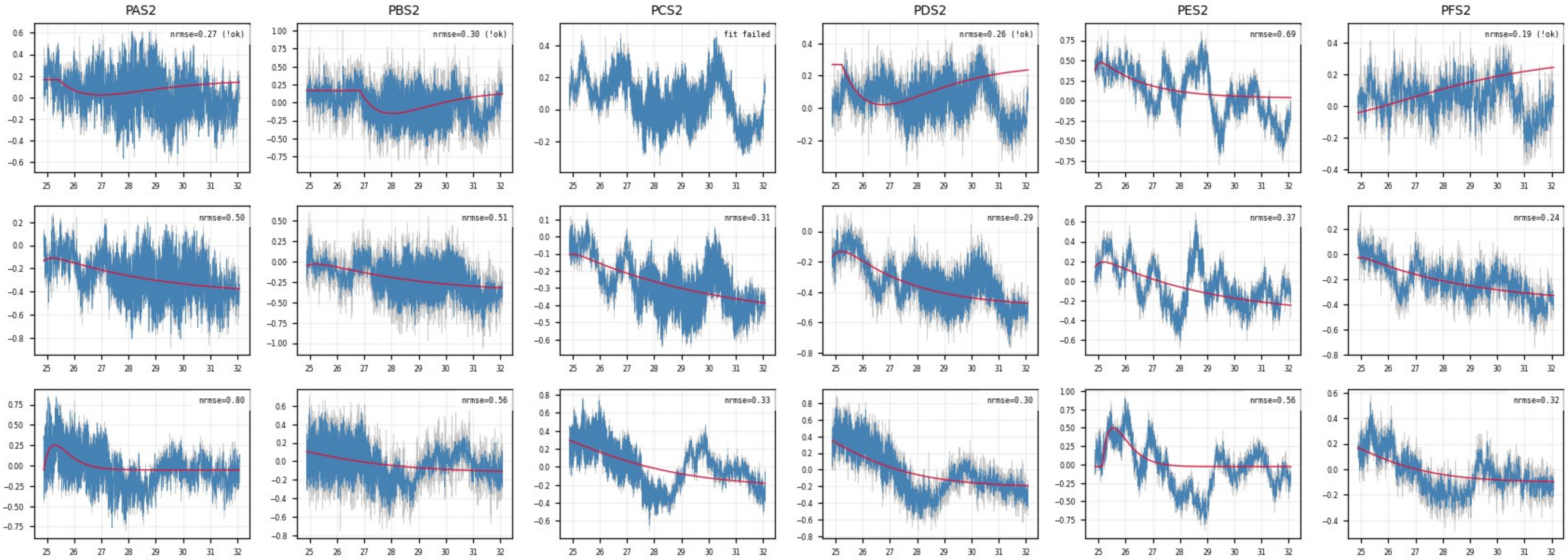
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (2/10)

figure: zip24_slow_rise_events.png — event rows 1–3 of 15, right half of the channels

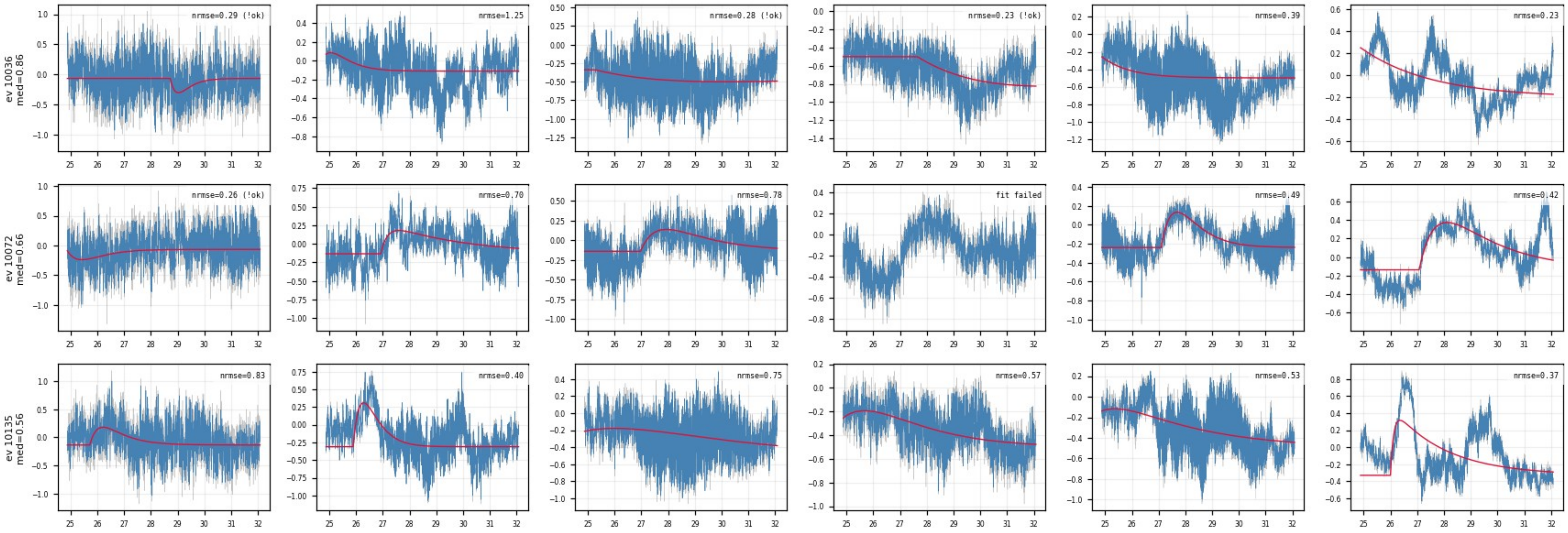
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (3/10)

figure: zip24_slow_rise_events.png — event rows 4–6 of 15, left half of the channels

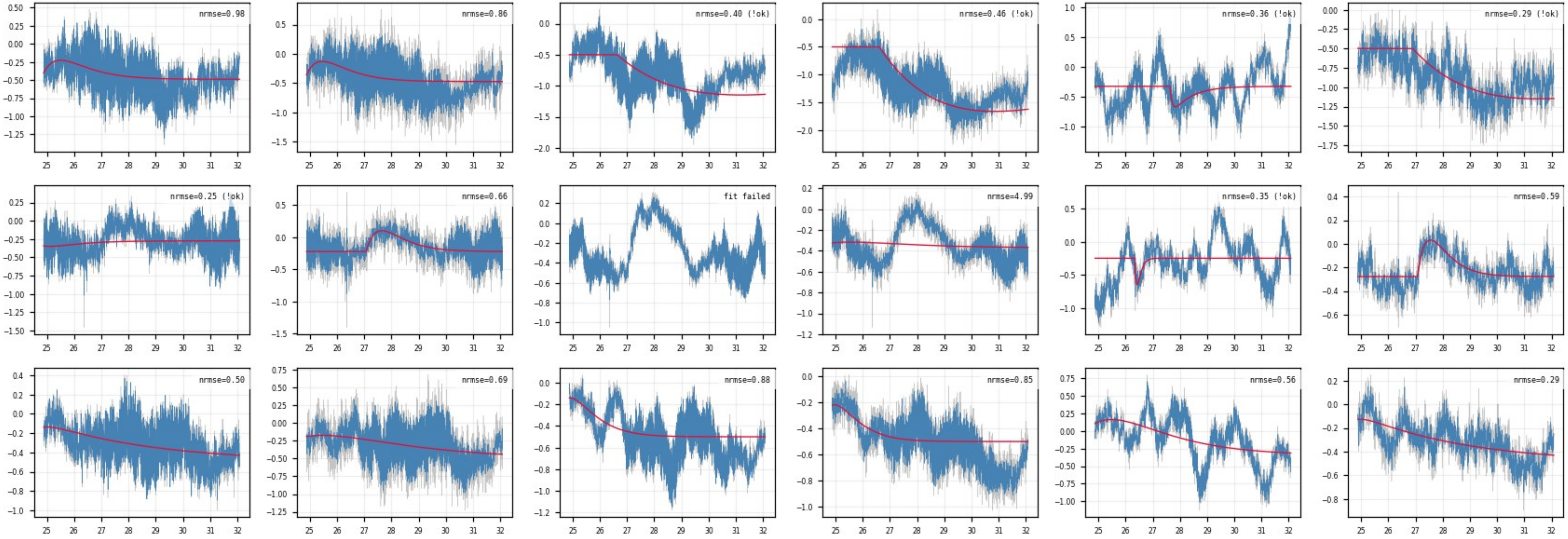
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (4/10)

figure: zip24_slow_rise_events.png — event rows 4–6 of 15, right half of the channels

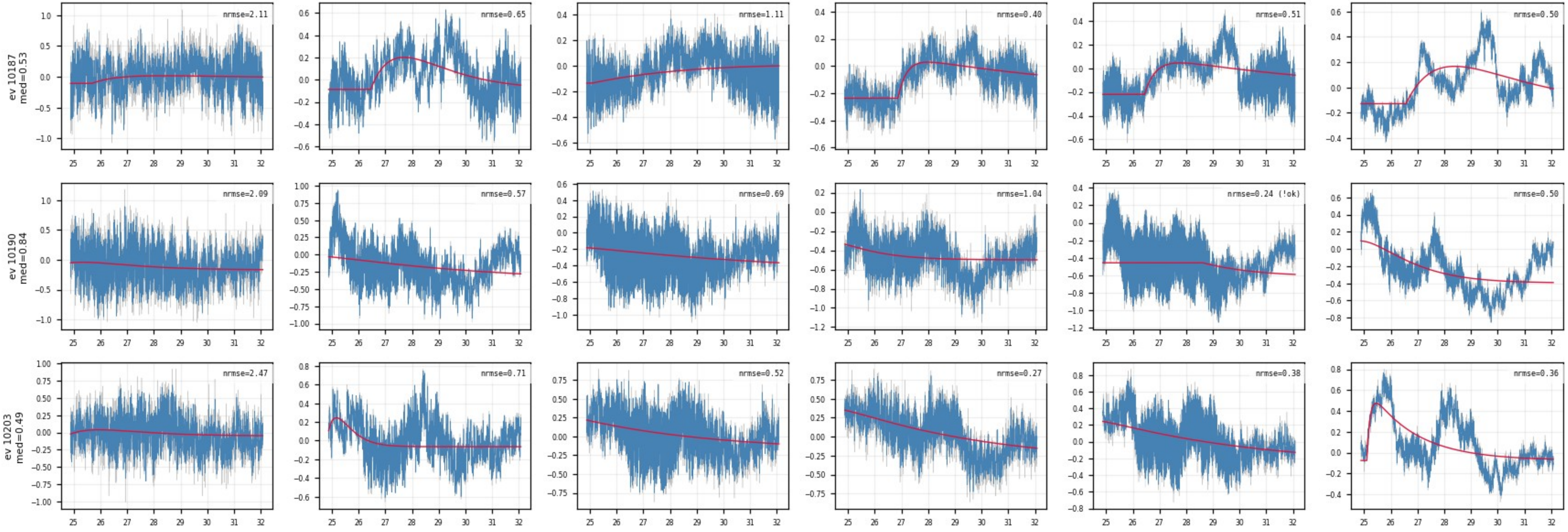
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (5/10)

figure: zip24_slow_rise_events.png — event rows 7–9 of 15, left half of the channels

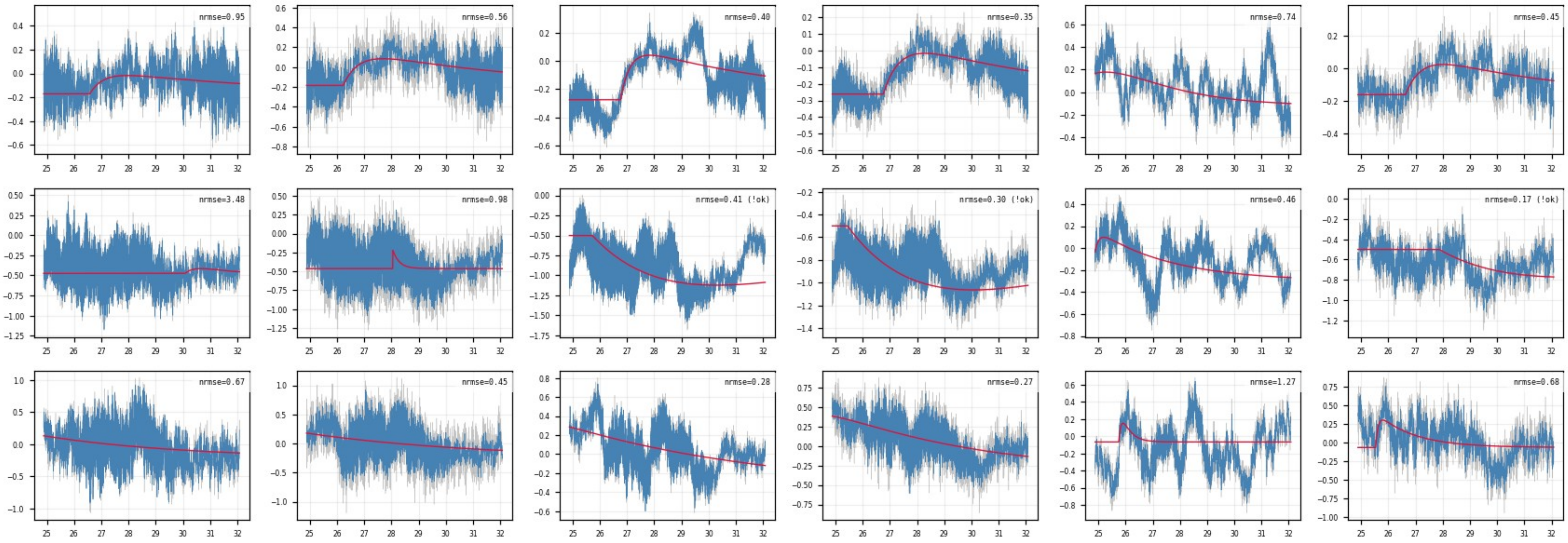
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (6/10)

figure: zip24_slow_rise_events.png — event rows 7–9 of 15, right half of the channels

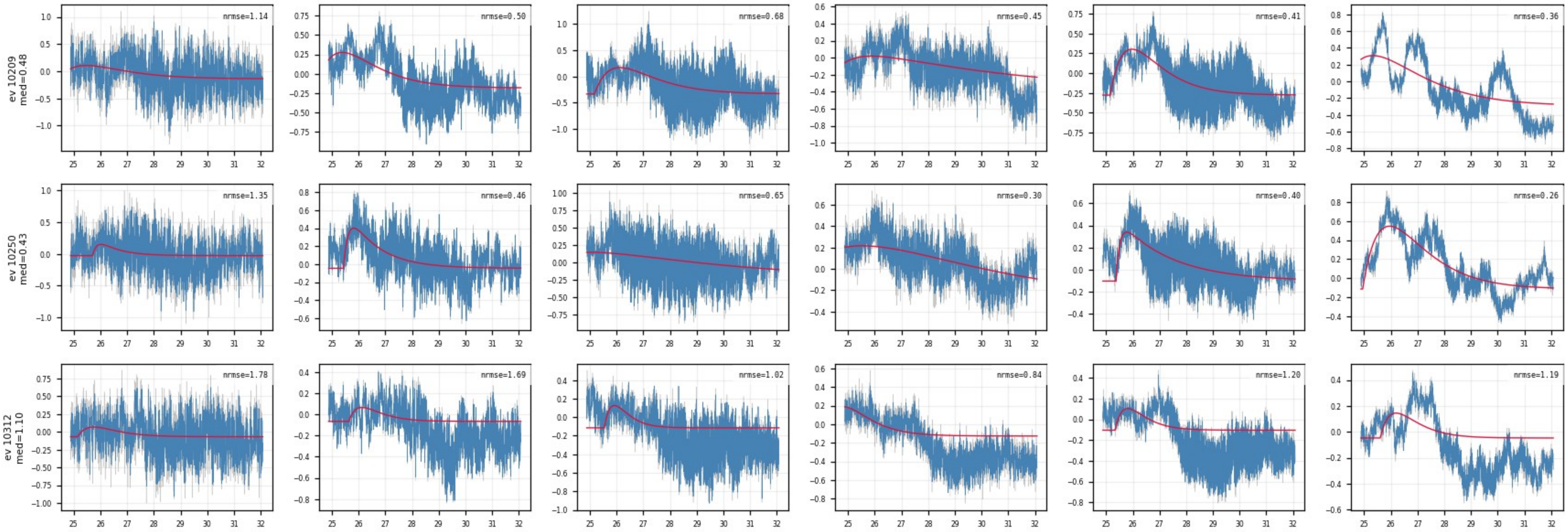
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (7/10)

figure: zip24_slow_rise_events.png — event rows 10–12 of 15, left half of the channels

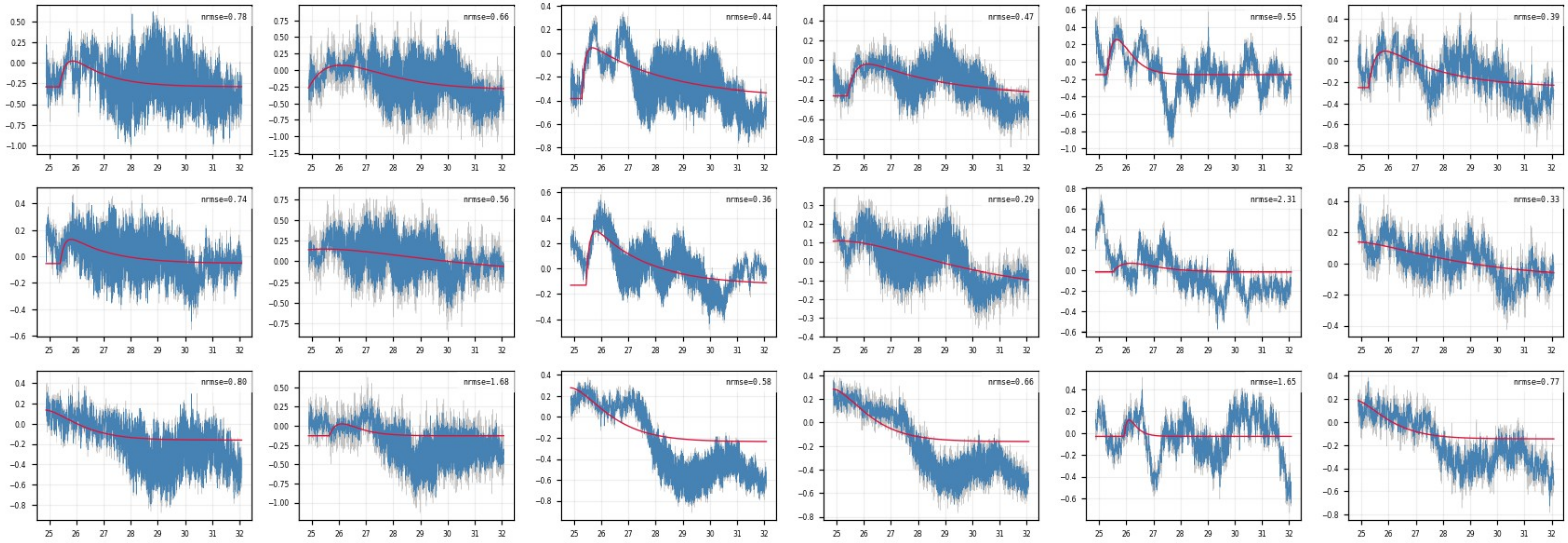
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (8/10)

figure: zip24_slow_rise_events.png — event rows 10–12 of 15, right half of the channels

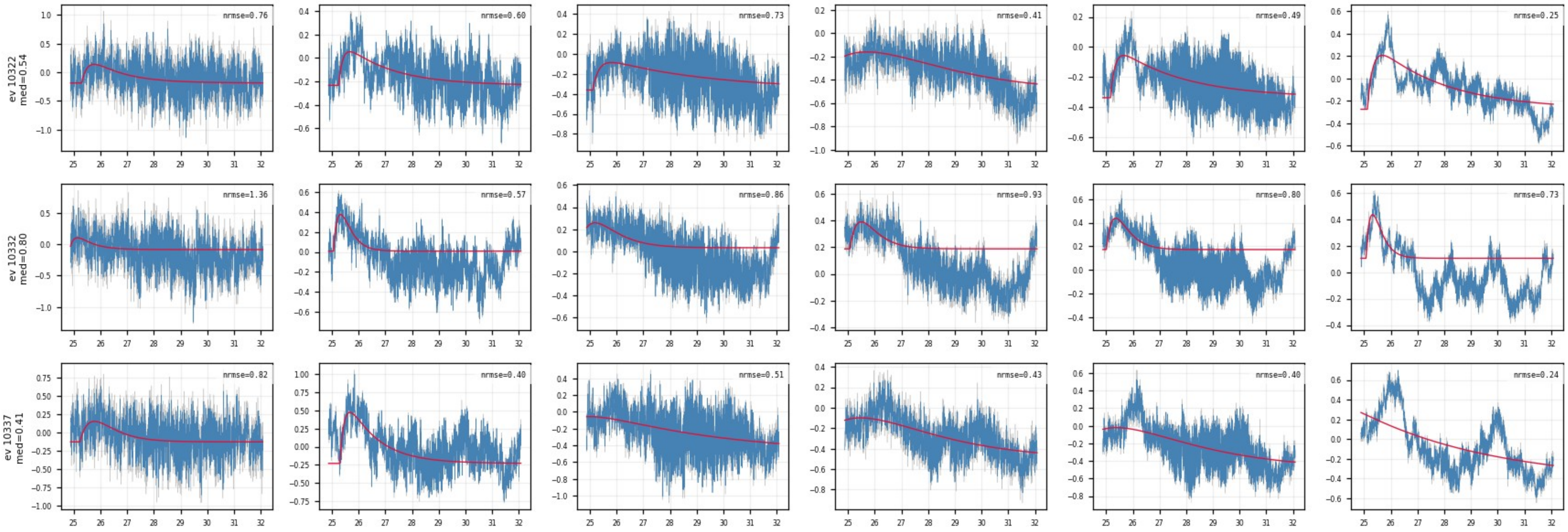
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (9/10)

figure: zip24_slow_rise_events.png — event rows 13–15 of 15, left half of the channels

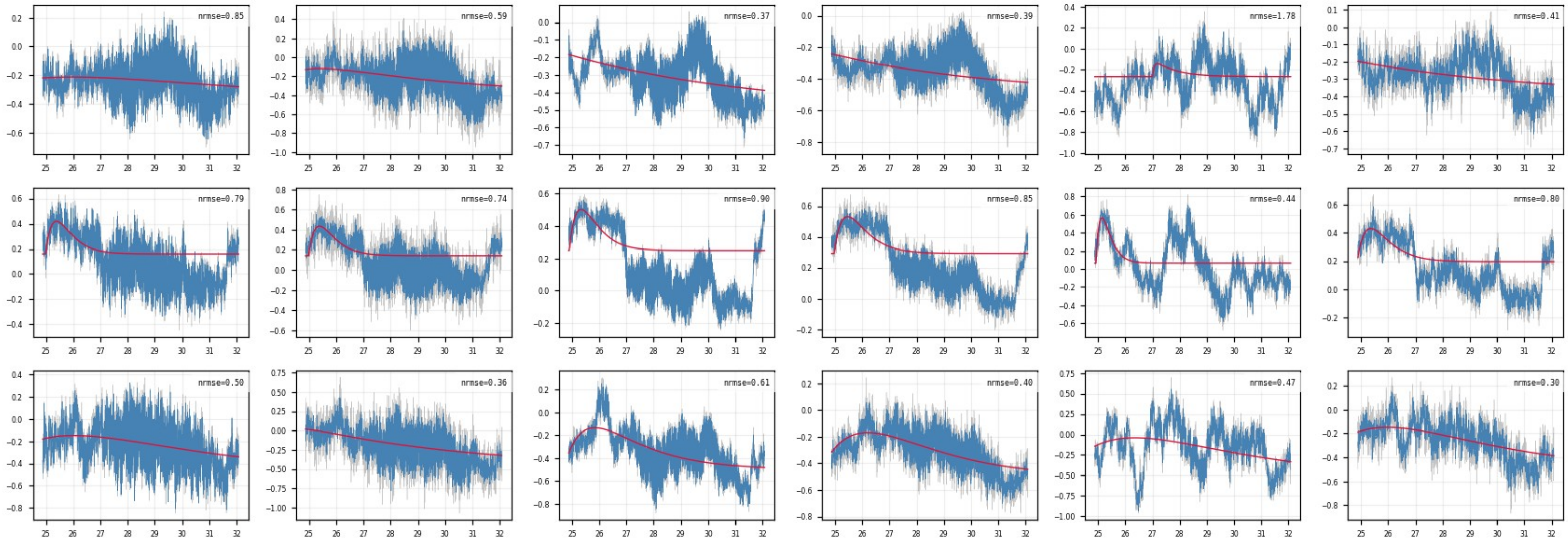
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · slow_rise_events — NRMSE-rejected events (10/10)

figure: zip24_slow_rise_events.png — event rows 13–15 of 15, right half of the channels

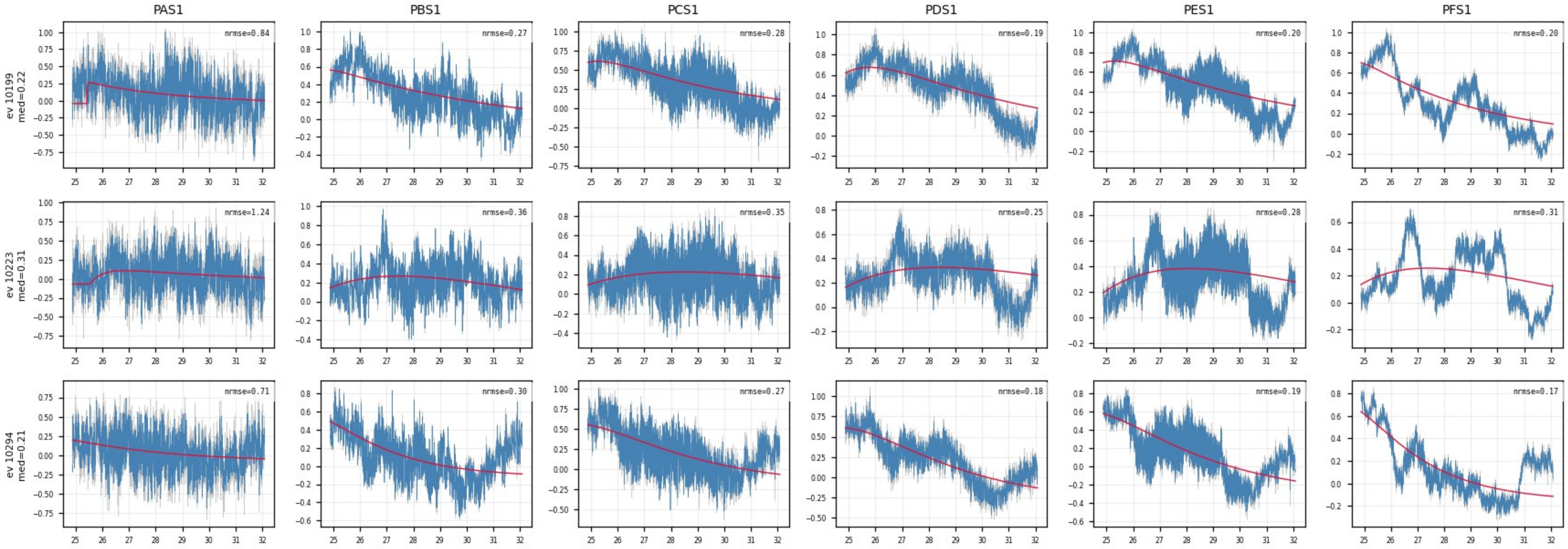
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



Z24 · shadow_events — well-fit slow-rise events (1/10)

figure: zip24_shadow_events.png — event rows 1–3 of 15, left half of the channels

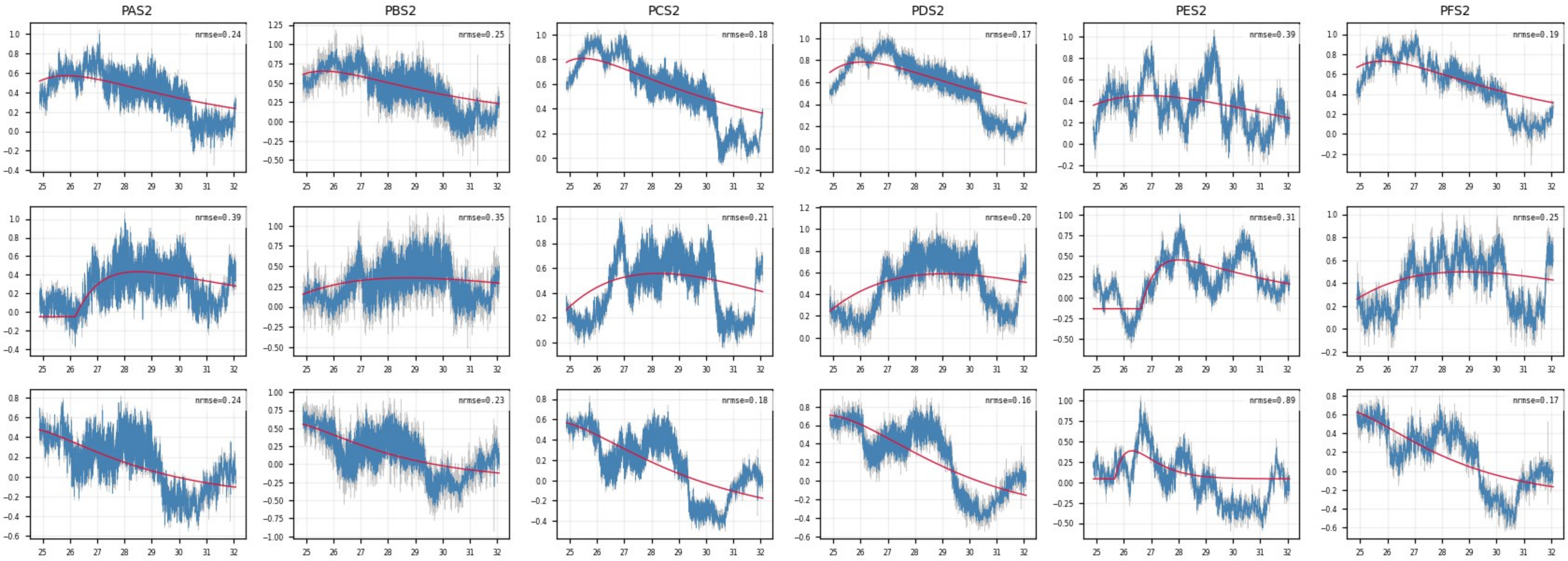
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (2/10)

figure: zip24_shadow_events.png — event rows 1–3 of 15, right half of the channels

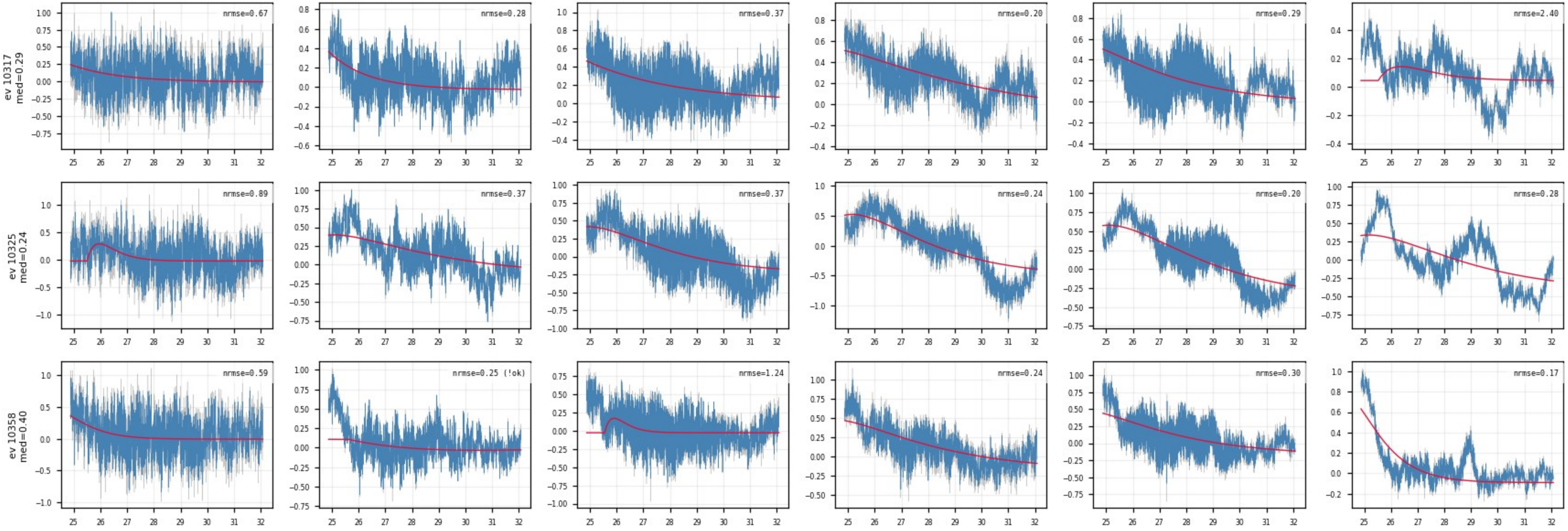
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (3/10)

figure: zip24_shadow_events.png — event rows 4–6 of 15, left half of the channels

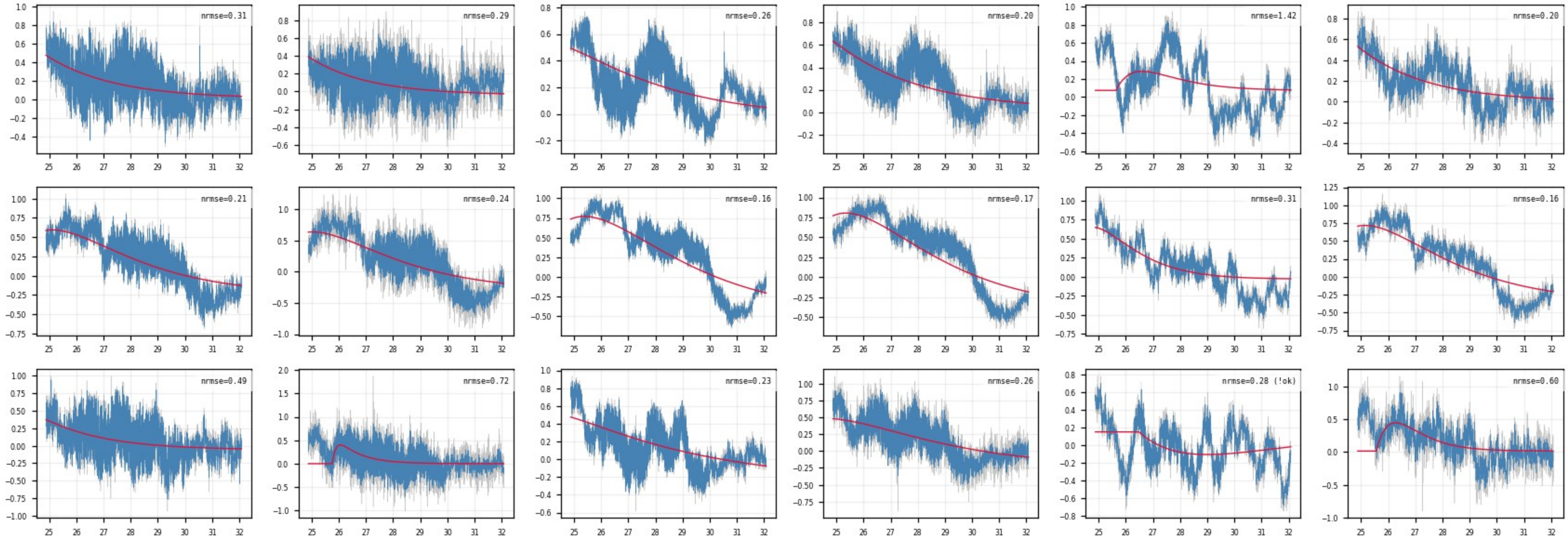
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (4/10)

figure: zip24_shadow_events.png — event rows 4–6 of 15, right half of the channels

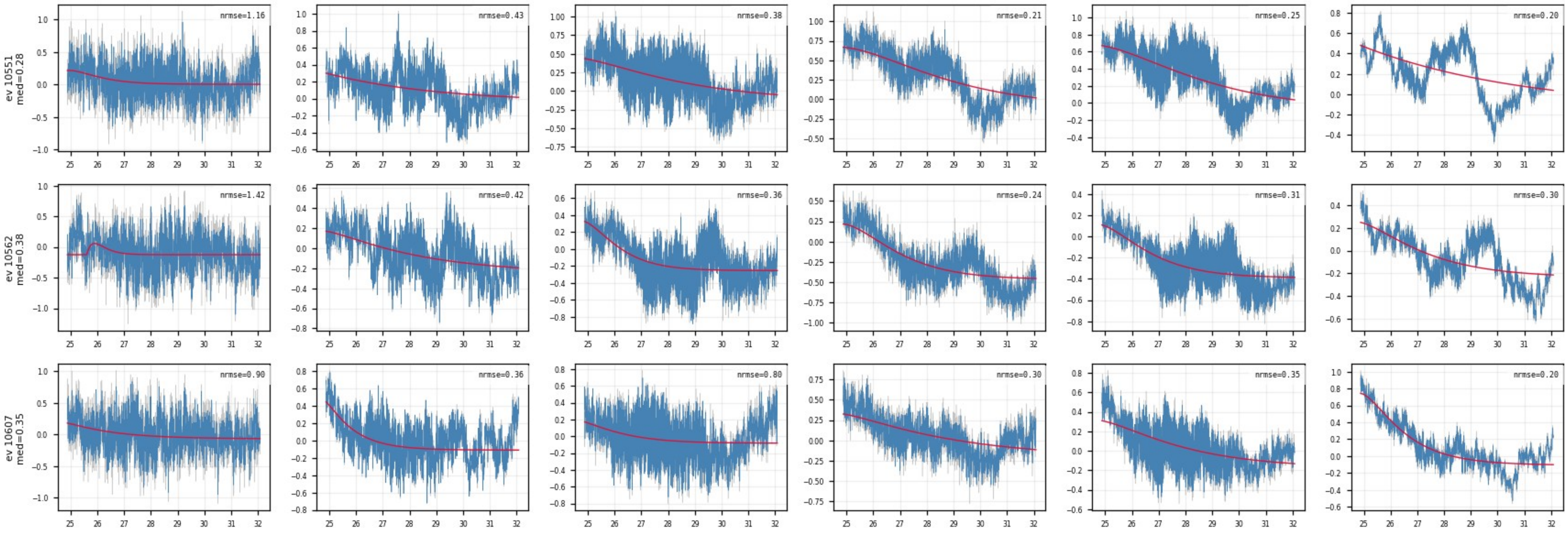
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (5/10)

figure: zip24_shadow_events.png — event rows 7–9 of 15, left half of the channels

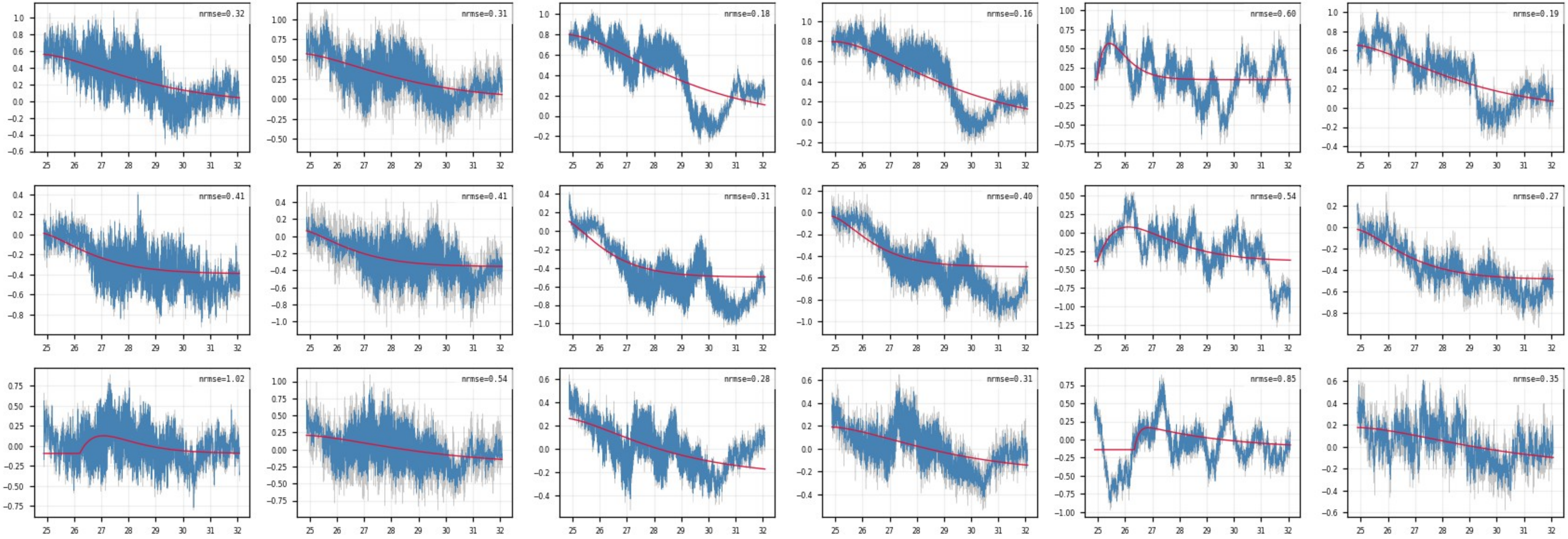
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (6/10)

figure: zip24_shadow_events.png — event rows 7–9 of 15, right half of the channels

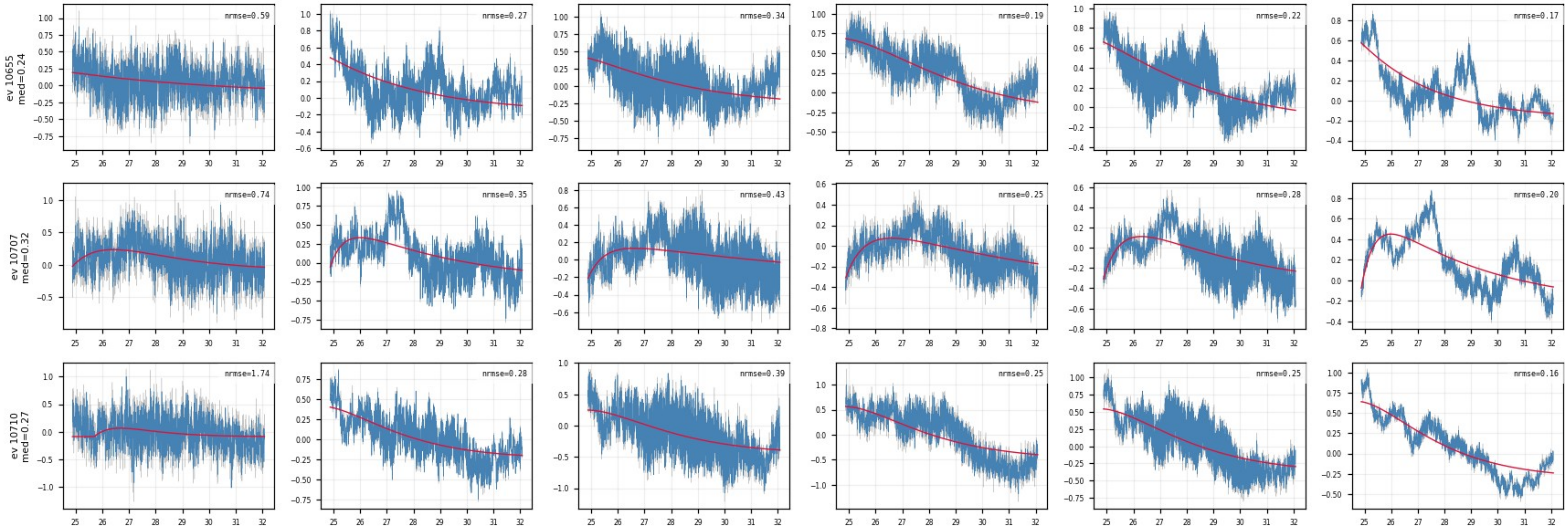
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (7/10)

figure: zip24_shadow_events.png — event rows 10–12 of 15, left half of the channels

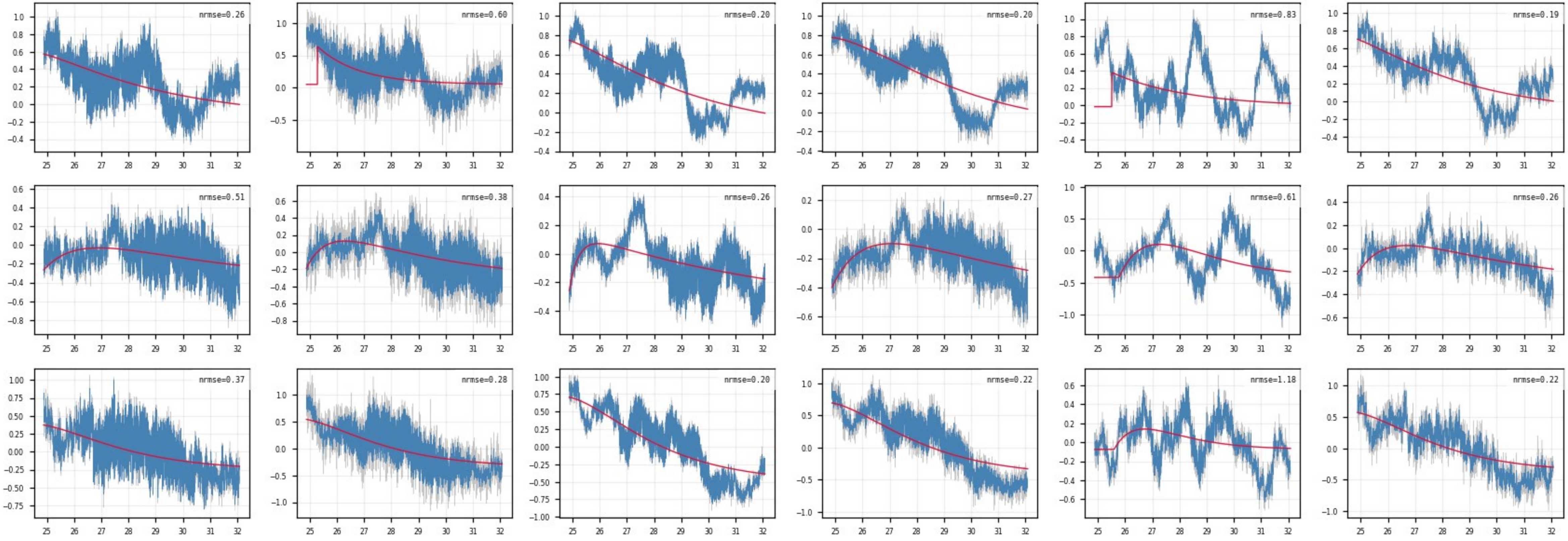
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (8/10)

figure: zip24_shadow_events.png — event rows 10–12 of 15, right half of the channels

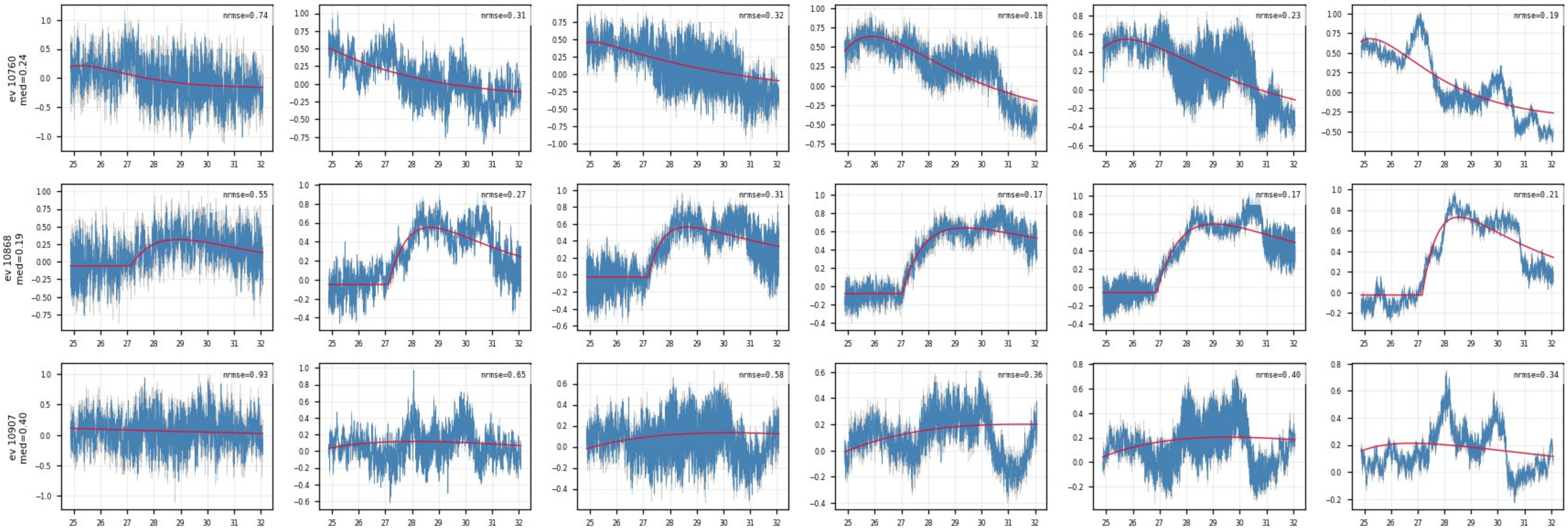
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (9/10)

figure: zip24_shadow_events.png — event rows 13–15 of 15, left half of the channels

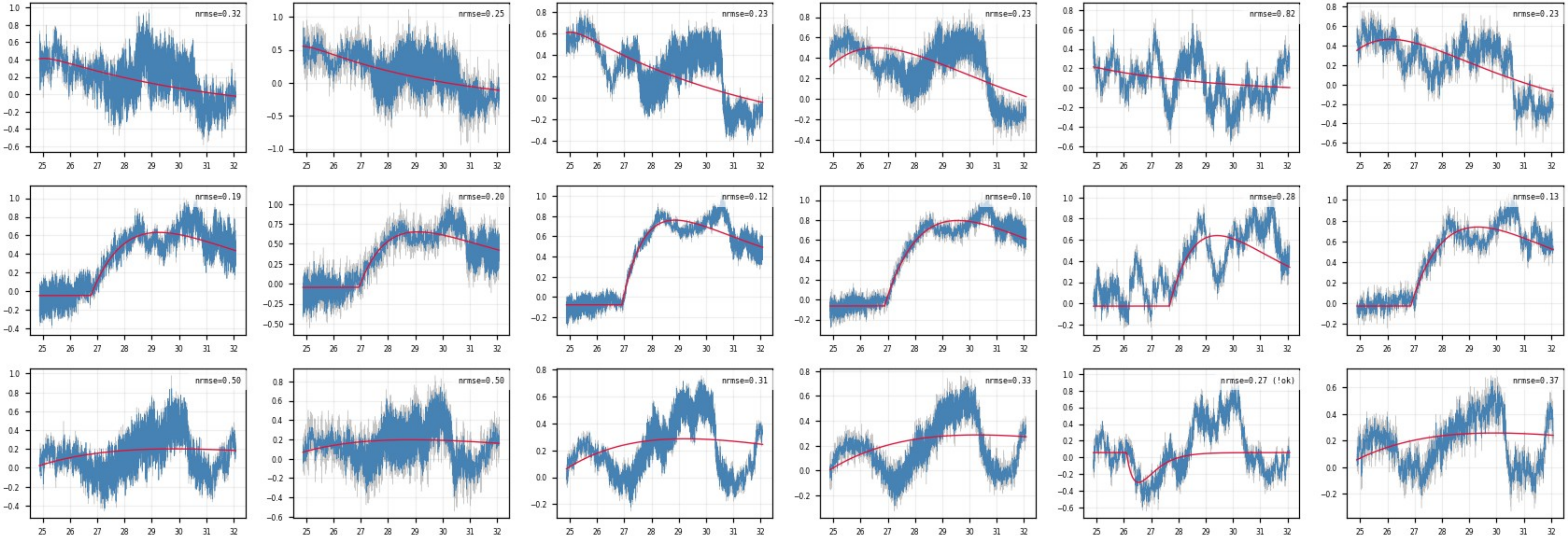
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · shadow_events — well-fit slow-rise events (10/10)

figure: zip24_shadow_events.png — event rows 13–15 of 15, right half of the channels

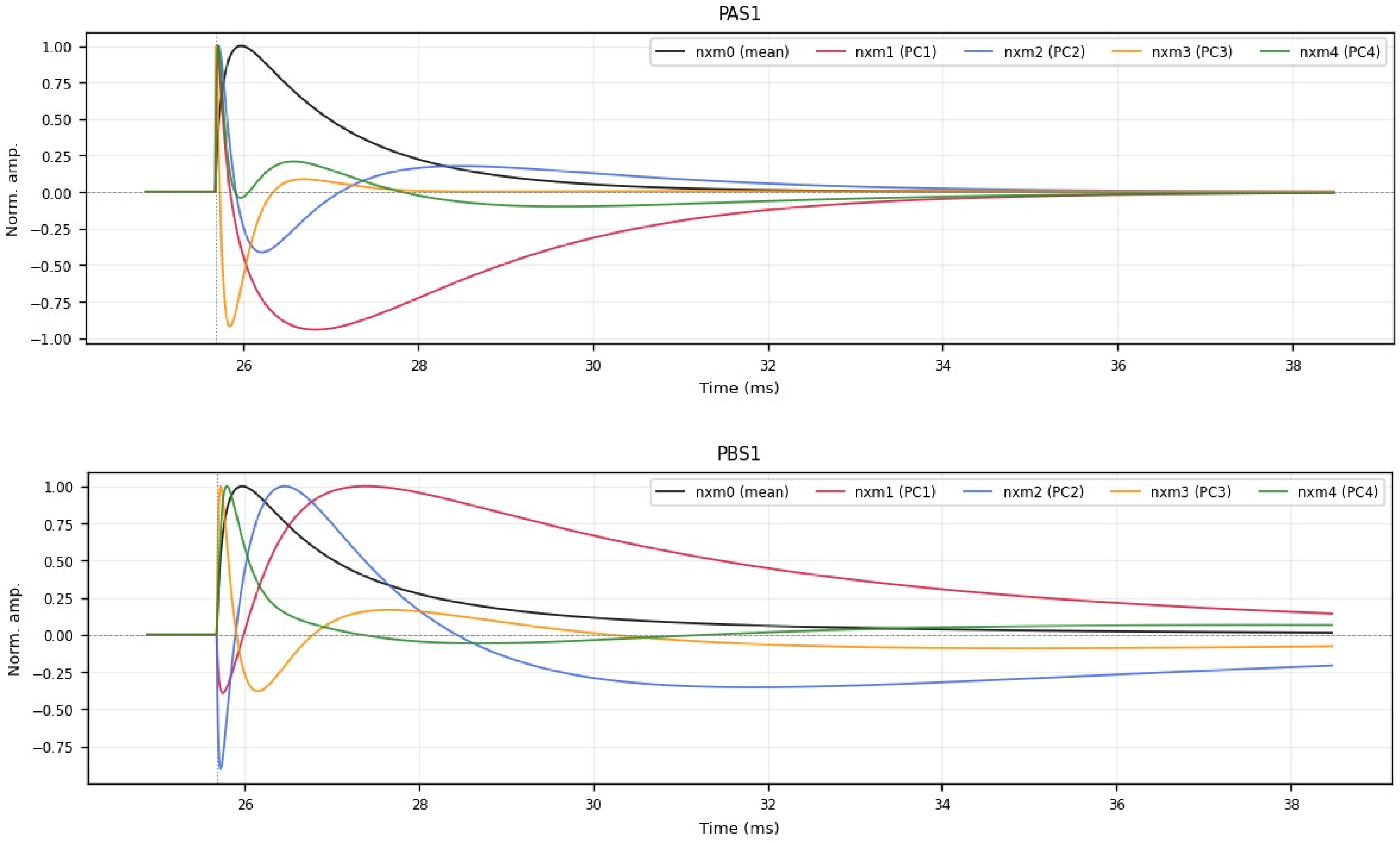
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



Z24 · deliverables/nxm — PCA templates nxm0–4 (1/6)

figure: zip24_pca_templates.png — channel panels 1–2 of 12, top to bottom

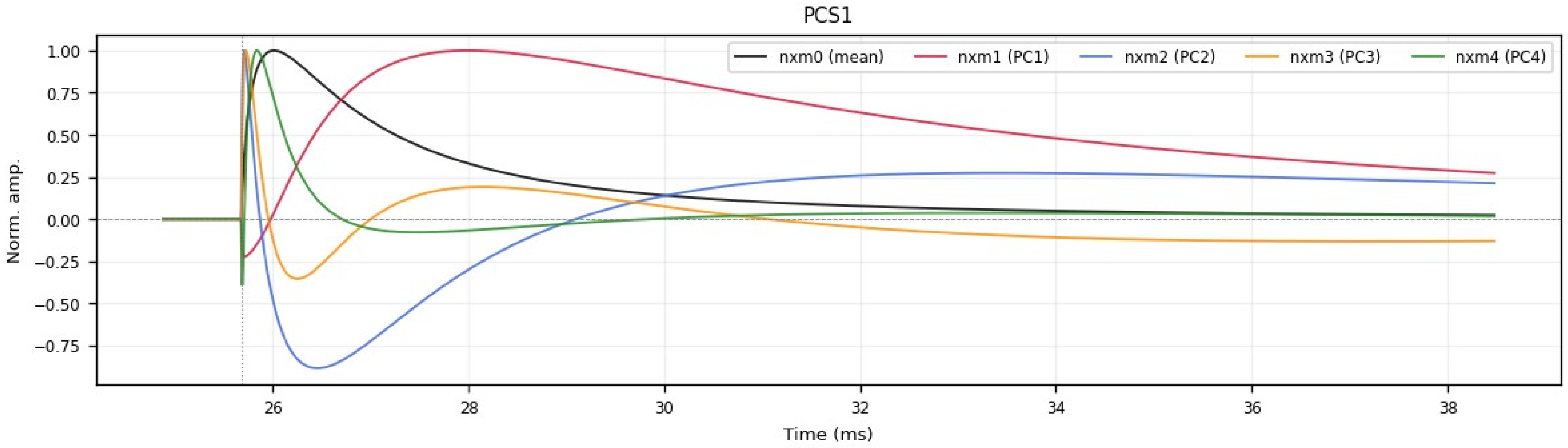
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms.

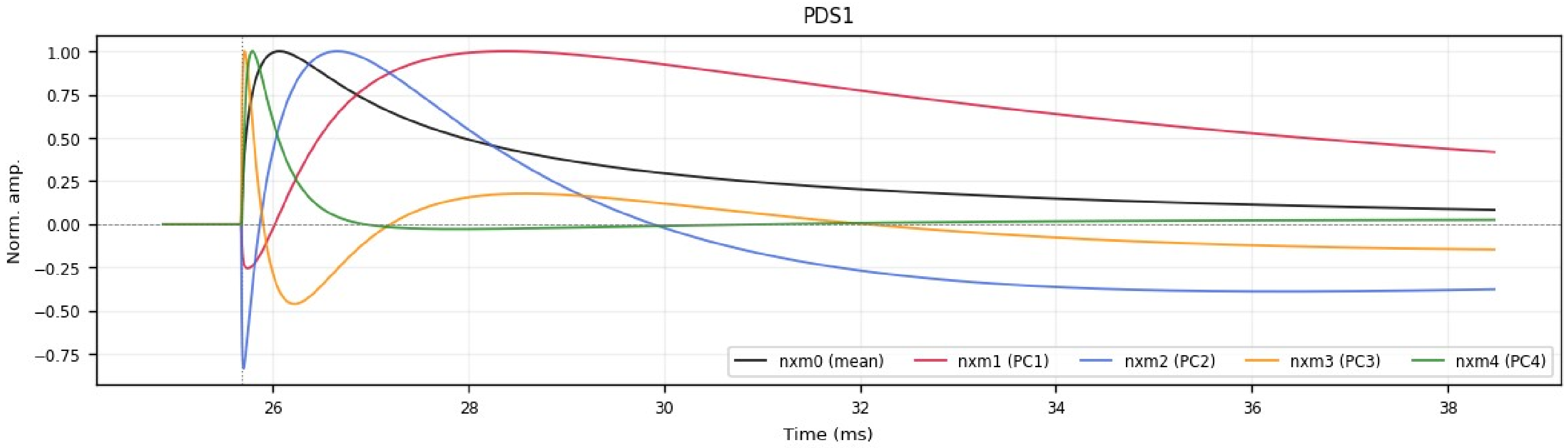


Z24 · deliverables/nxm — PCA templates nxm0–4 (2/6)

figure: zip24_pca_templates.png — channel panels 3–4 of 12, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms.

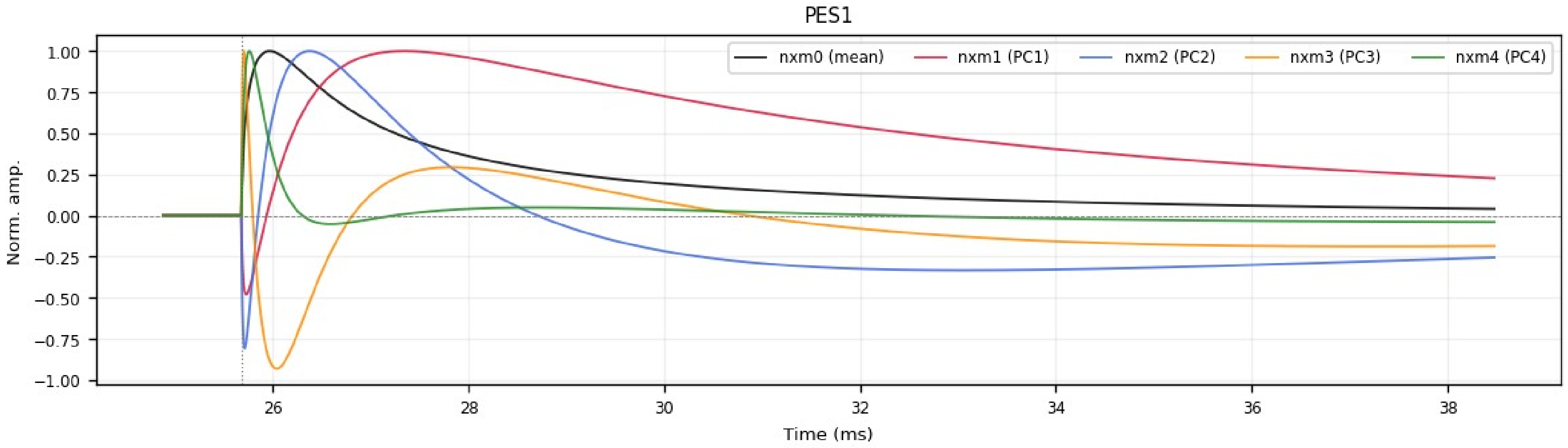


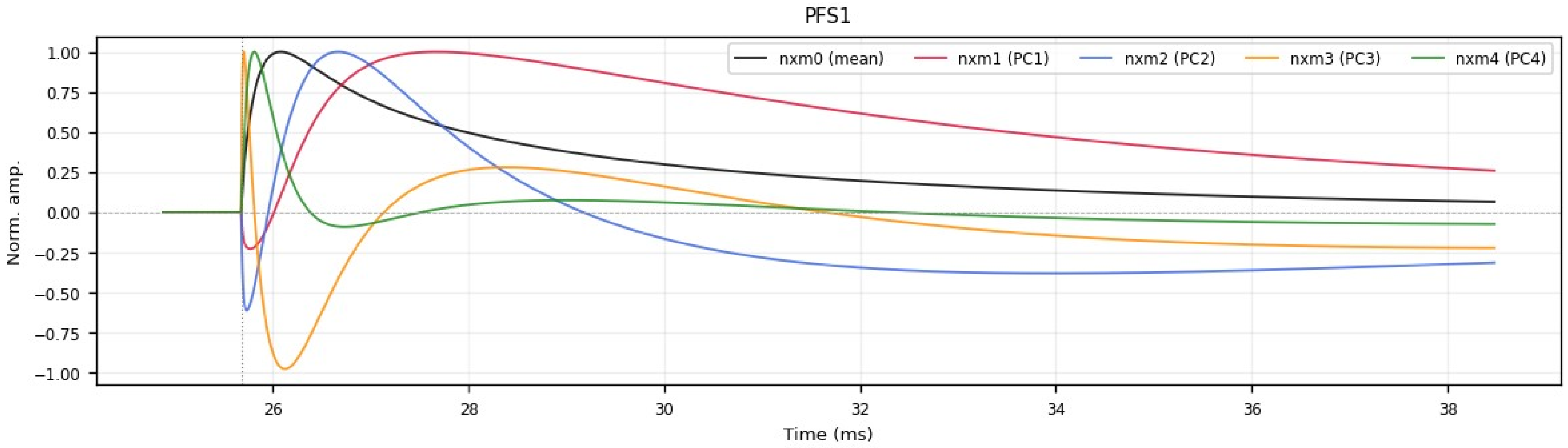


Z24 · deliverables/nxm — PCA templates nxm0–4 (3/6)

figure: zip24_pca_templates.png — channel panels 5–6 of 12, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms.

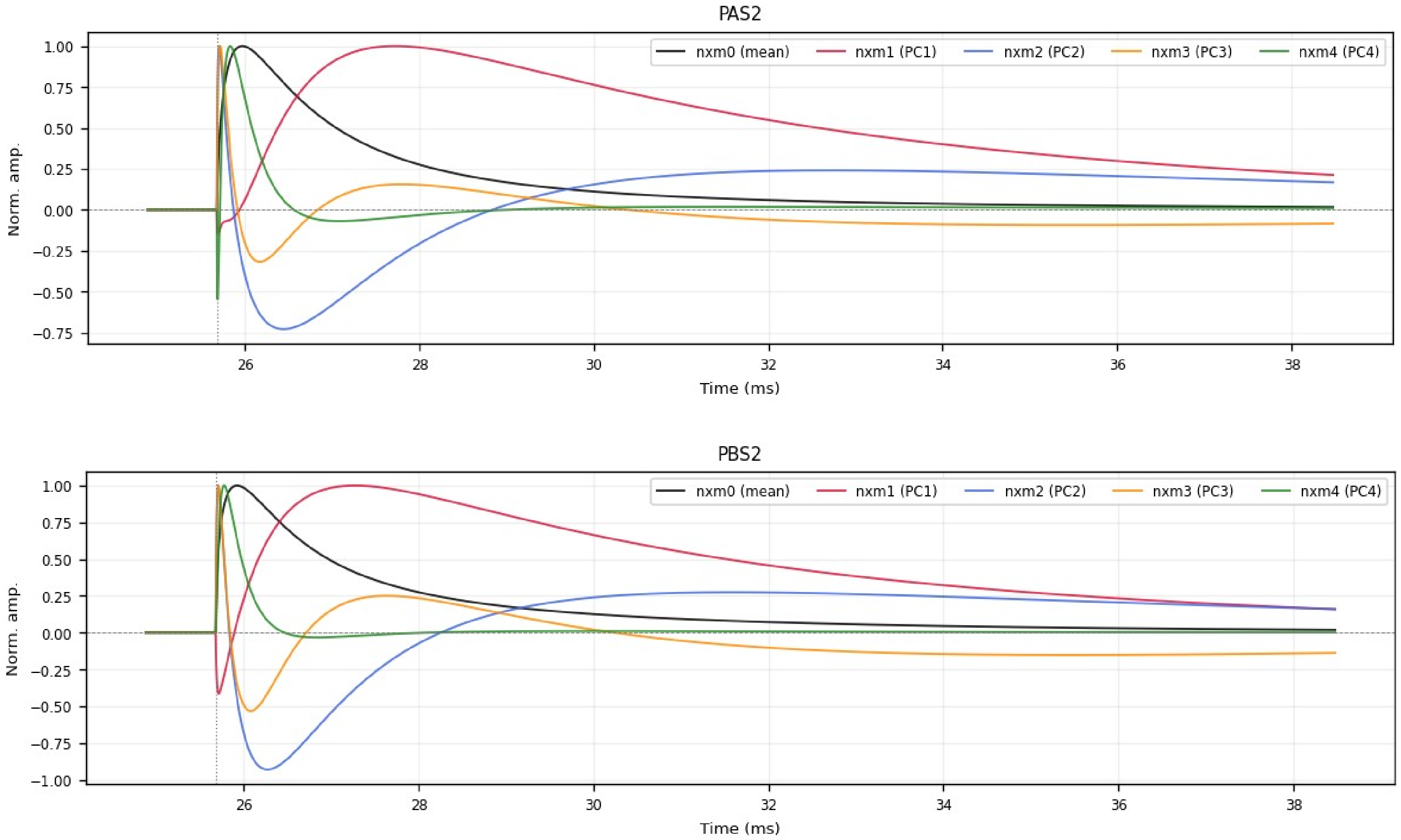




Z24 · deliverables/nxm — PCA templates nxm0–4 (4/6)

figure: zip24_pca_templates.png — channel panels 7–8 of 12, top to bottom

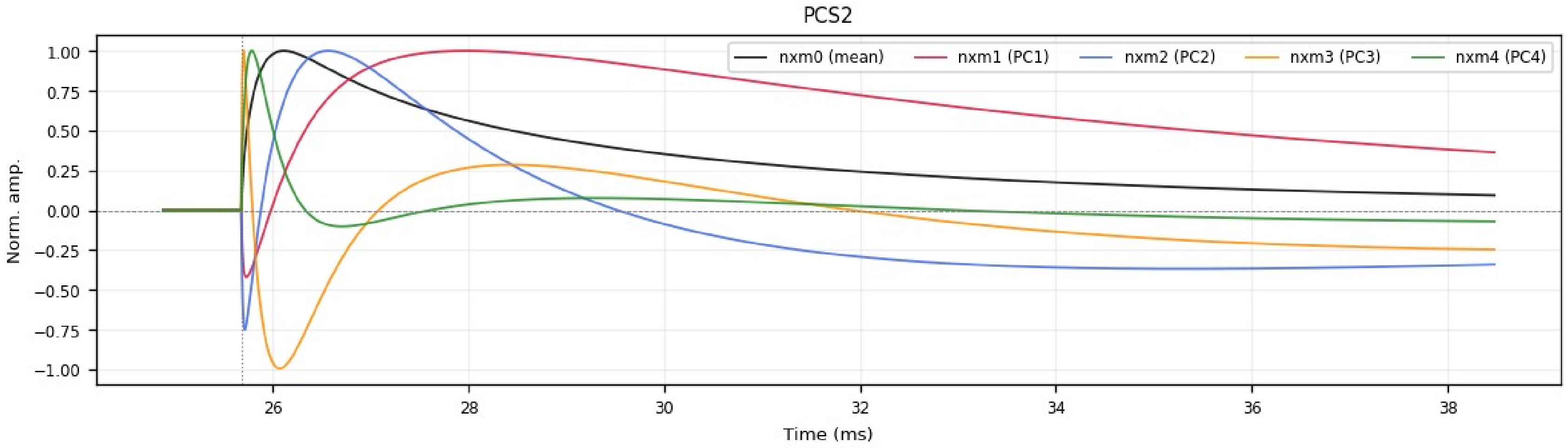
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms.

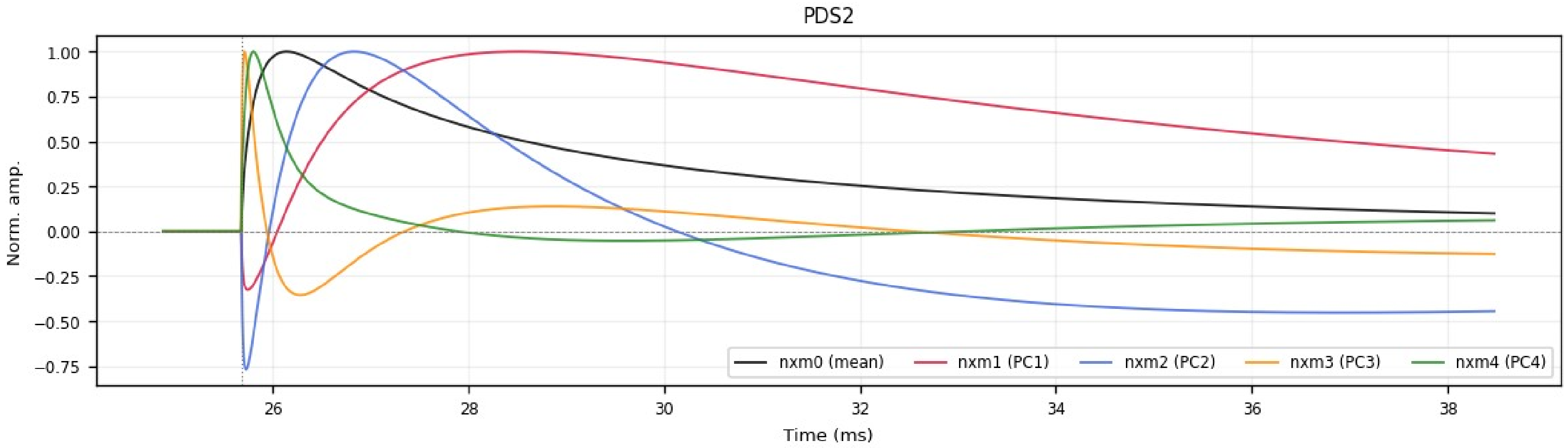


Z24 · deliverables/nxm — PCA templates nxm0–4 (5/6)

figure: zip24_pca_templates.png — channel panels 9–10 of 12, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms.

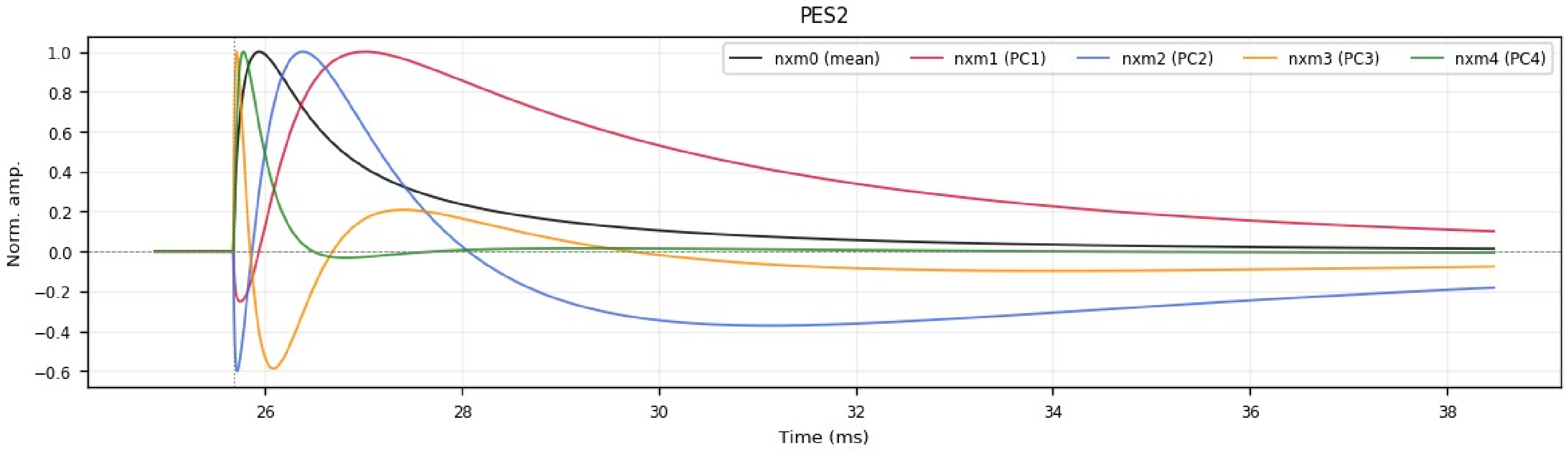


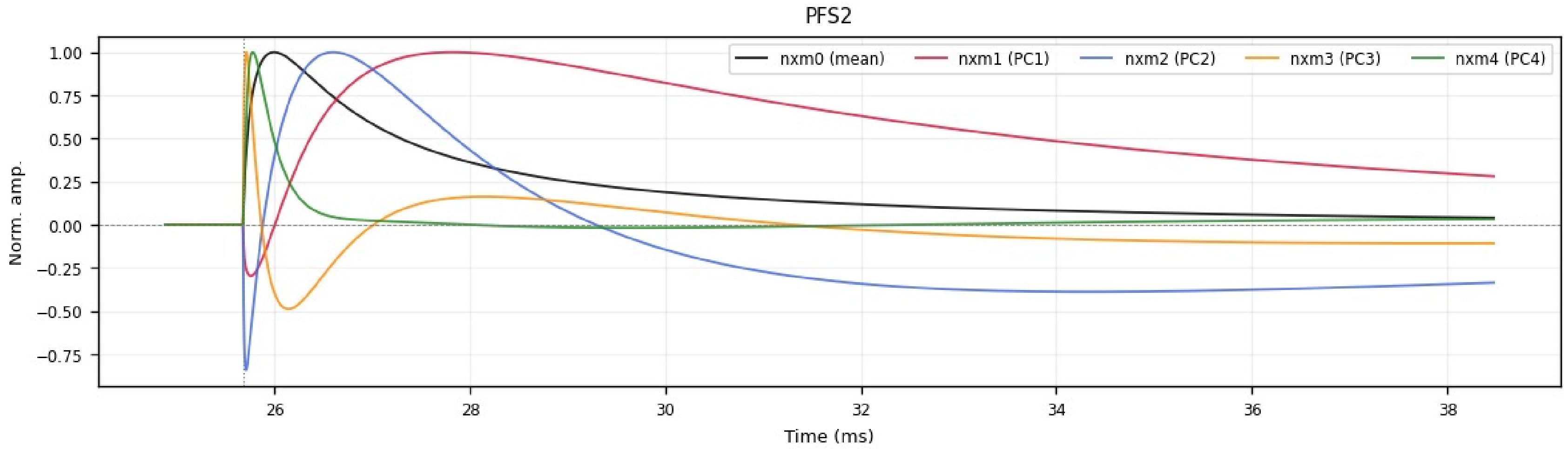


Z24 · deliverables/nxm — PCA templates nxm0–4 (6/6)

figure: zip24_pca_templates.png — channel panels 11–12 of 12, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms.

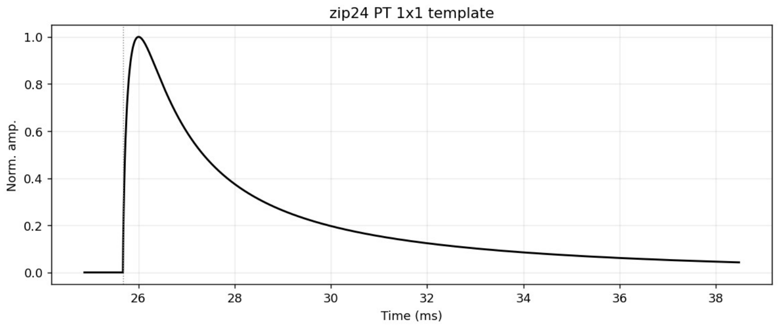




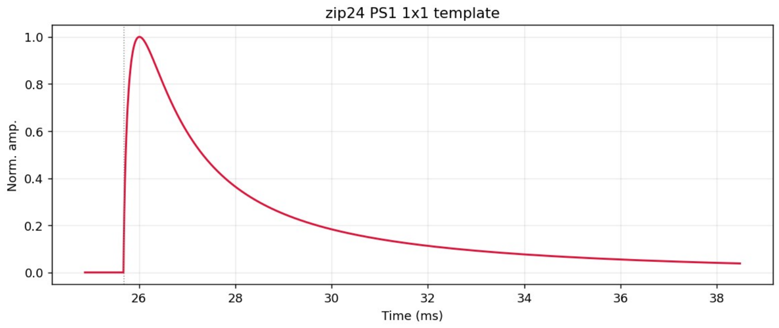
Z24 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip24_*.png

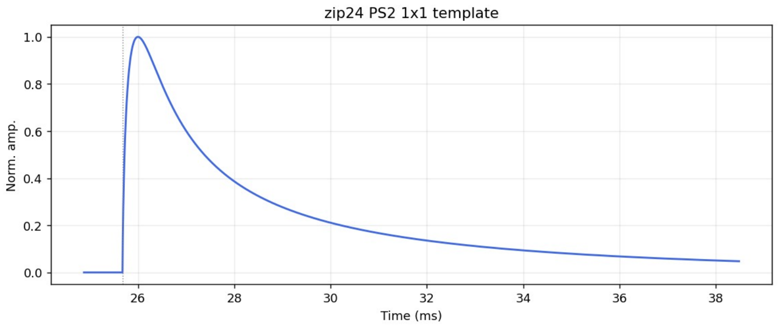
Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates — PT = all channels, PS1 / PS2 = side-1 / side-2 only.



Z24 PT



Z24 PS1



Z24 PS2